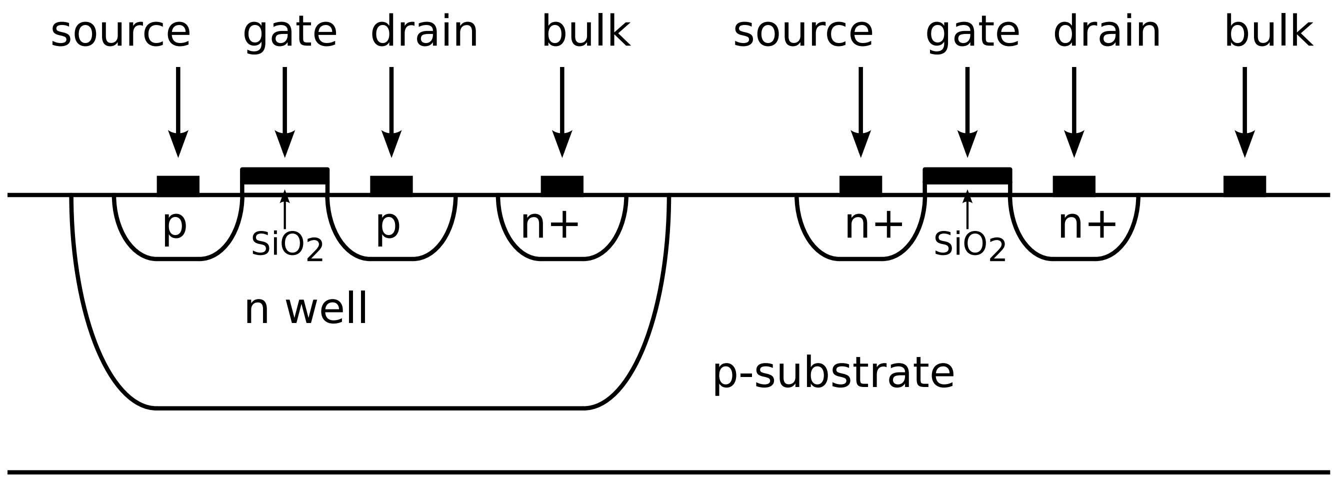
Structured Electronic Design

MOS EKV model

Anton J.M. Montagne

MOS operation

MOS operation



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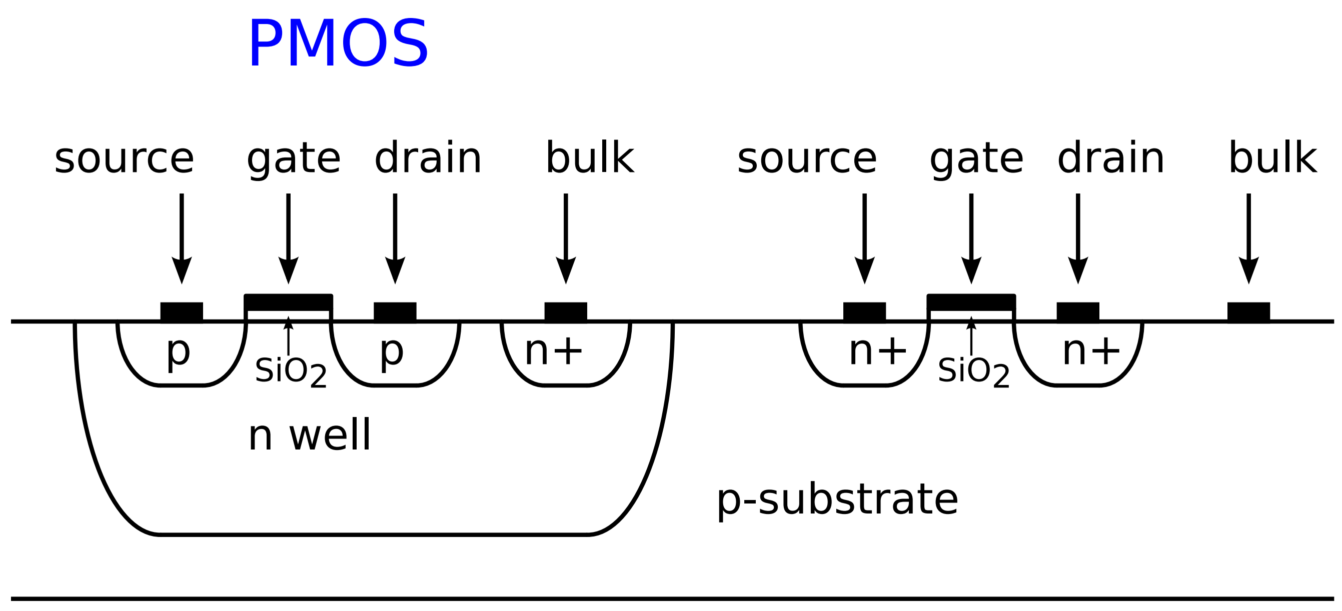

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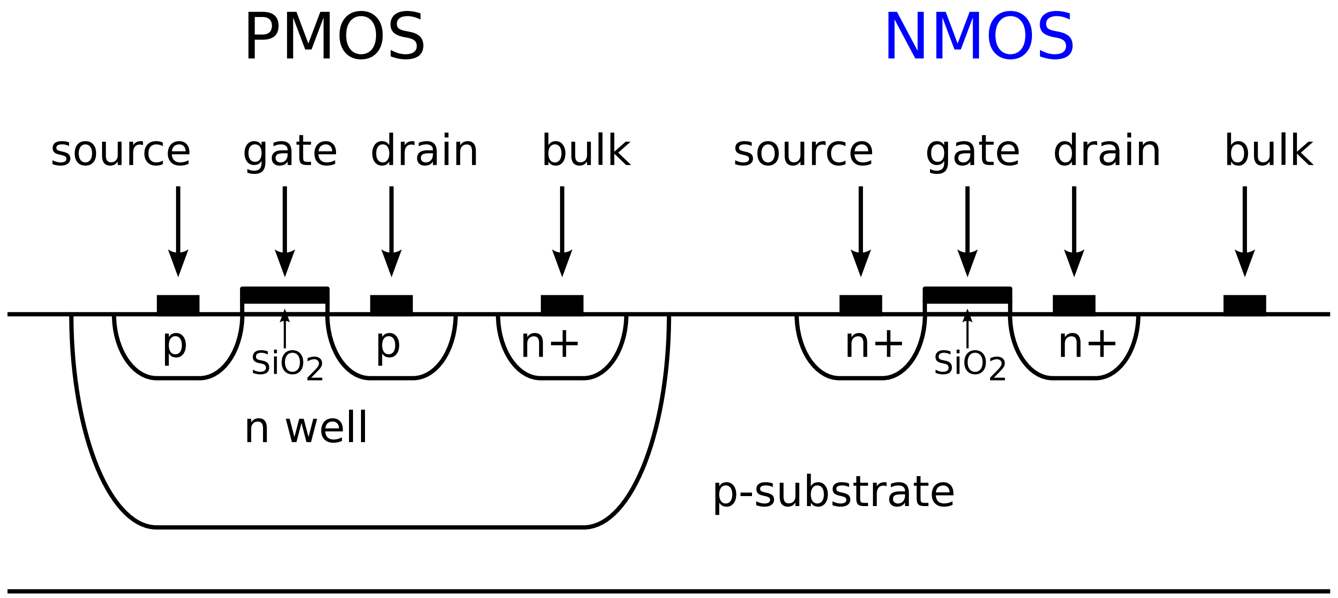

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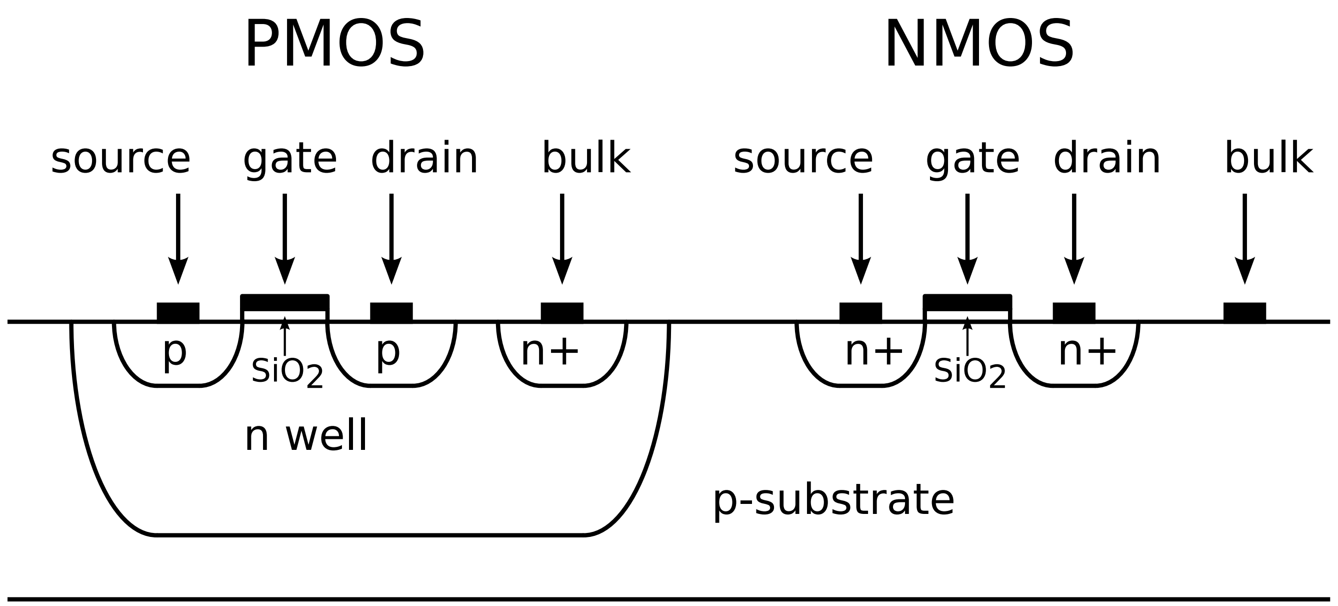
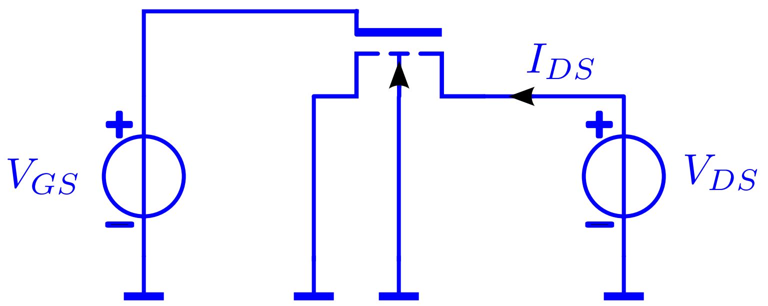
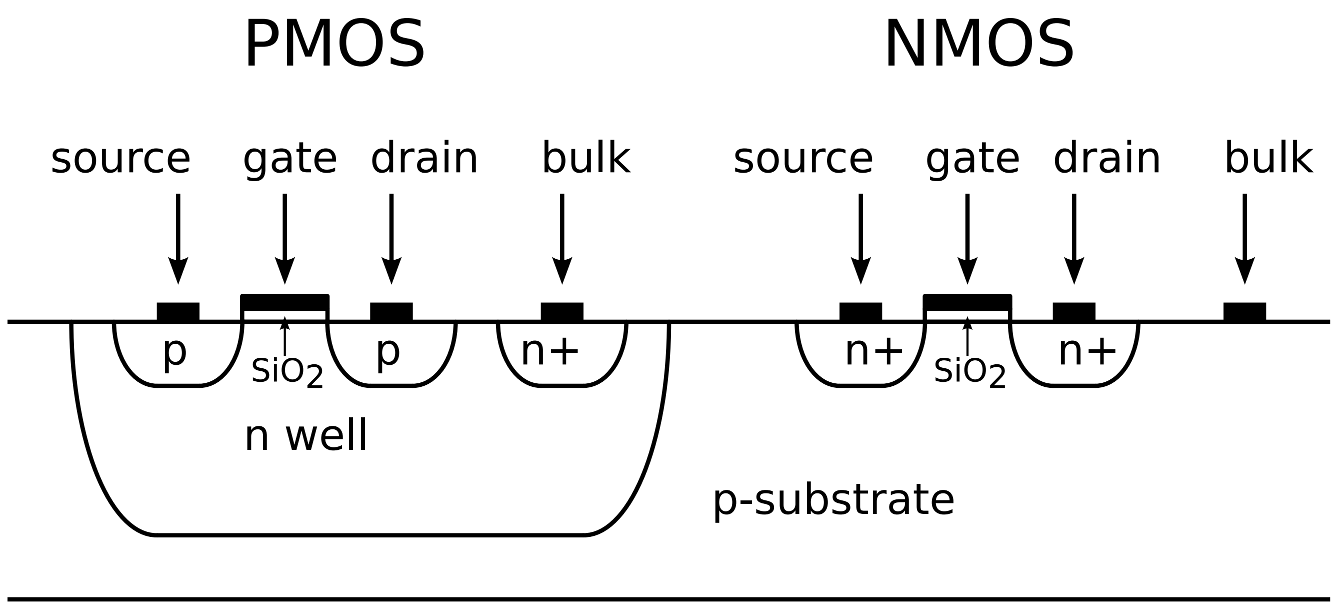
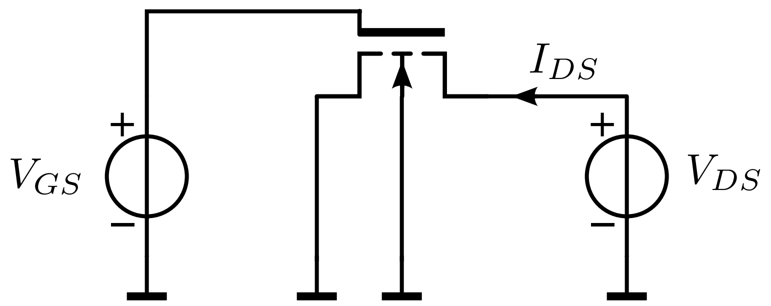




MOS operation

$V_{GS} = 0$, $V_{DS} > 0$





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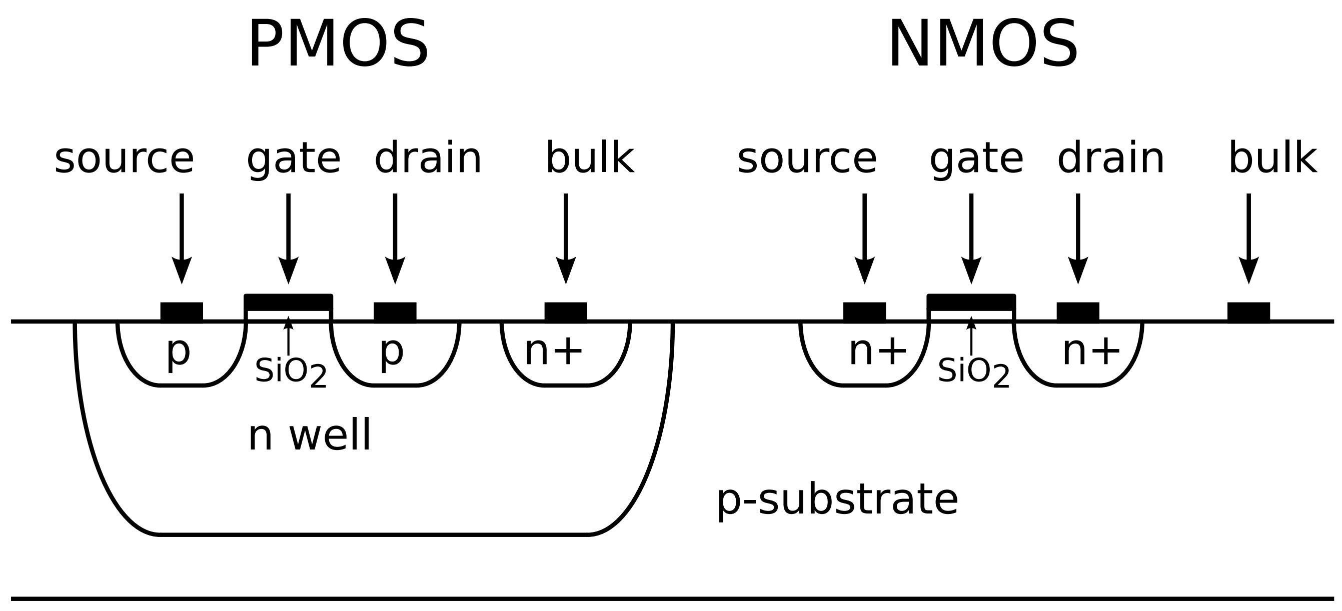
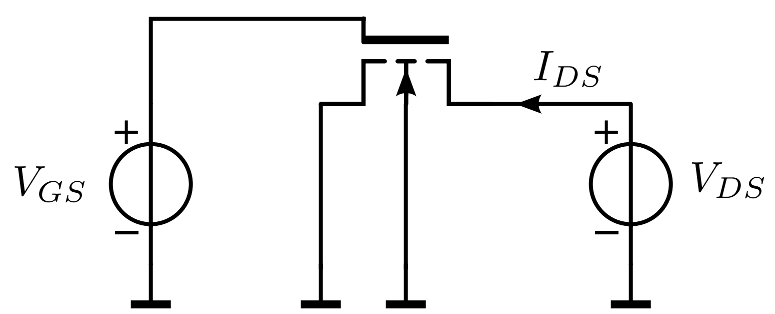
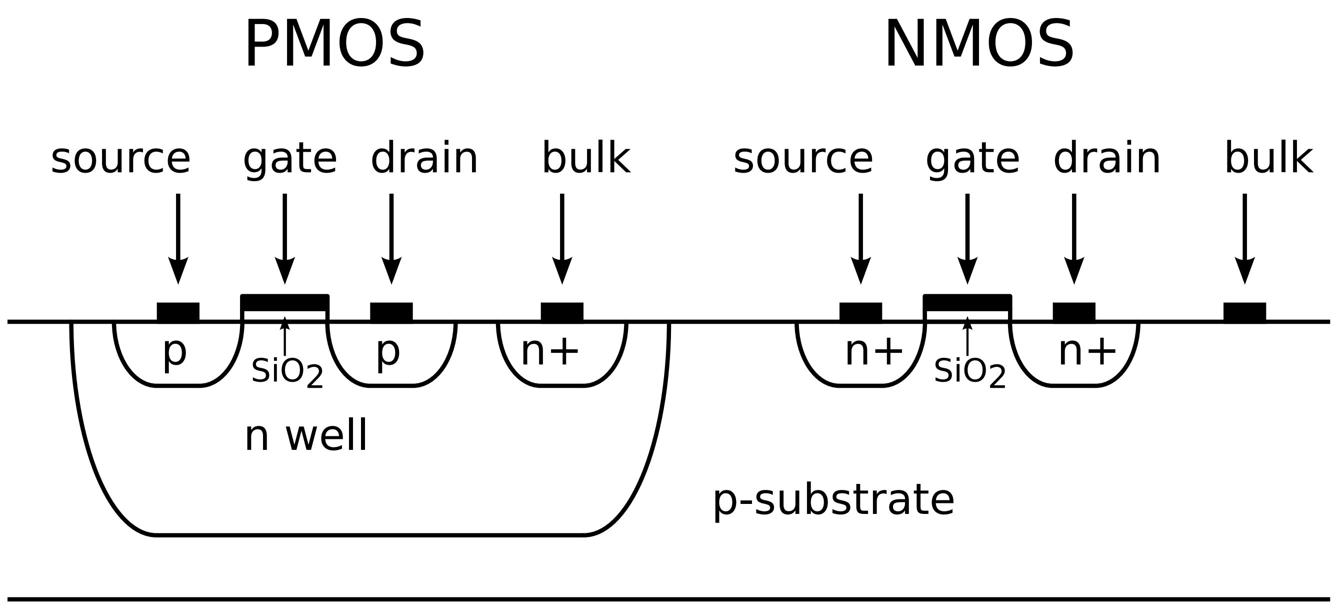
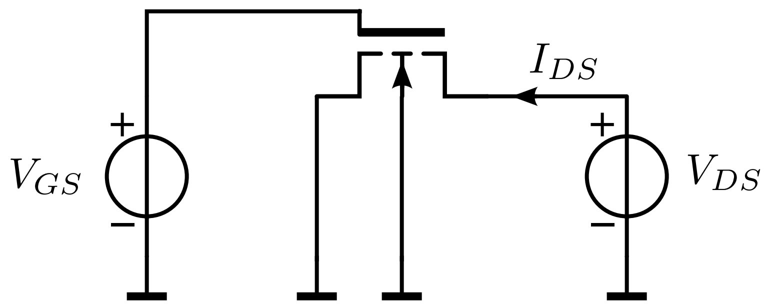




$V_{GS} = 0,\, V_{DS} > 0$ No current flow

MOS operation





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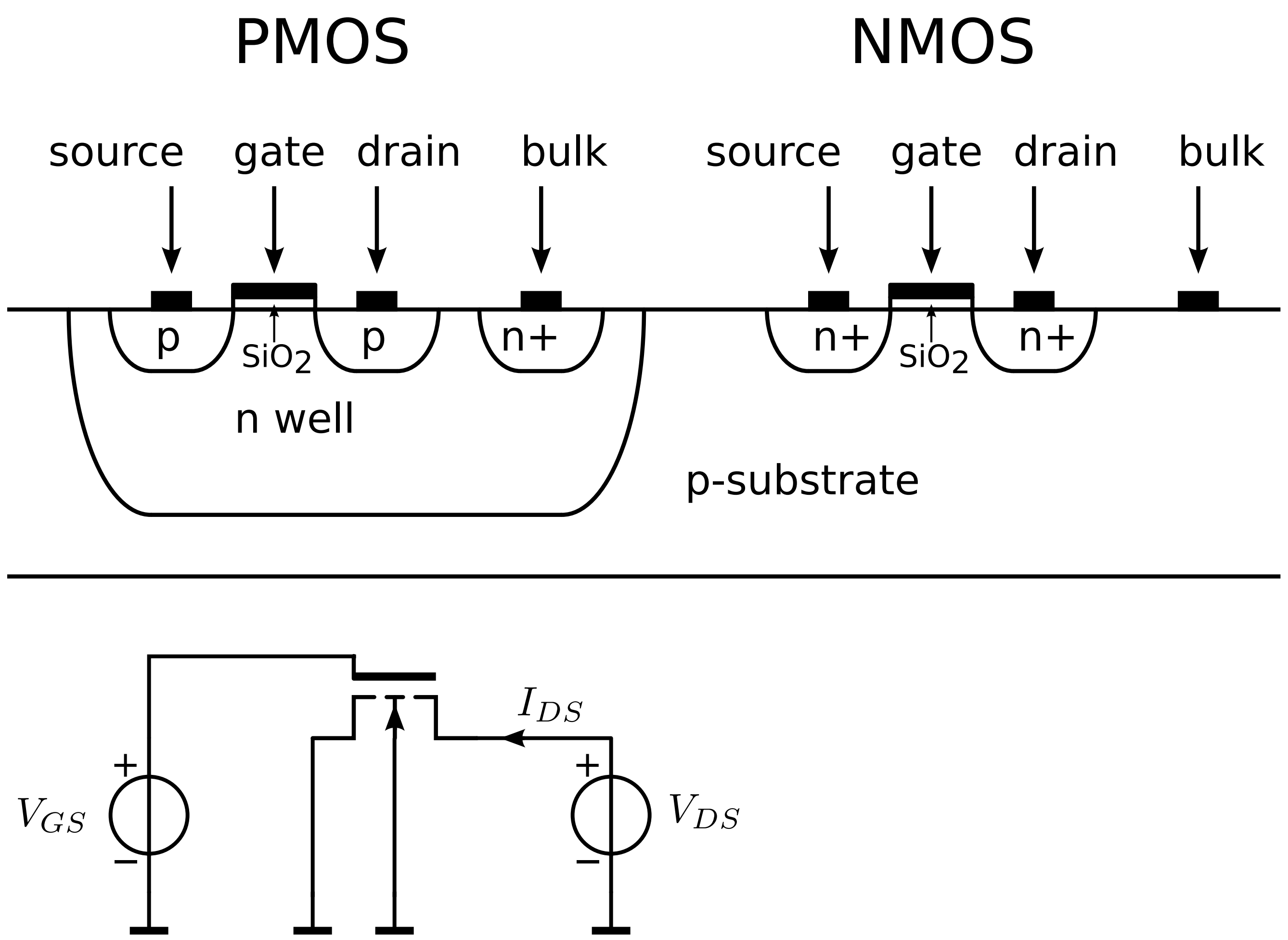

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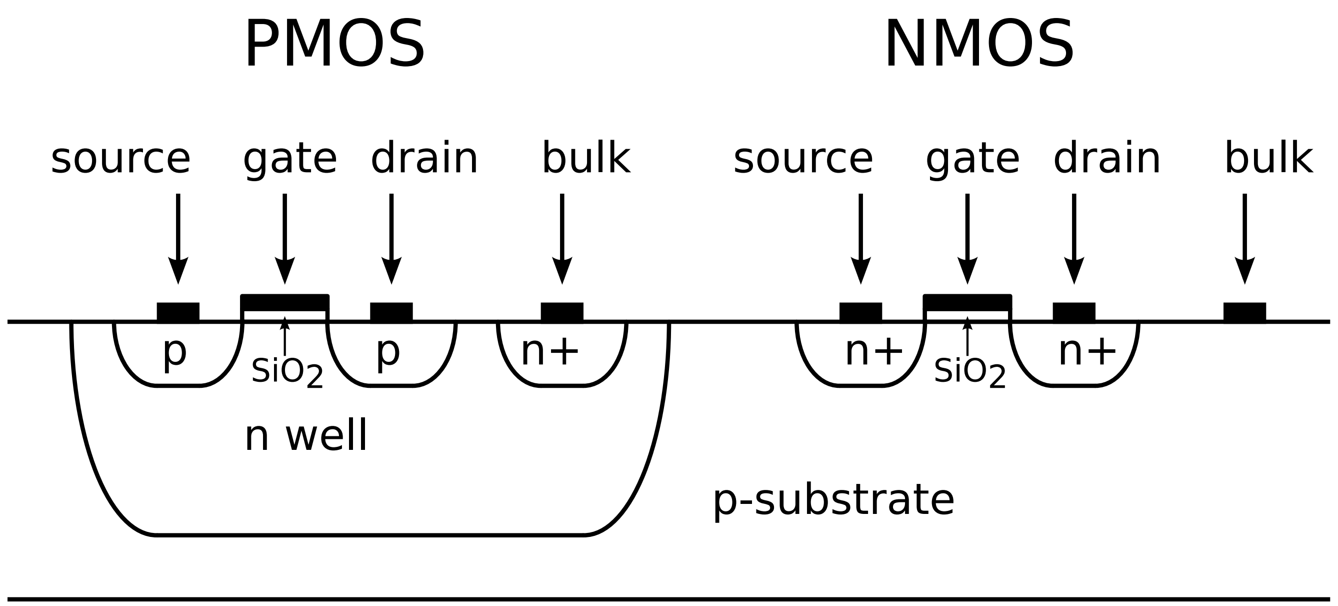
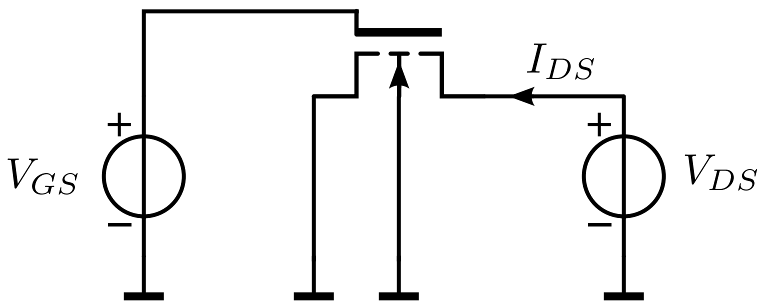
Surface potential at oxide-Si interface rises

MOS operation





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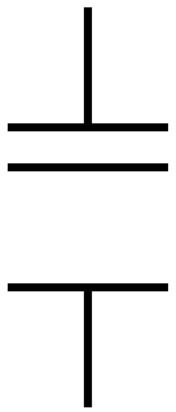
capacitive coupling



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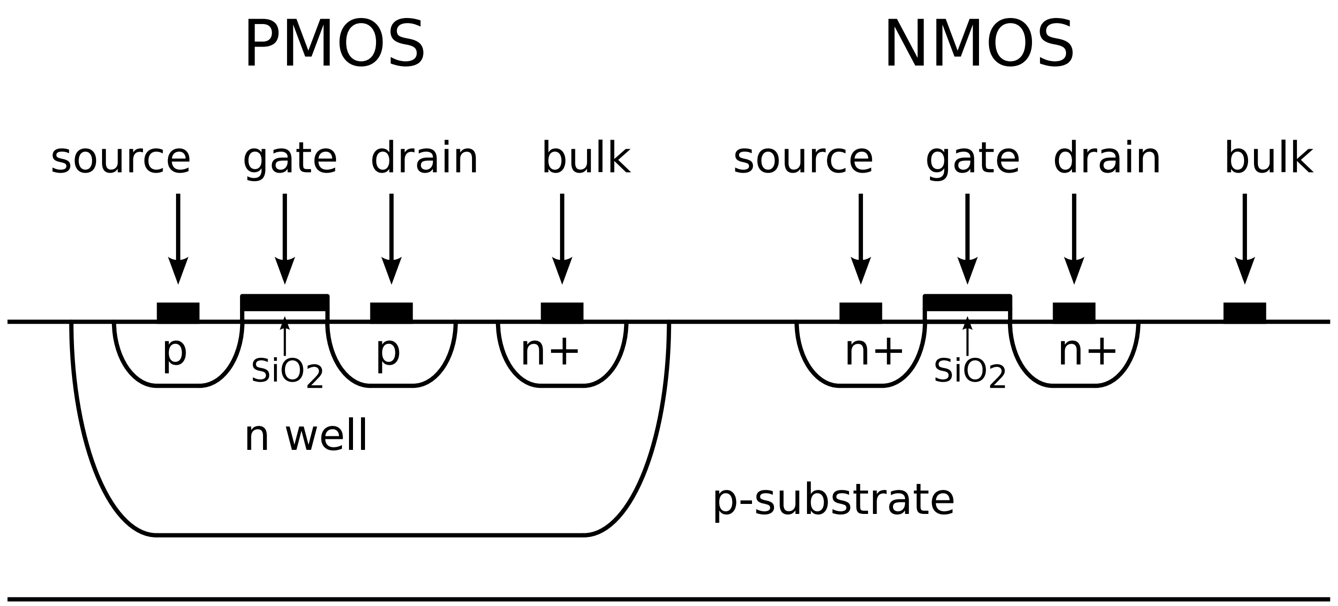

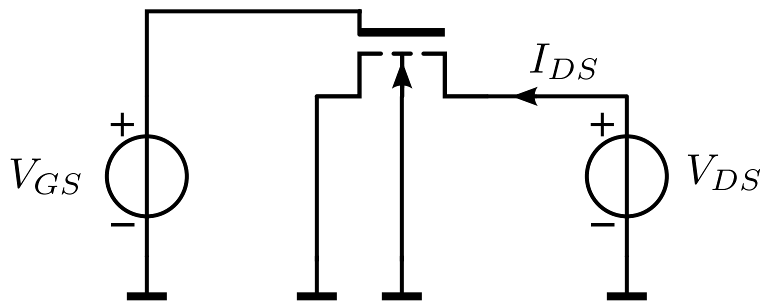

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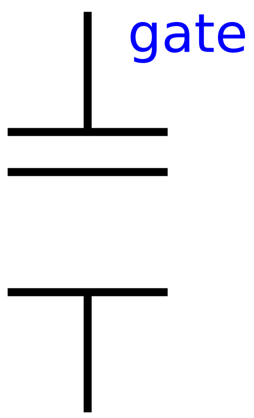

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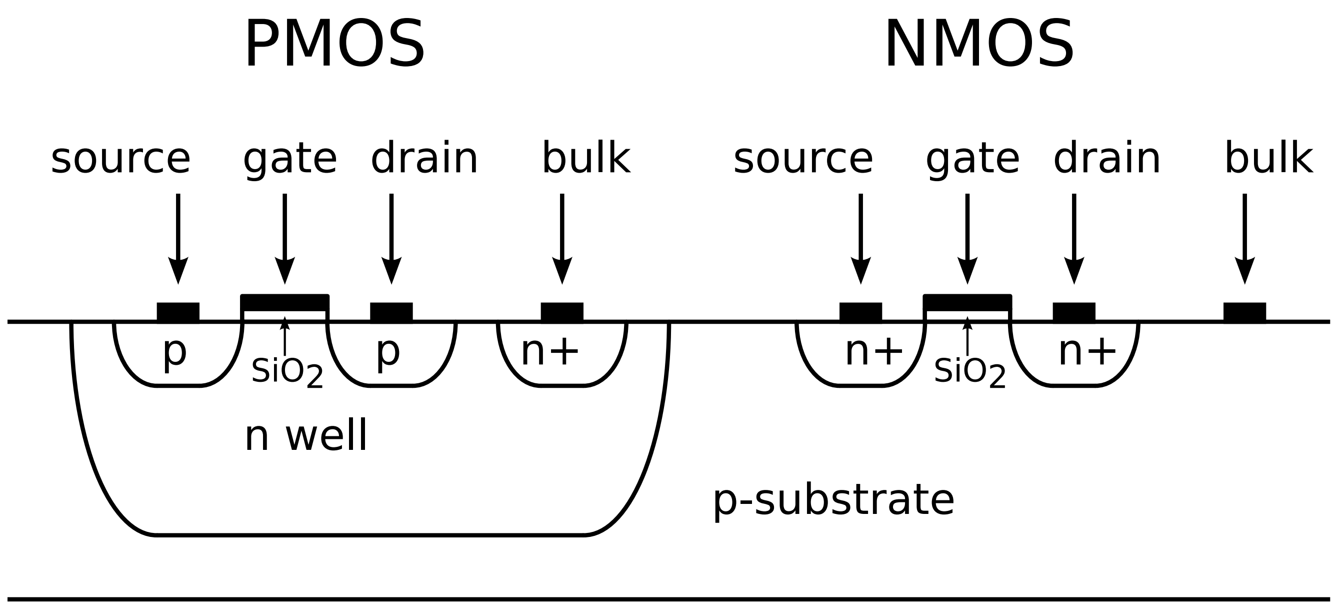
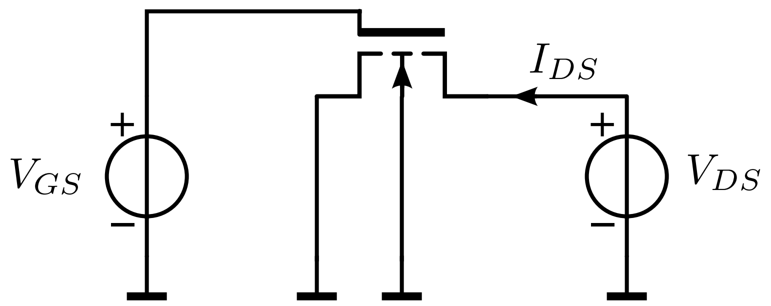
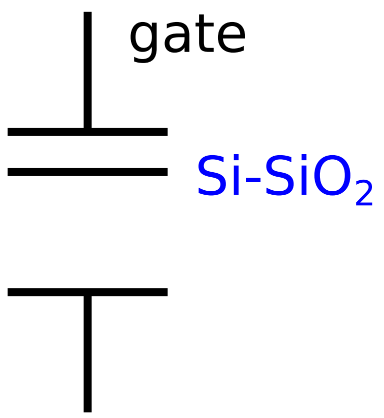




capacitive
coupling



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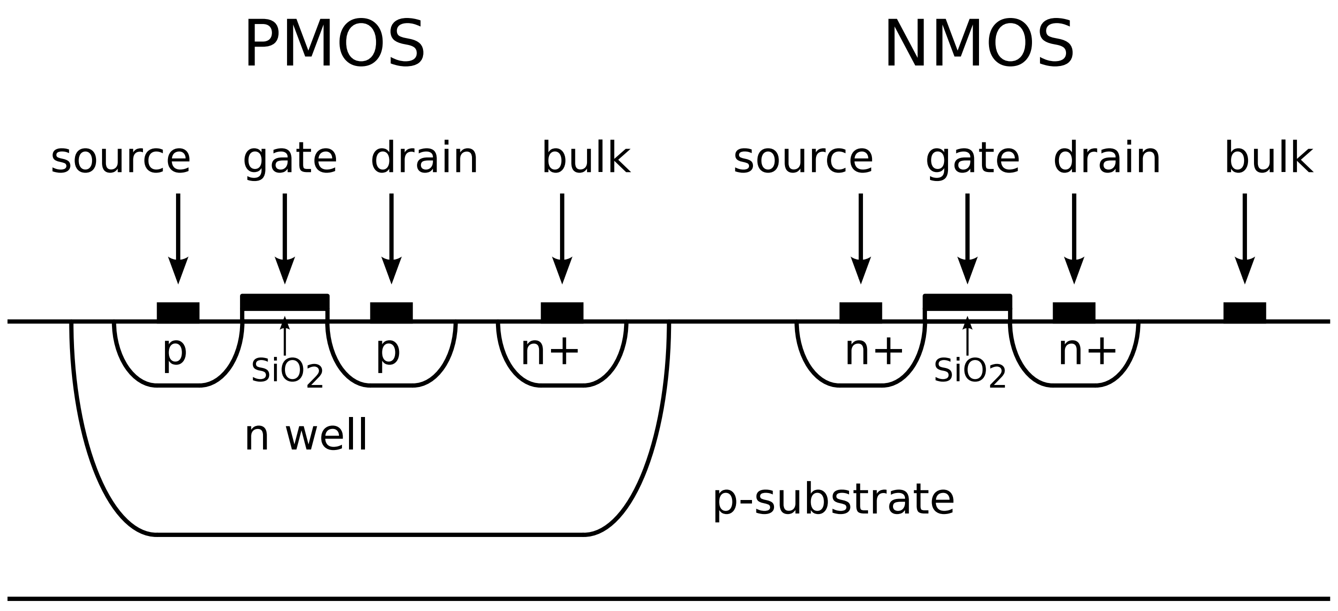

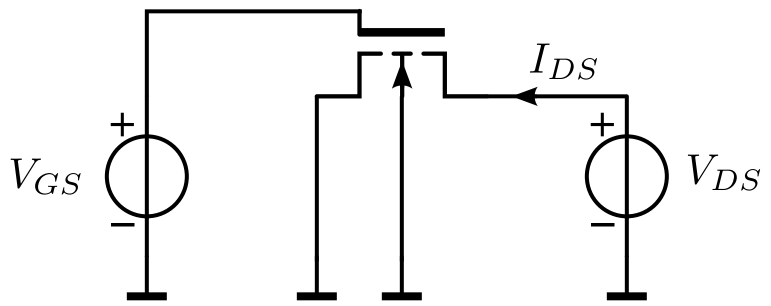

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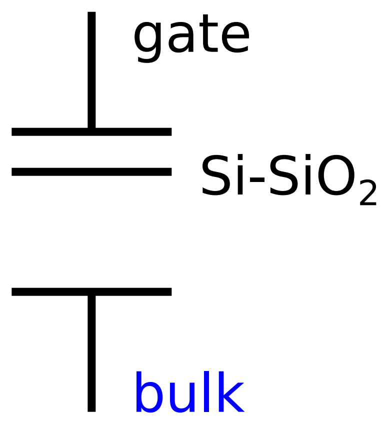
Surface potential at oxide-Si interface rises
Source injects electrons in p region

capacitive coupling



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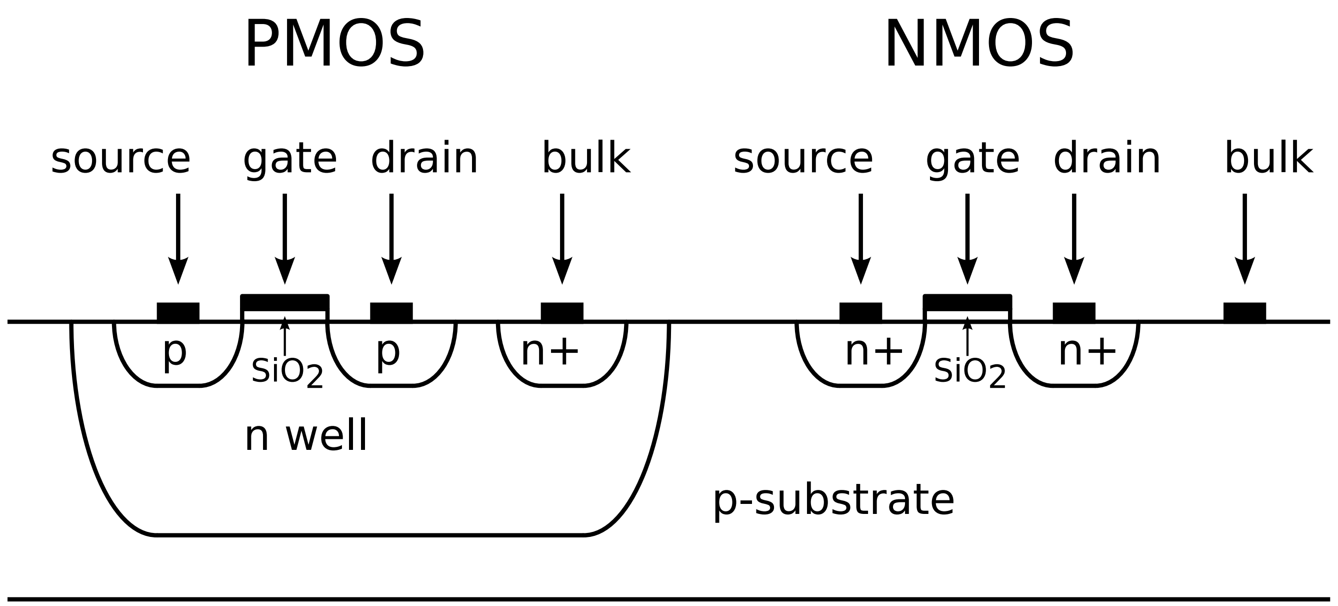

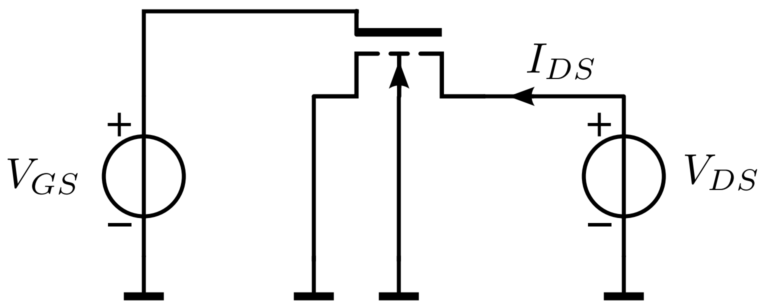

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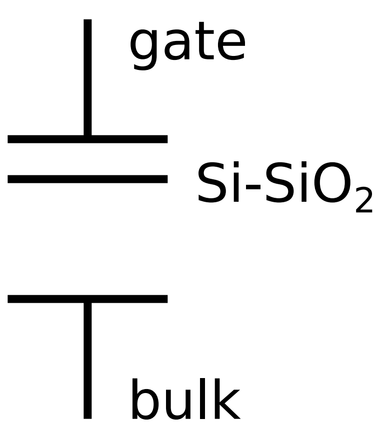
Surface potential at oxide-Si interface rises
Source injects electrons in p region
Weak inversion

capacitive coupling



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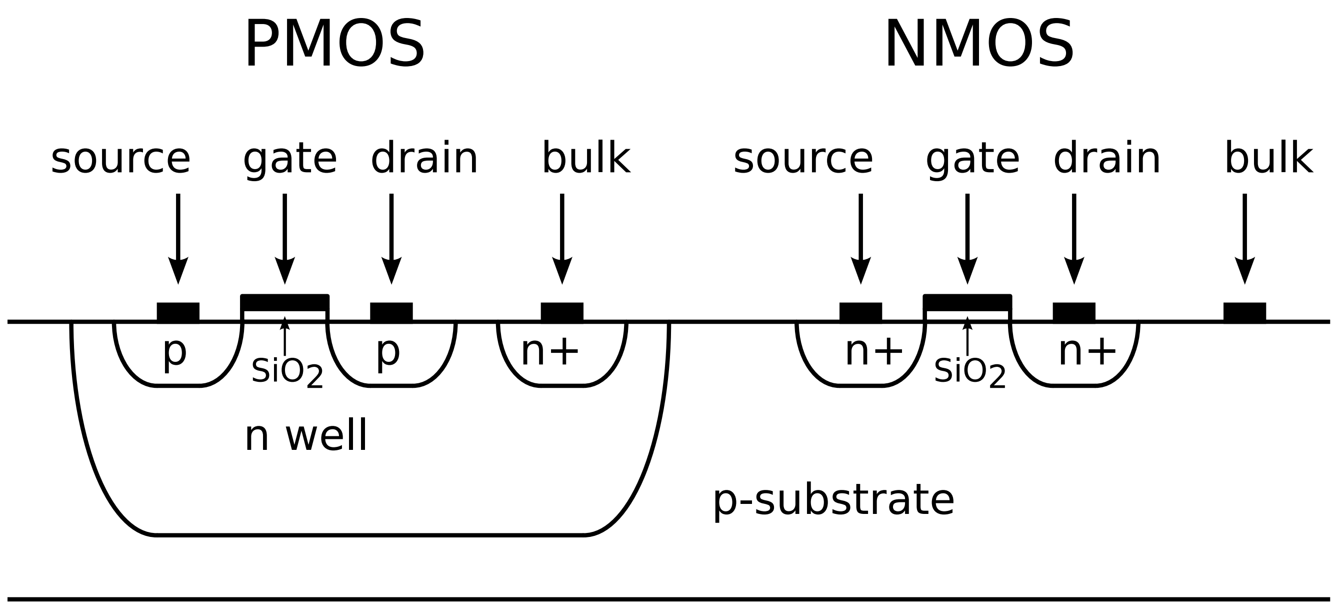

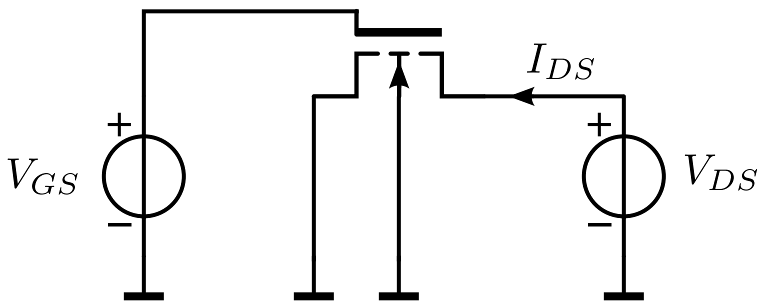

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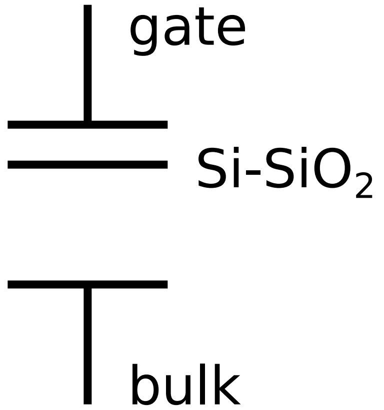
Surface potential at oxide-Si interface rises
Source injects electrons in p region
Weak inversion

Drain current increases exponentially with the gate-source voltage

capacitive coupling



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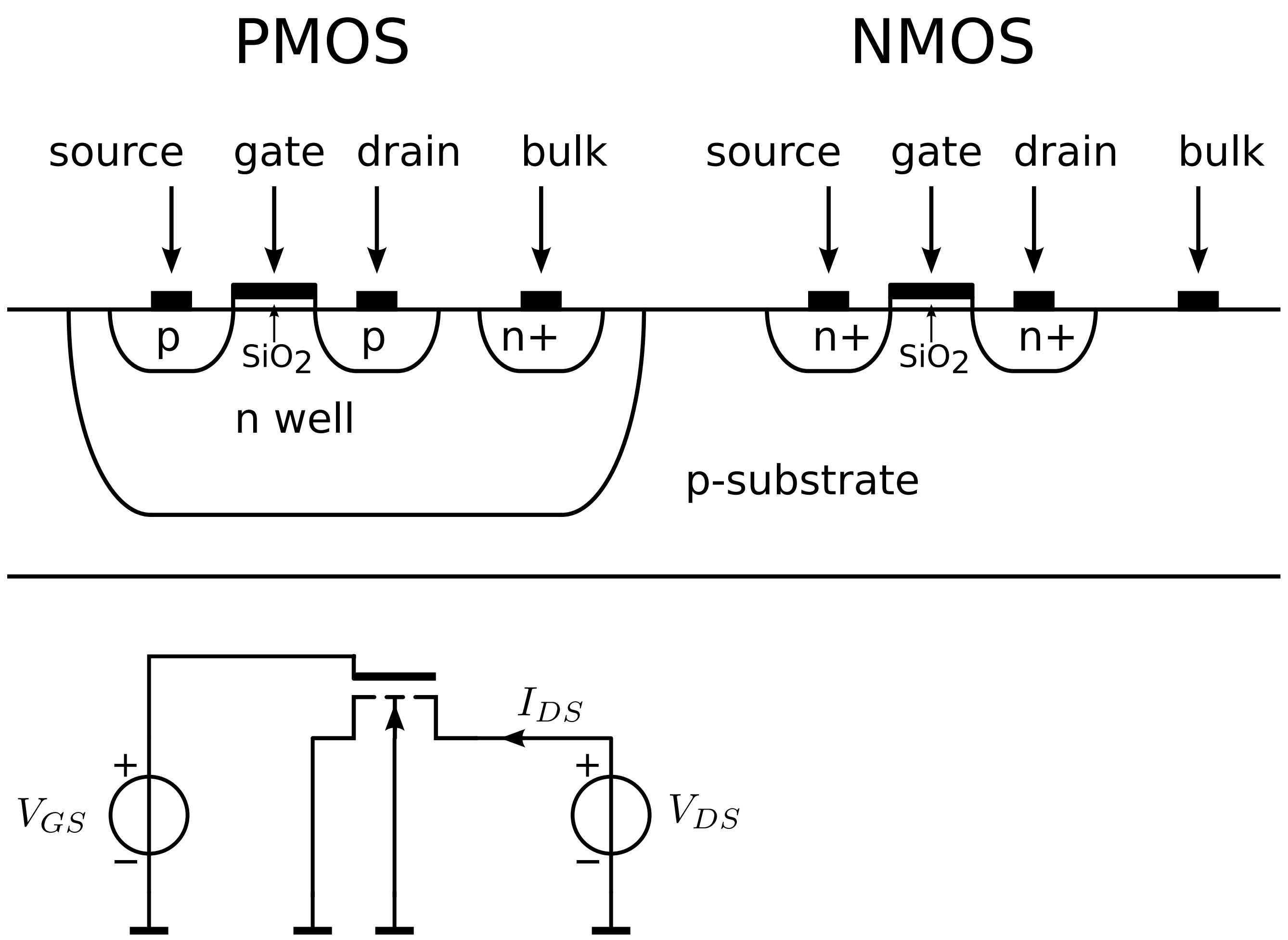

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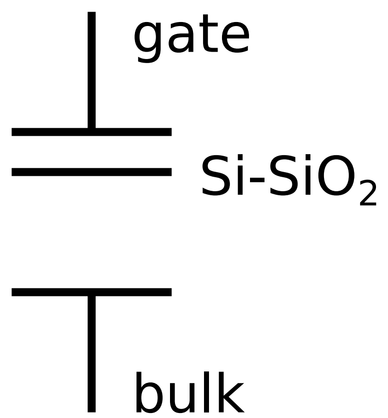
$V_{GS} > V_{th},\, V_{DS} > V_{GS} - V_{th}$

capacitive coupling



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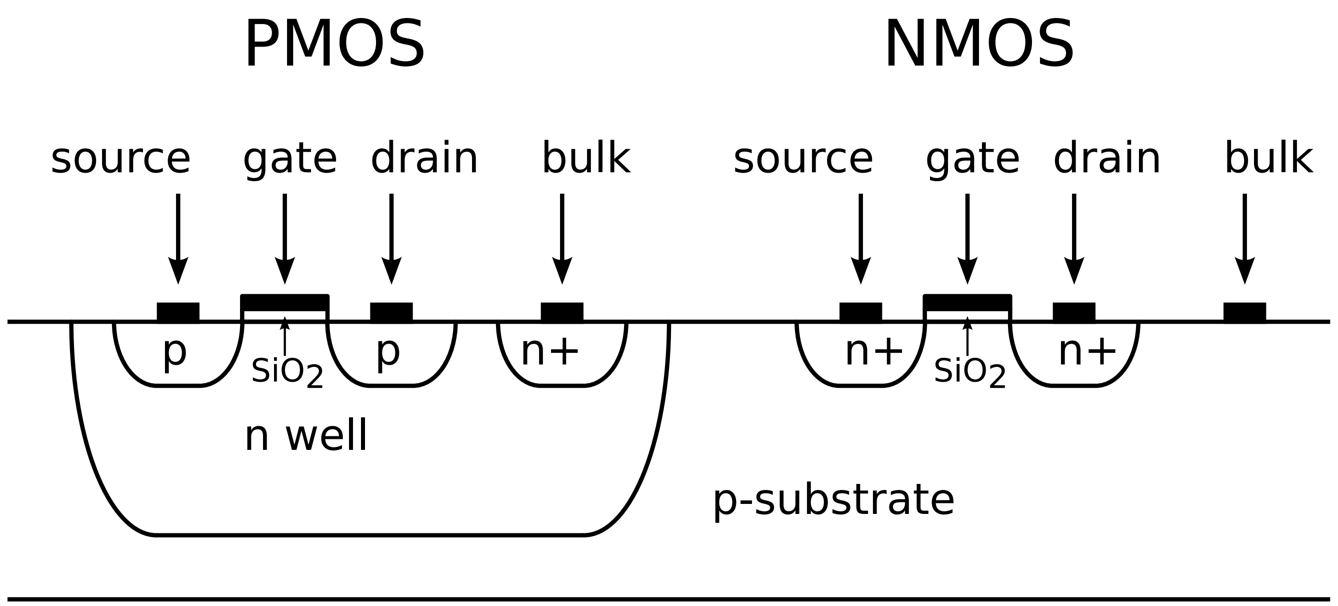

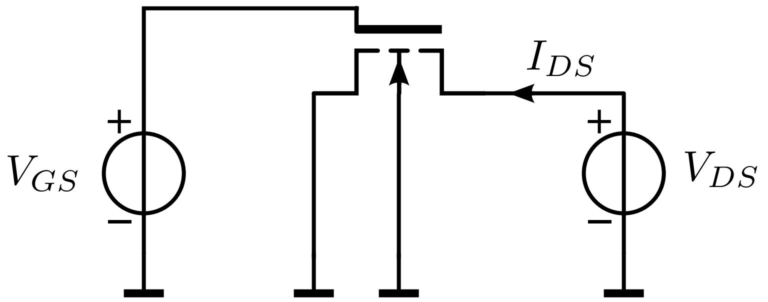

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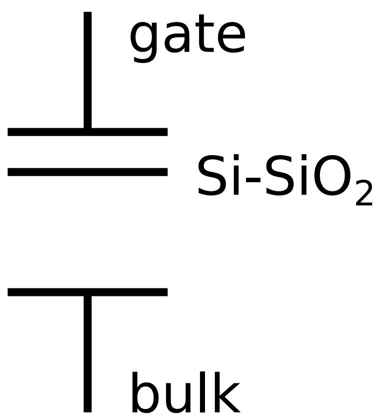
An n-channel is established between source and drain

capacitive coupling



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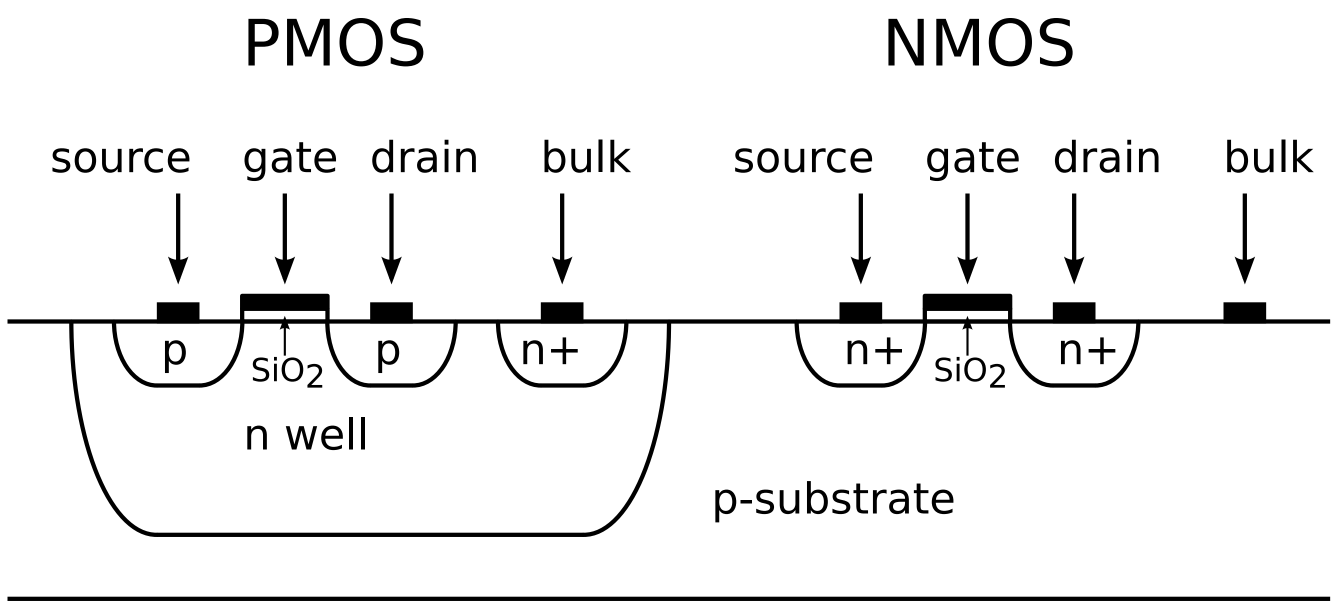

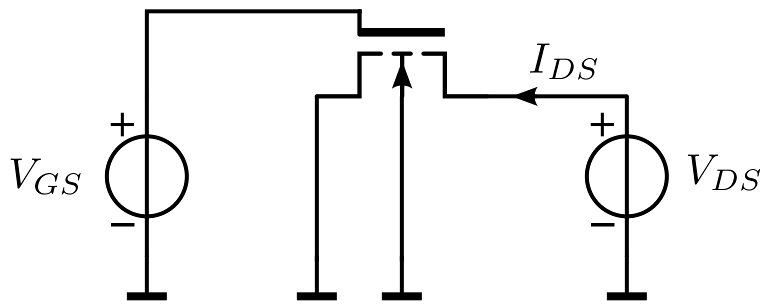

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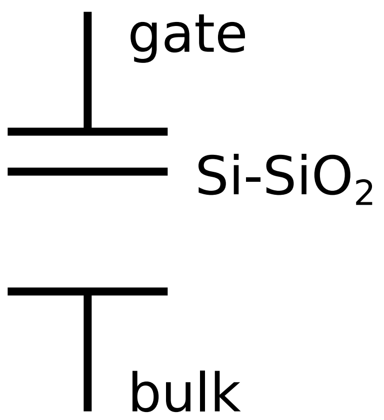

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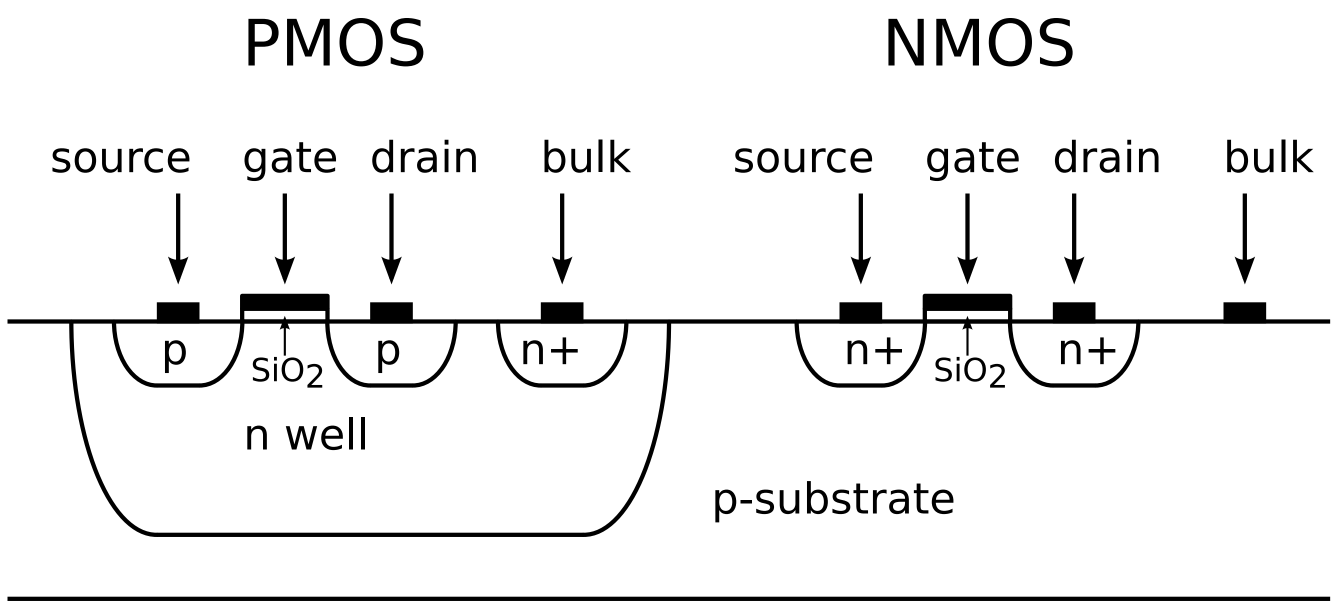
Drain current increases exponentially with the gate-source voltage

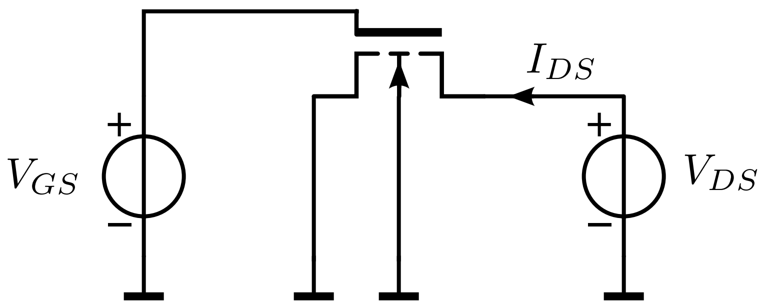
$V_{GS} > V_{th},\ V_{DS} > V_{GS} - V_{th}$

An n-channel is established between source and drain

Drain current increases quadratically with the gate-source voltage

MOS operation





Drain-source voltage dependency

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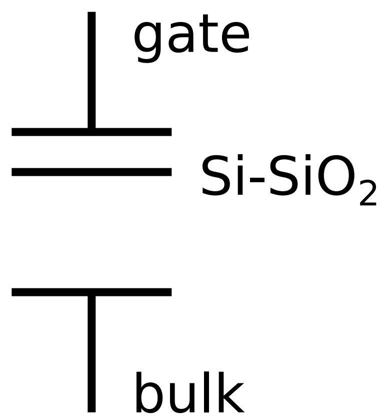

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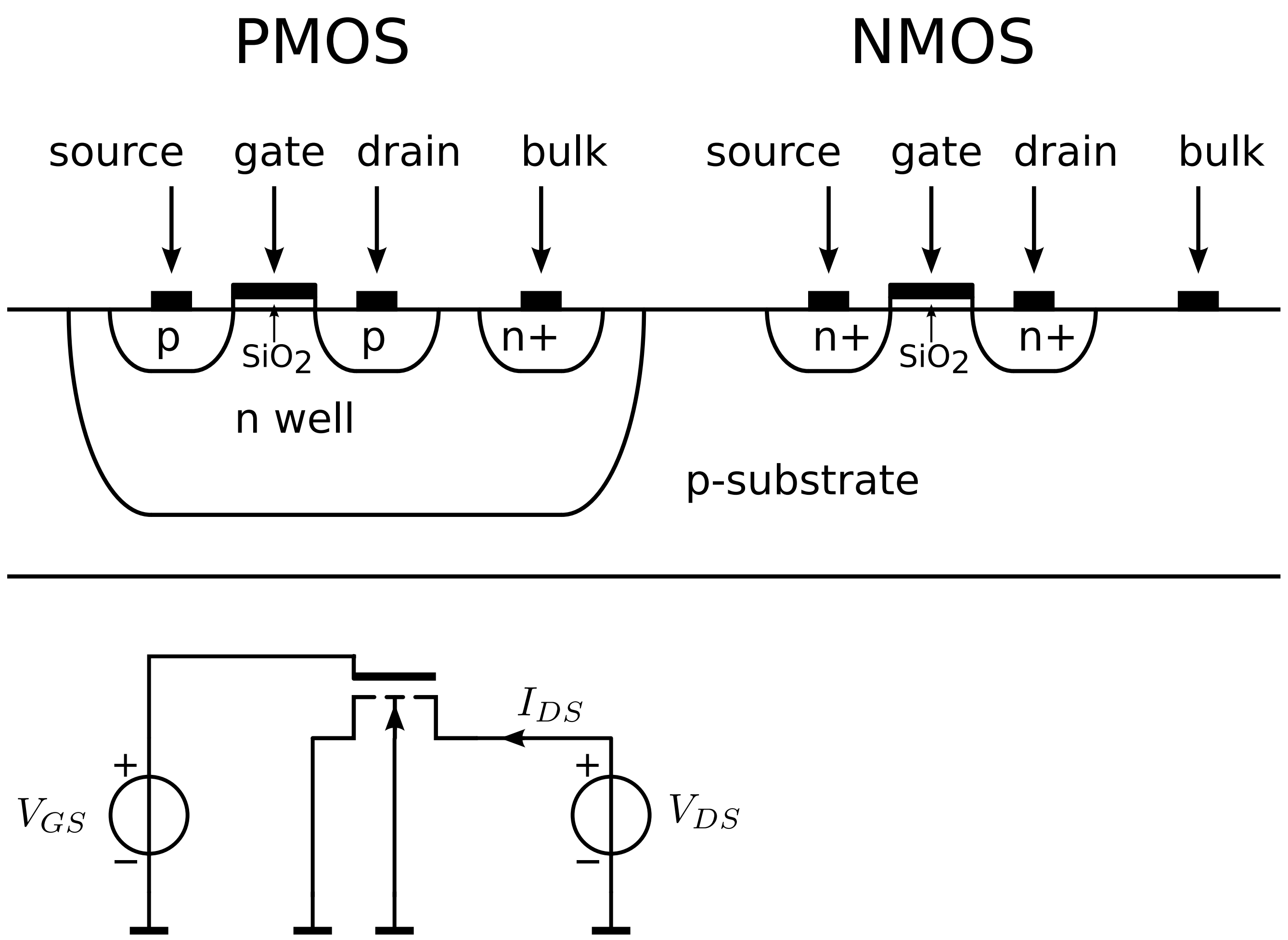

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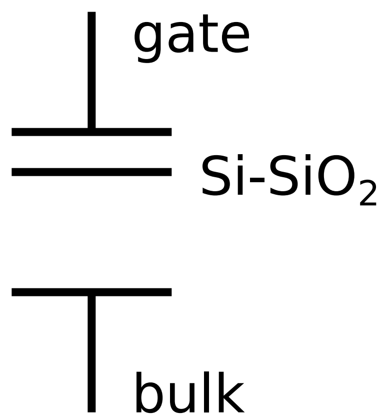

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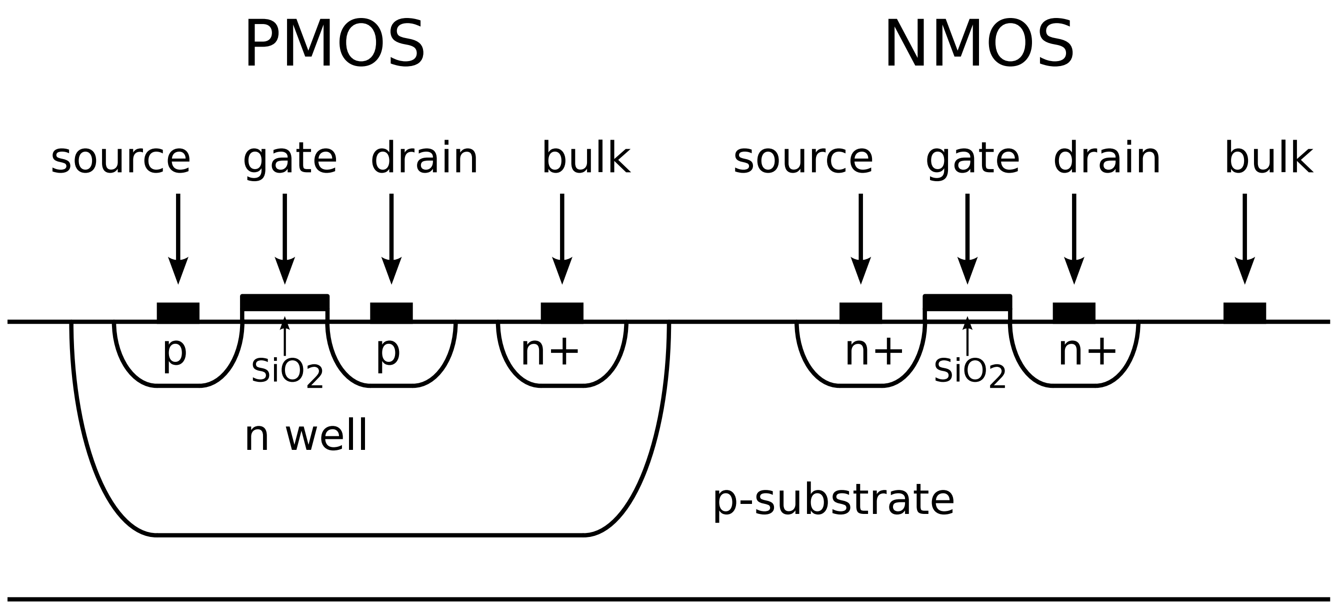
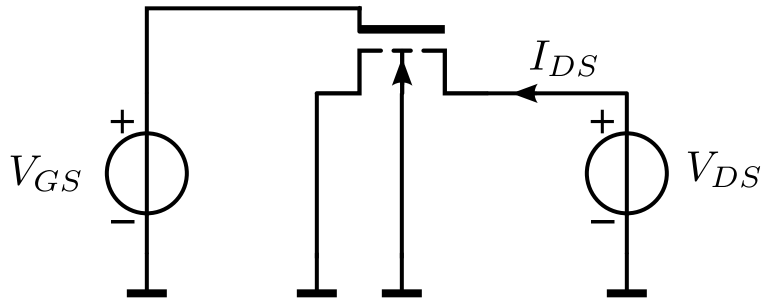
Drain-source voltage dependency

Channel length modulation (CLM)

MOS operation





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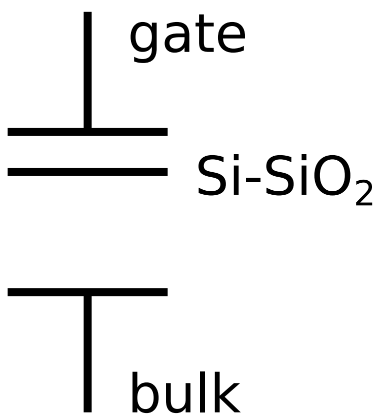

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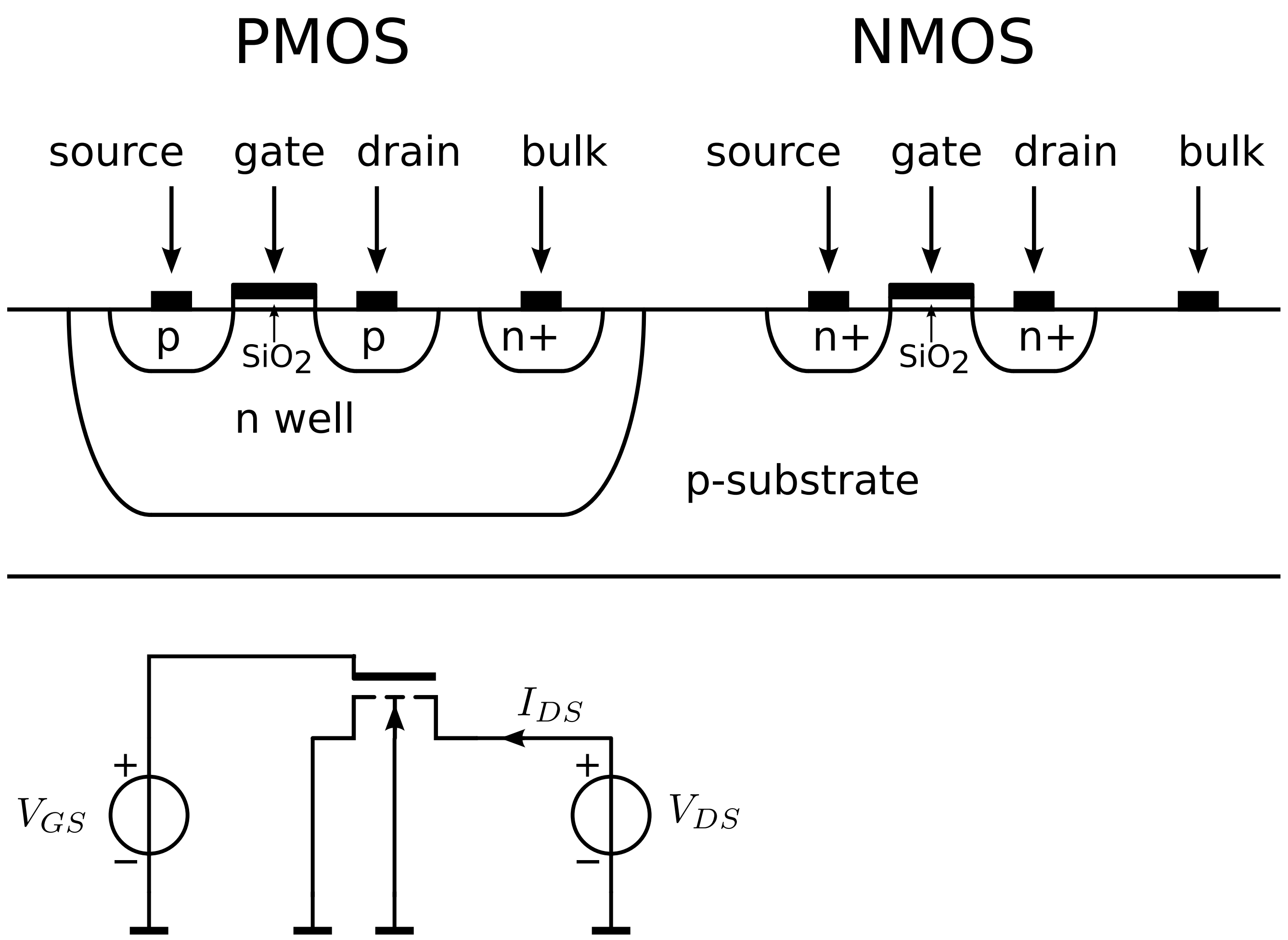
Drain-source voltage dependency

Channel length modulation (CLM)

Drain current increases with drain-source voltage

MOS operation



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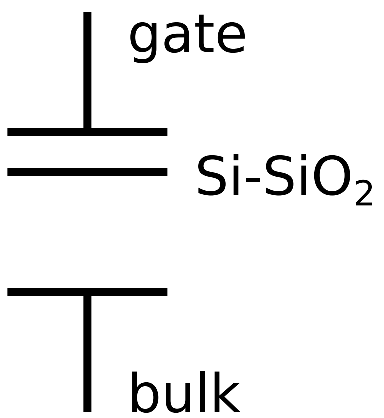

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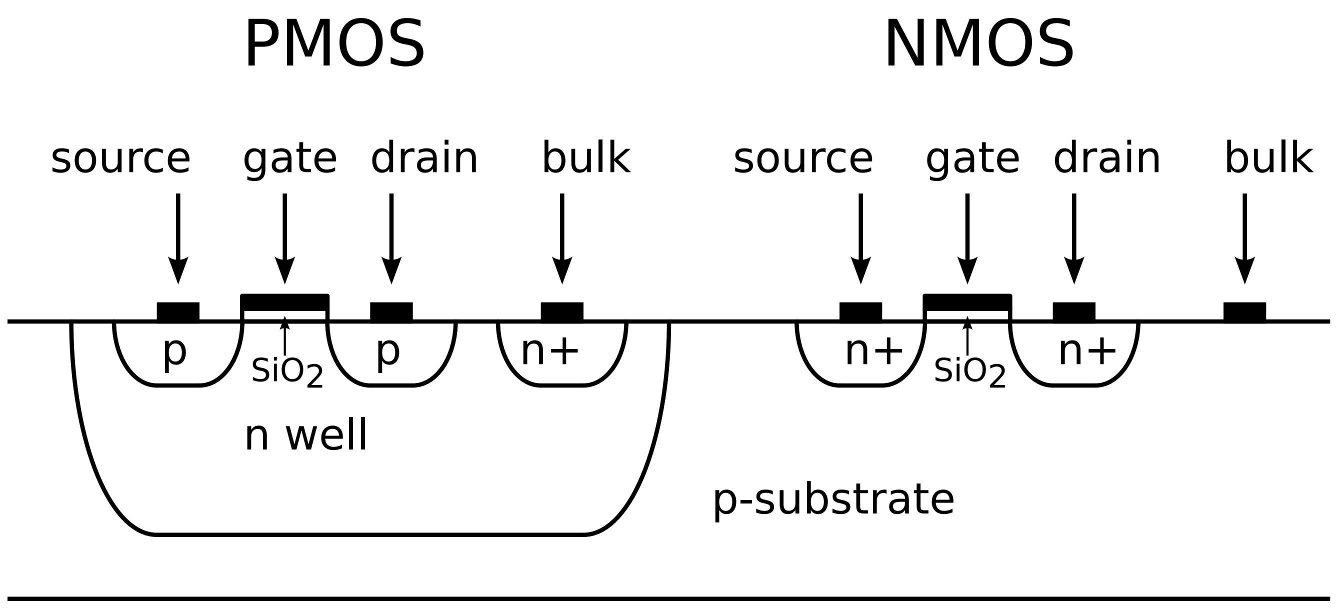
An n-channel is established between source and drain

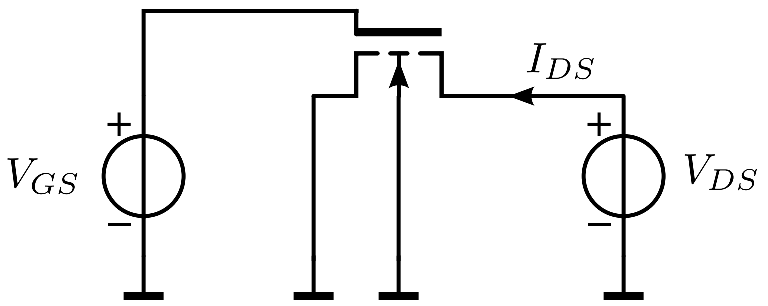
Drain current increases quadratically with the gate-source voltage

Drain-source voltage dependency

- Channel length modulation (CLM)
  - Drain current increases with drain-source voltage
- Breakdown

MOS operation





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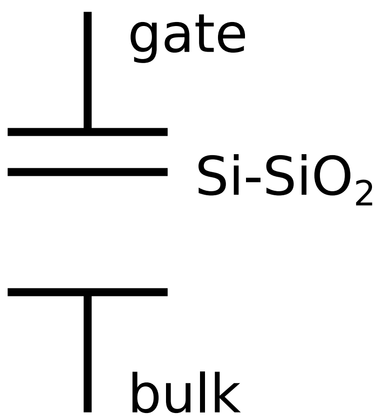

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Drain-source voltage dependency

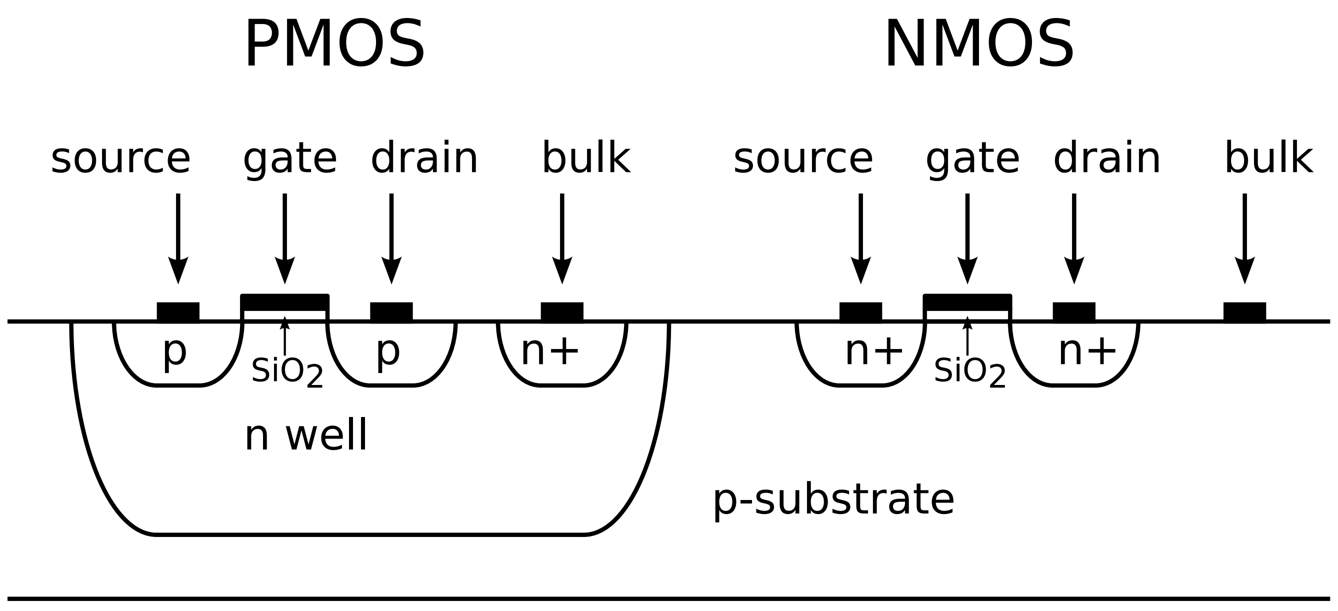
Channel length modulation (CLM)

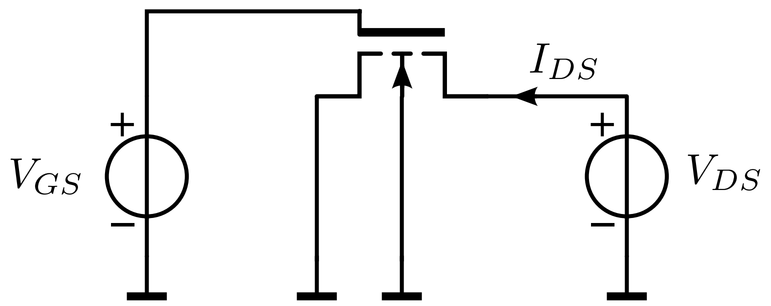
Drain current increases with drain-source voltage

Breakdown

Short channel effects

MOS operation





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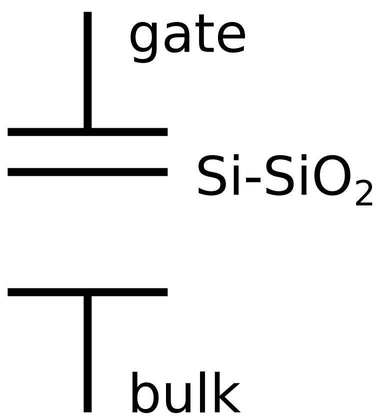

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An n-channel is established between source and drain

Drain current increases quadratically with the gate-source voltage

Drain-source voltage dependency

Channel length modulation (CLM)

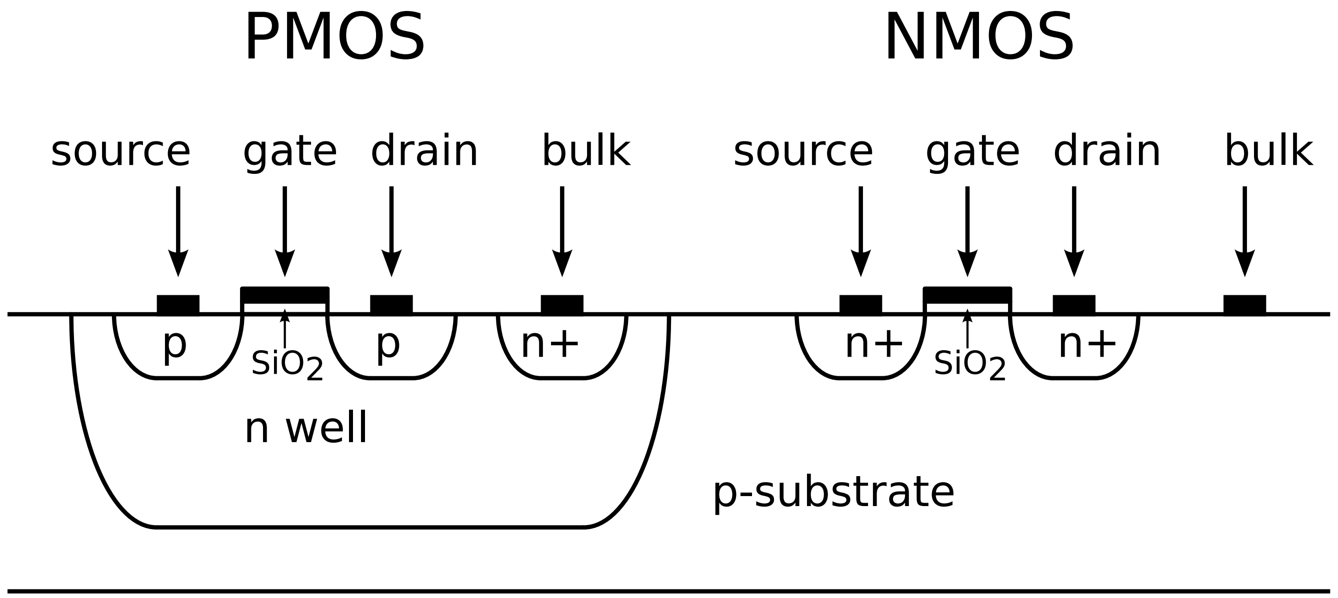
Drain current increases with drain-source voltage

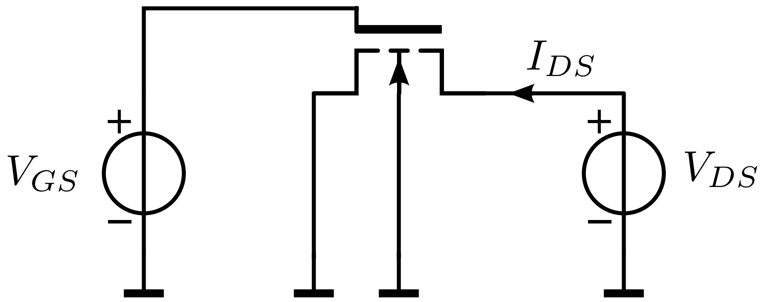
Breakdown

Short channel effects

Vertical field mobility reduction (VFMR)

MOS operation





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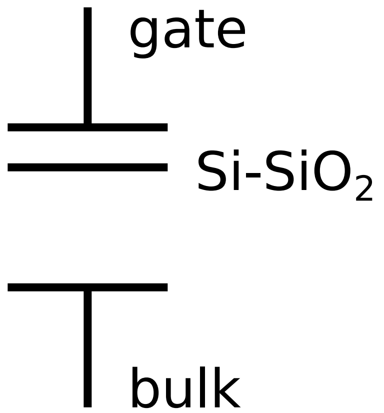


Drain-source voltage dependency

Channel length modulation (CLM)

Drain current increases with drain-source voltage

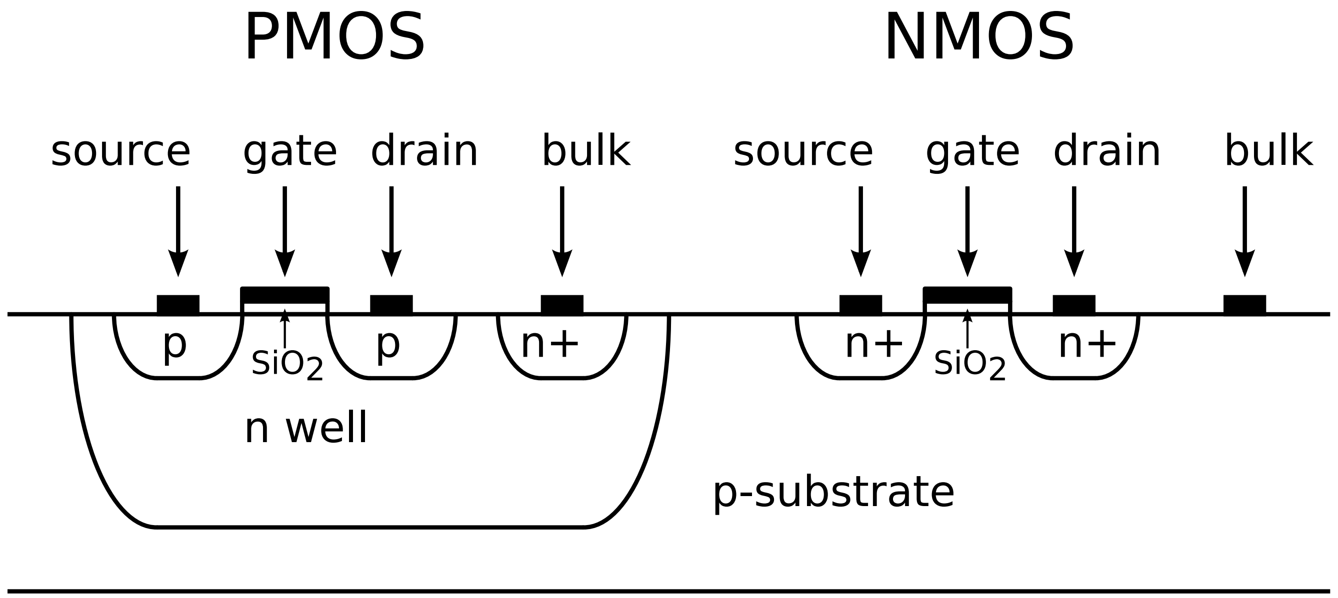
Breakdown

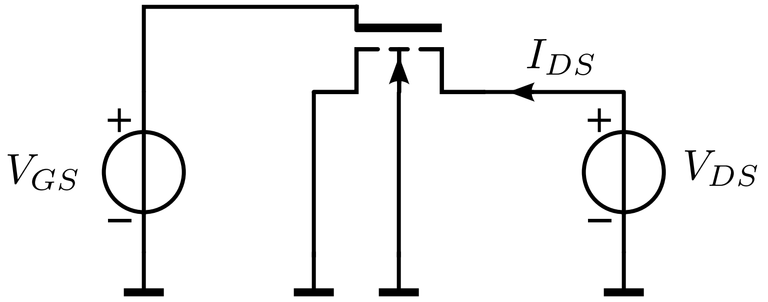
Short channel effects

Vertical field mobility reduction (VFMR)

Velocity saturation (VS)

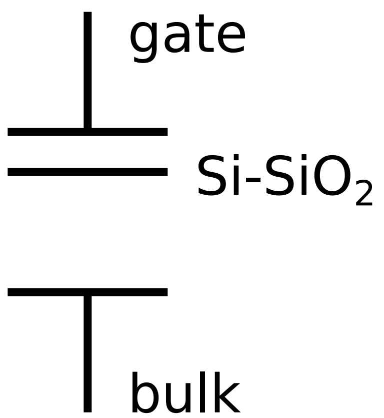
MOS operation





$V_{GS} = 0,\, V_{DS} > 0$ No current flow

capacitive coupling



$V_{GS} > 0,\, V_{DS} >> 0$

Surface potential at oxide-Si interface rises
Source injects electrons in p region
Weak inversion

Drain current increases exponentially with the gate-source voltage

$V_{GS} > V_{th},\, V_{DS} > V_{GS} - V_{th}$

An n-channel is established between source and drain

Drain current increases quadratically with the gate-source voltage

Drain-source voltage dependency

Channel length modulation (CLM)

Drain current increases with drain-source voltage

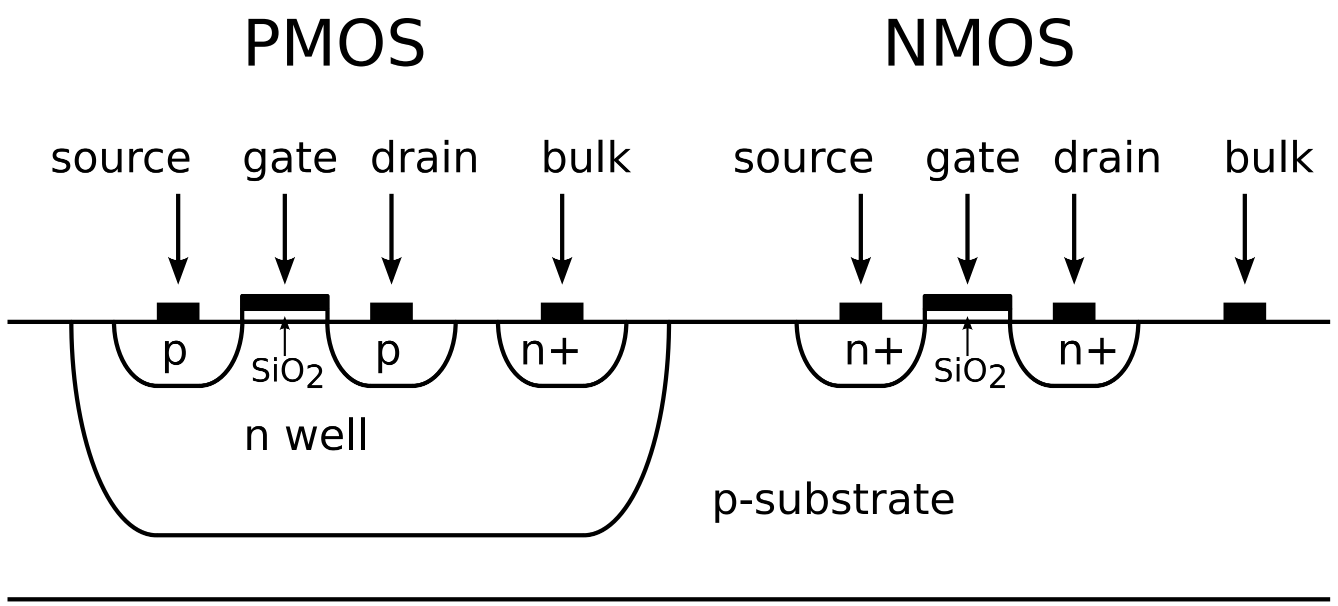
Breakdown

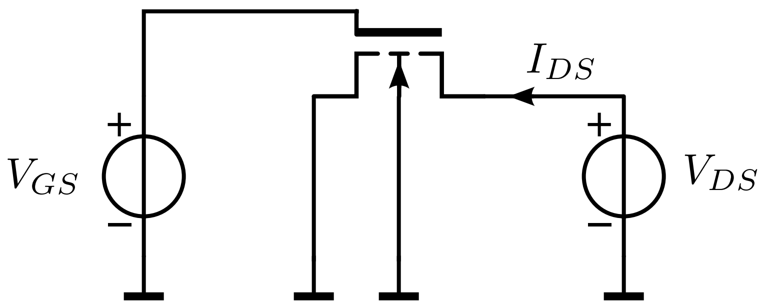
Short channel effects

Vertical field mobility reduction (VFMR)
Velocity saturation (VS)

Drain current increases not longer quadratically with the gate-source voltage

MOS operation





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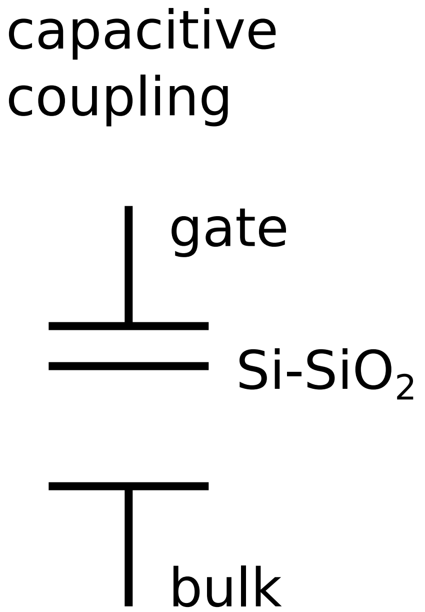

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  - Drain current increases with drain-source voltage
- Breakdown

Short channel effects

- Vertical field mobility reduction (VFMR)
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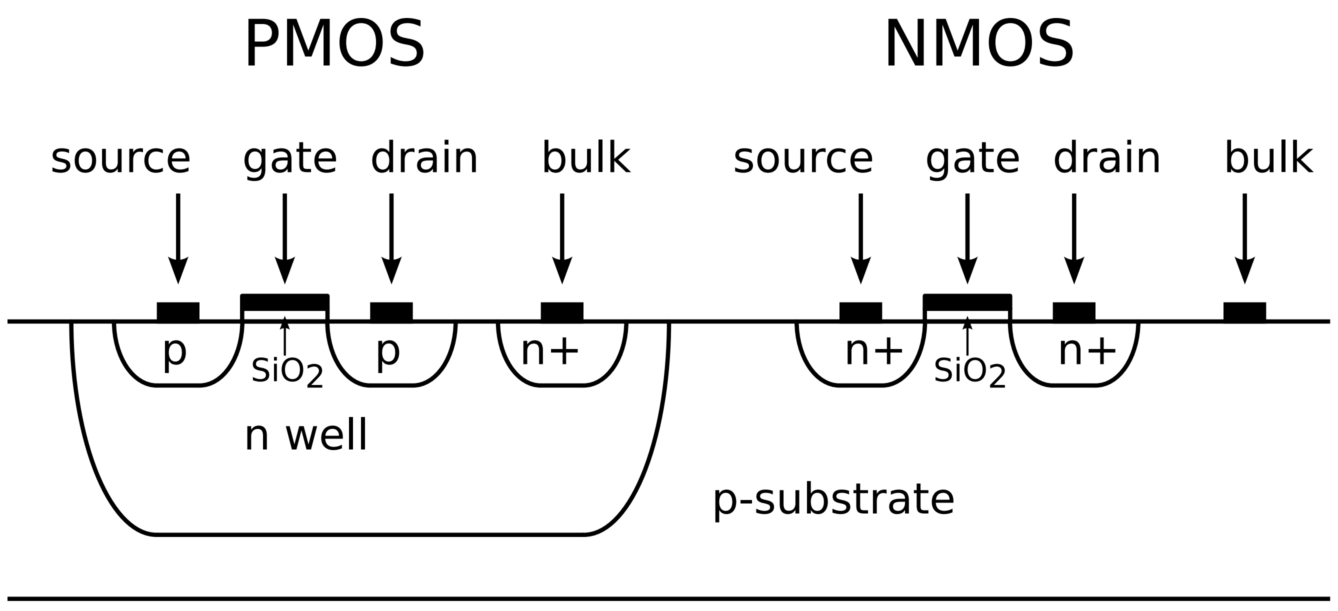
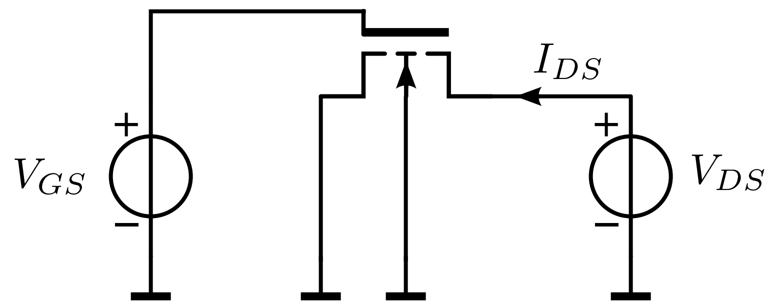
  - Drain current increases not longer quadratically with the gate-source voltage
- Drain-induced barrier lowering (DIBL)

MOS operation





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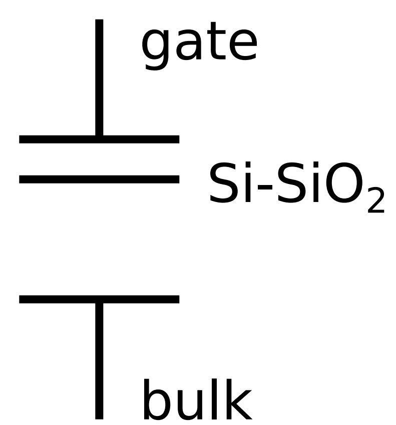

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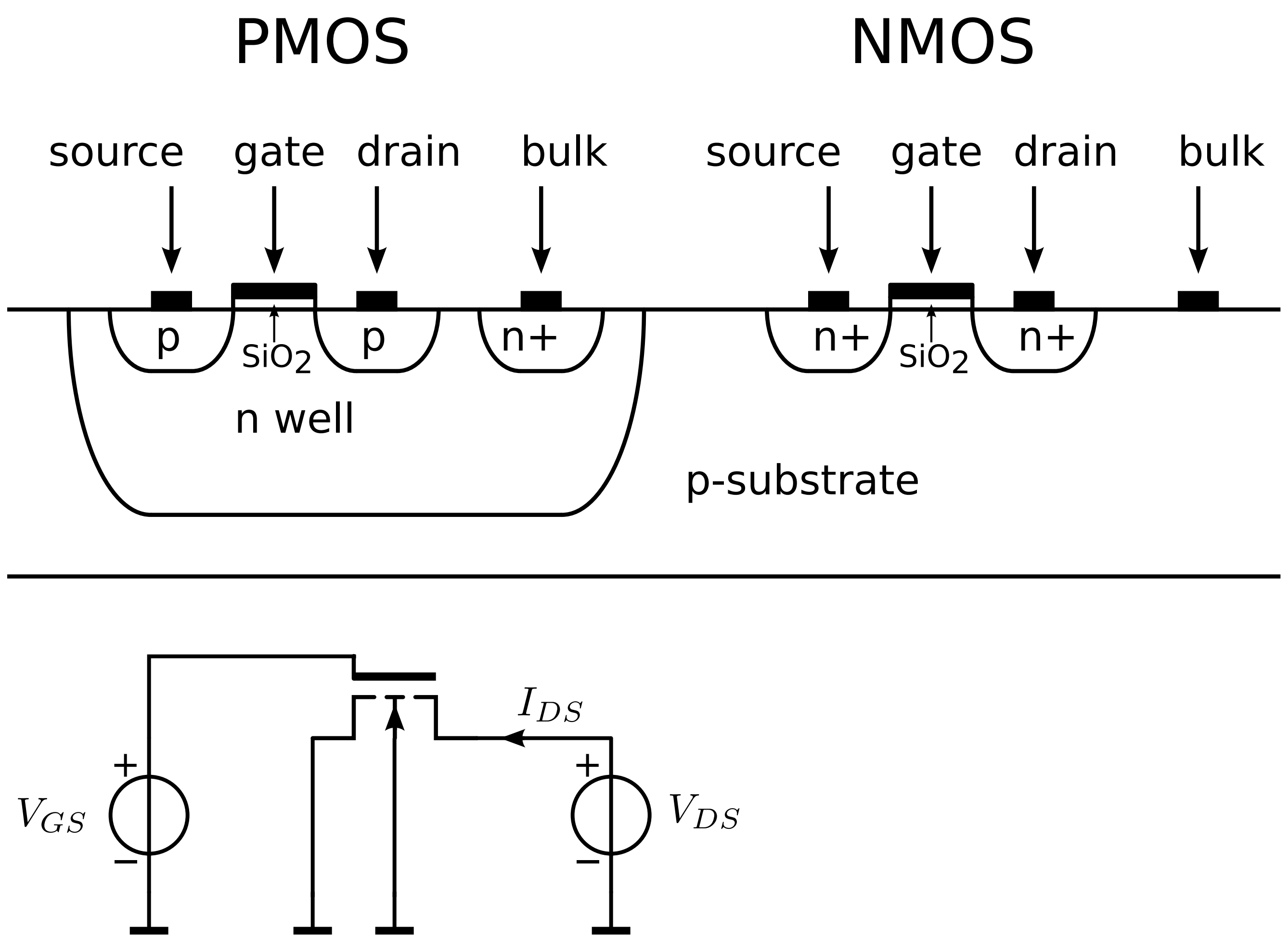
- Drain-induced barrier lowering (DIBL)
  - Capacitive coupling increases:

MOS operation



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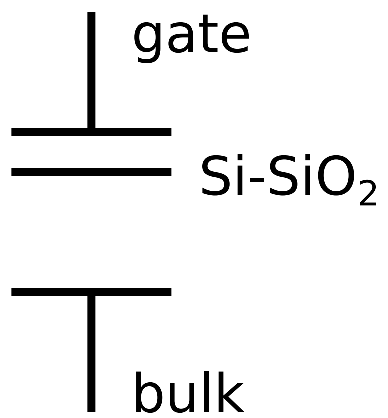

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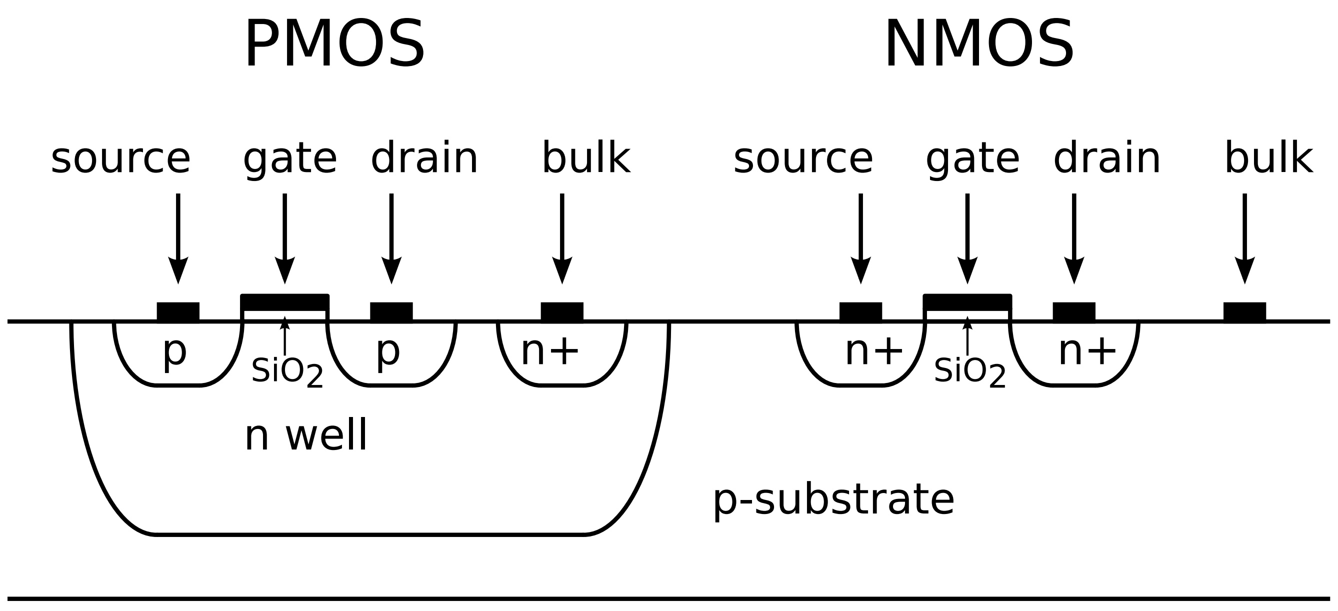

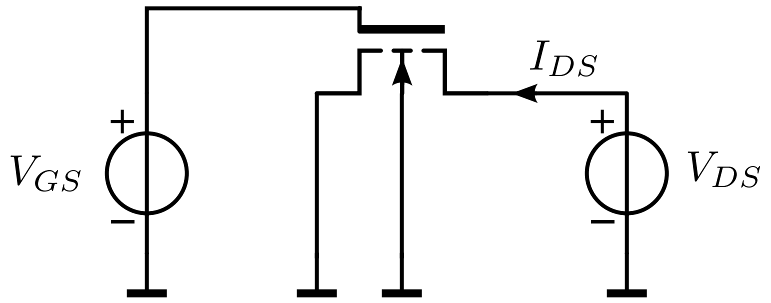

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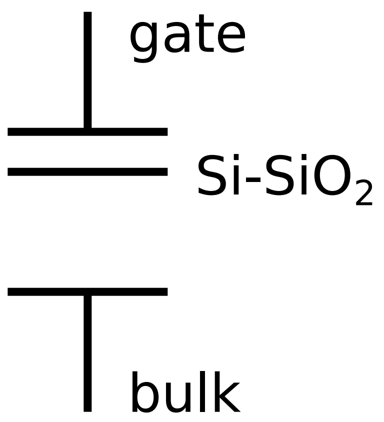

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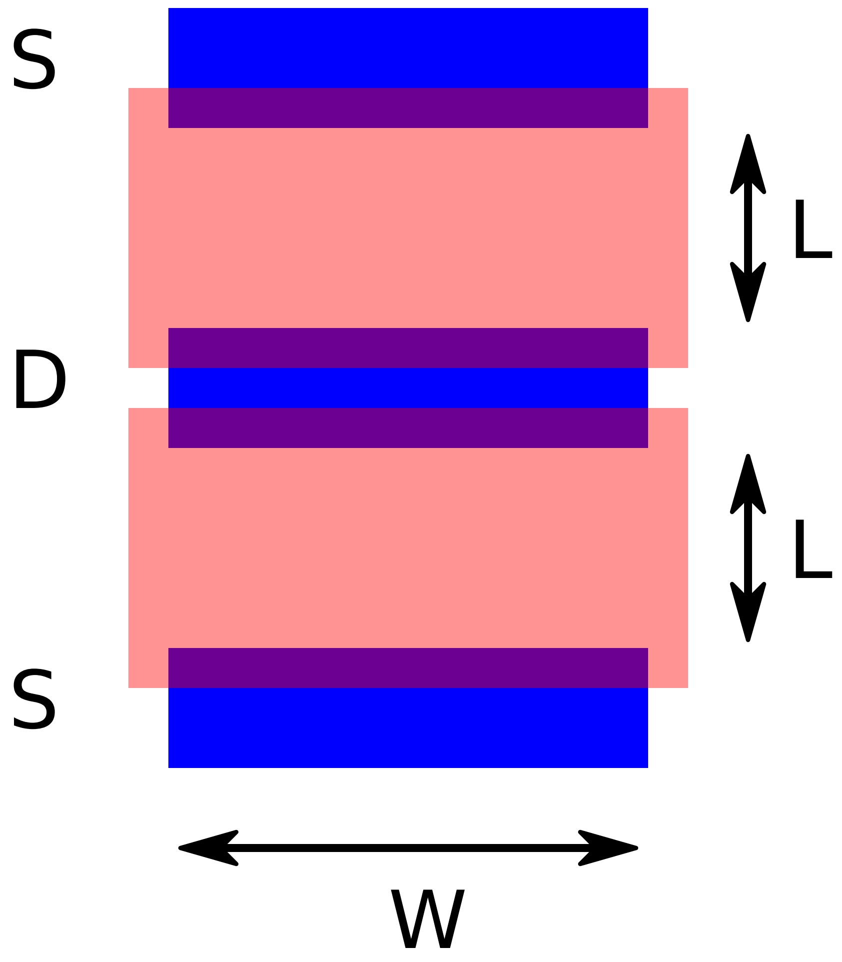
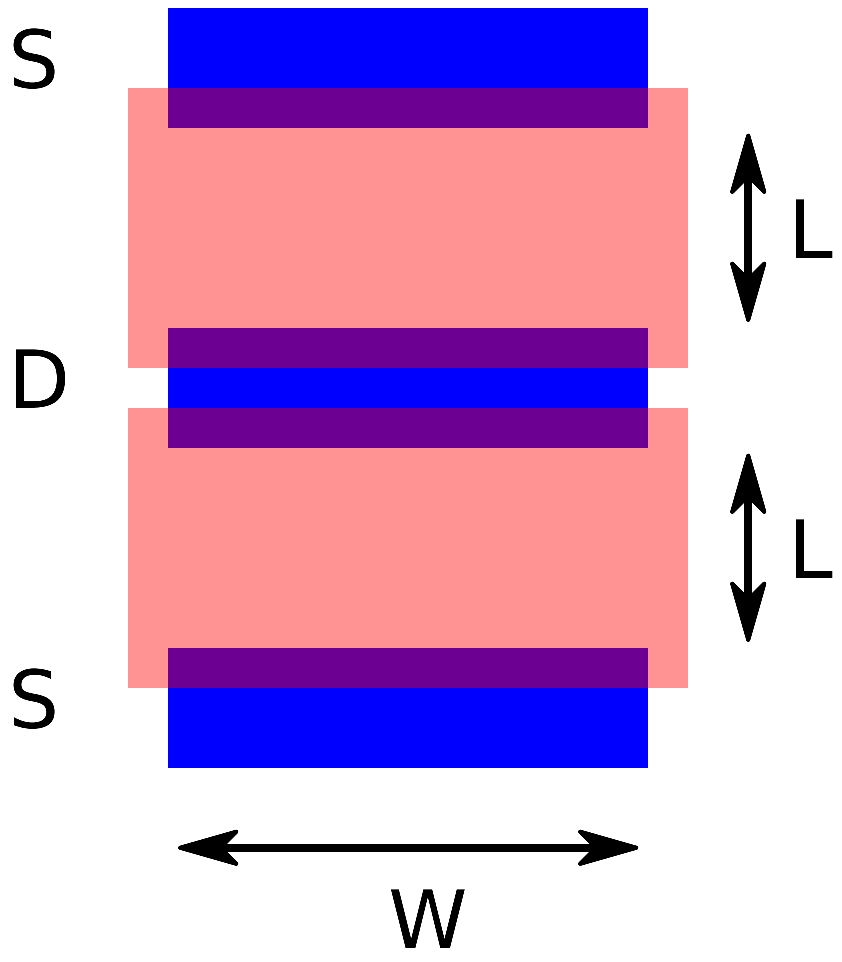
MOS design

MOS design

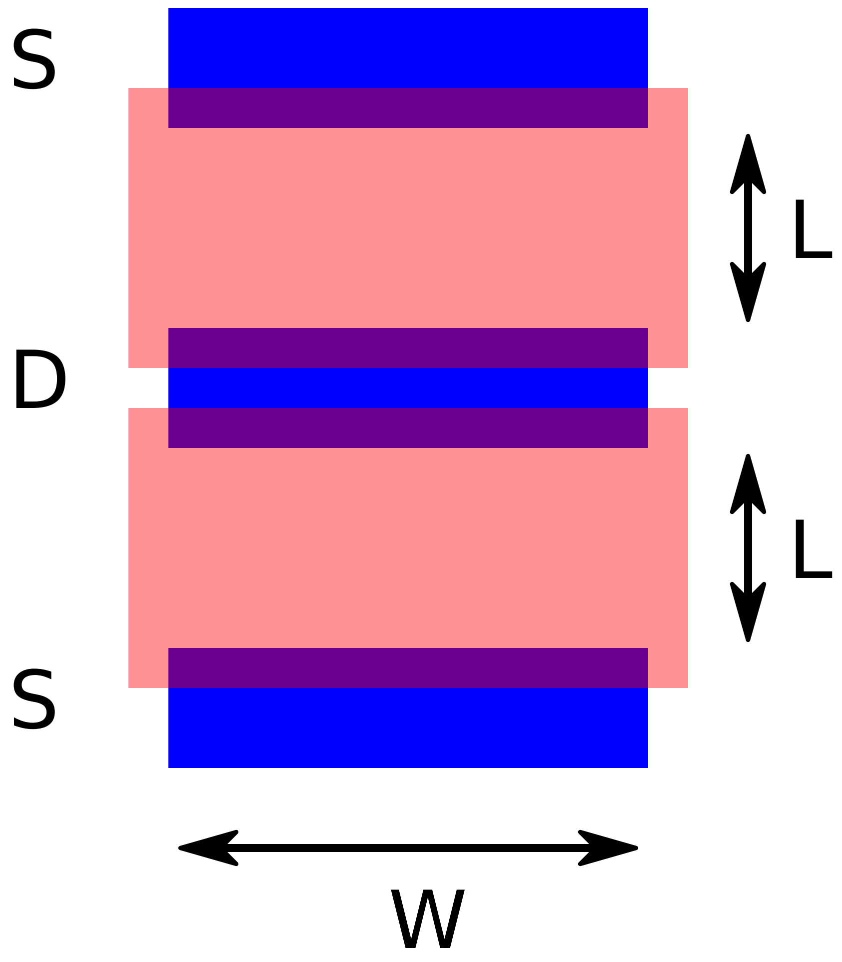


MOS design



Design question

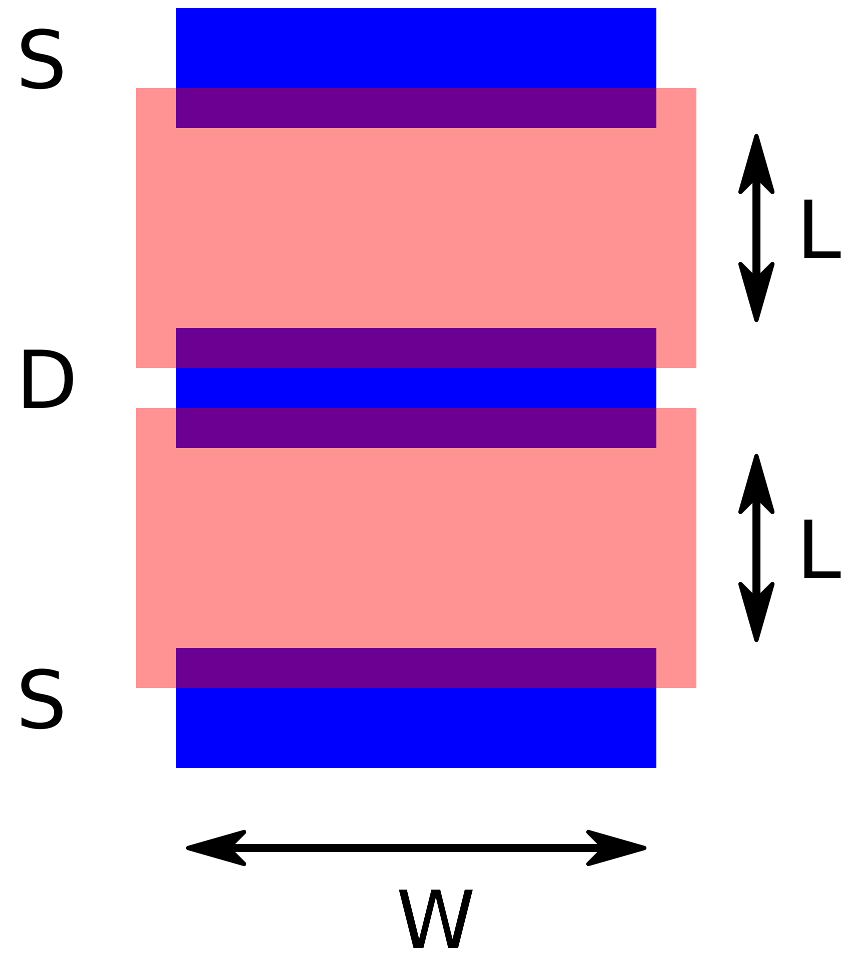
MOS design



Design question

In which way do the performance parameters of a MOS transistor depend on its design parameters

MOS design

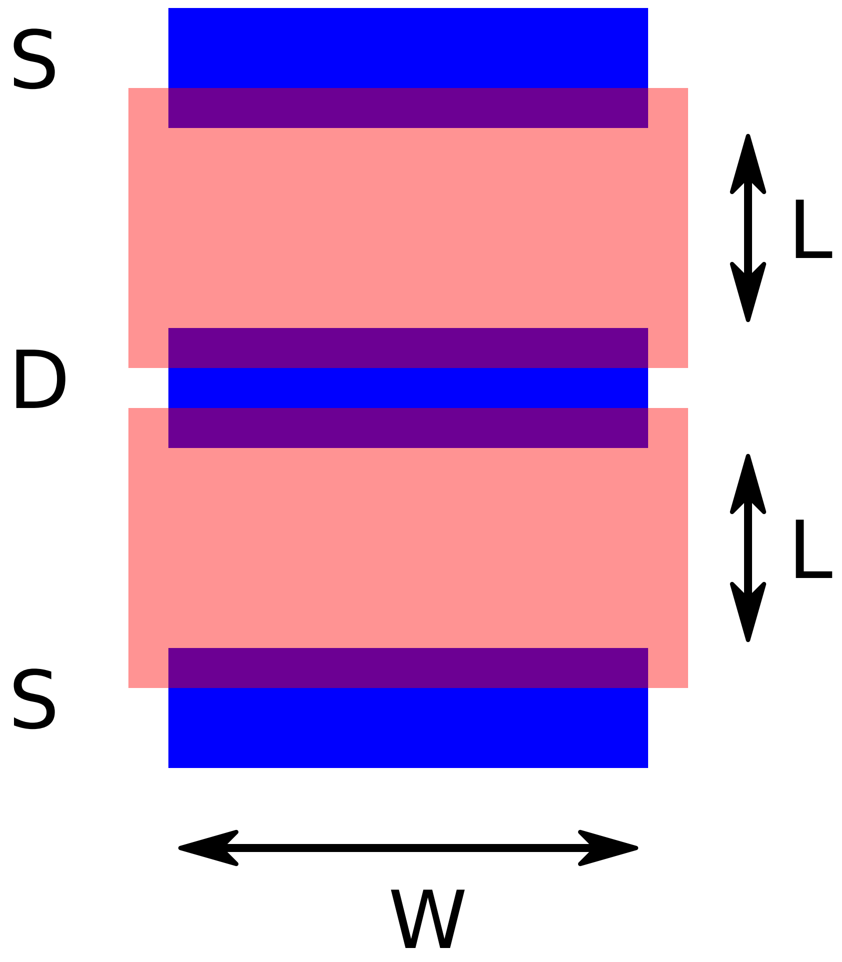


Design question

In which way do the performance parameters of a MOS transistor depend on its design parameters

Design parameters available to the designer

MOS design



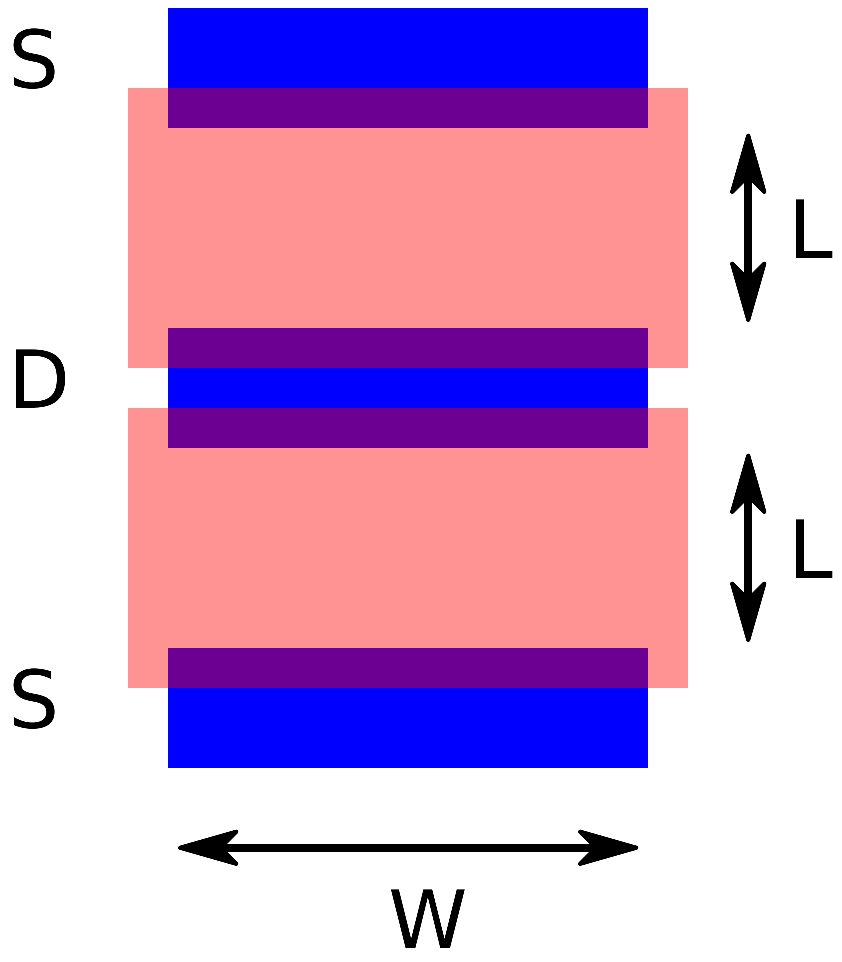
Design question

In which way do the performance parameters of a MOS transistor depend on its design parameters

Design parameters available to the designer

Channel width

MOS design



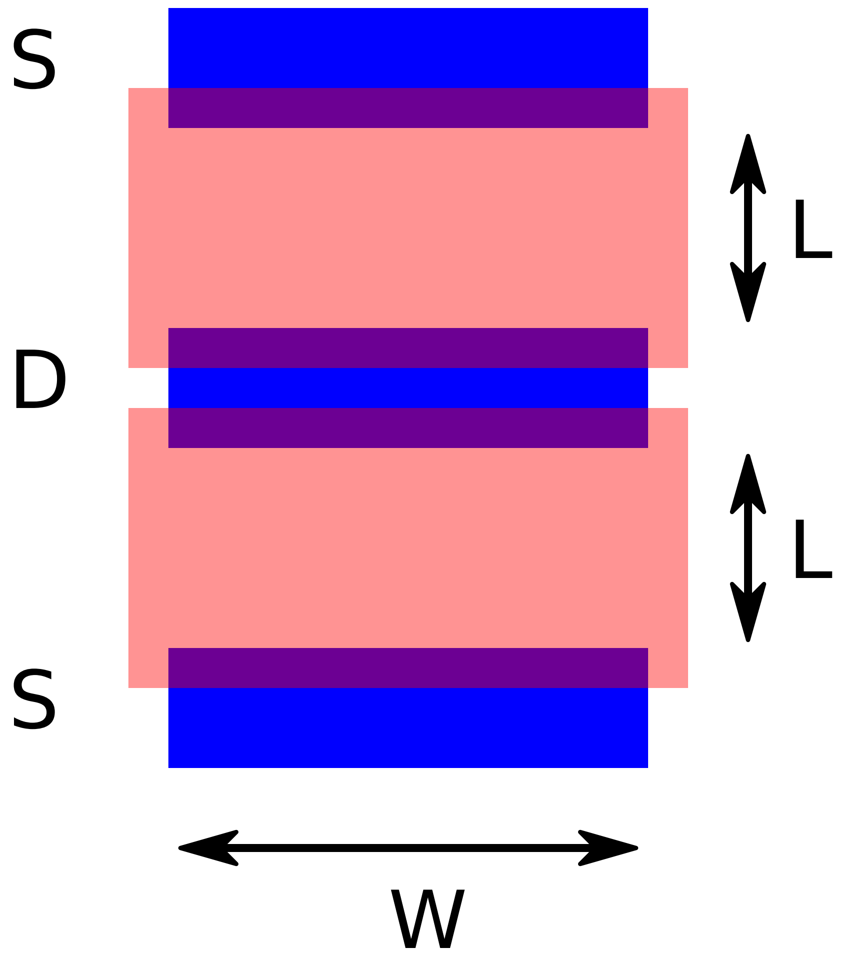
Design question

In which way do the performance parameters of a MOS transistor depend on its design parameters

Design parameters available to the designer

Channel width
Channel length

MOS design



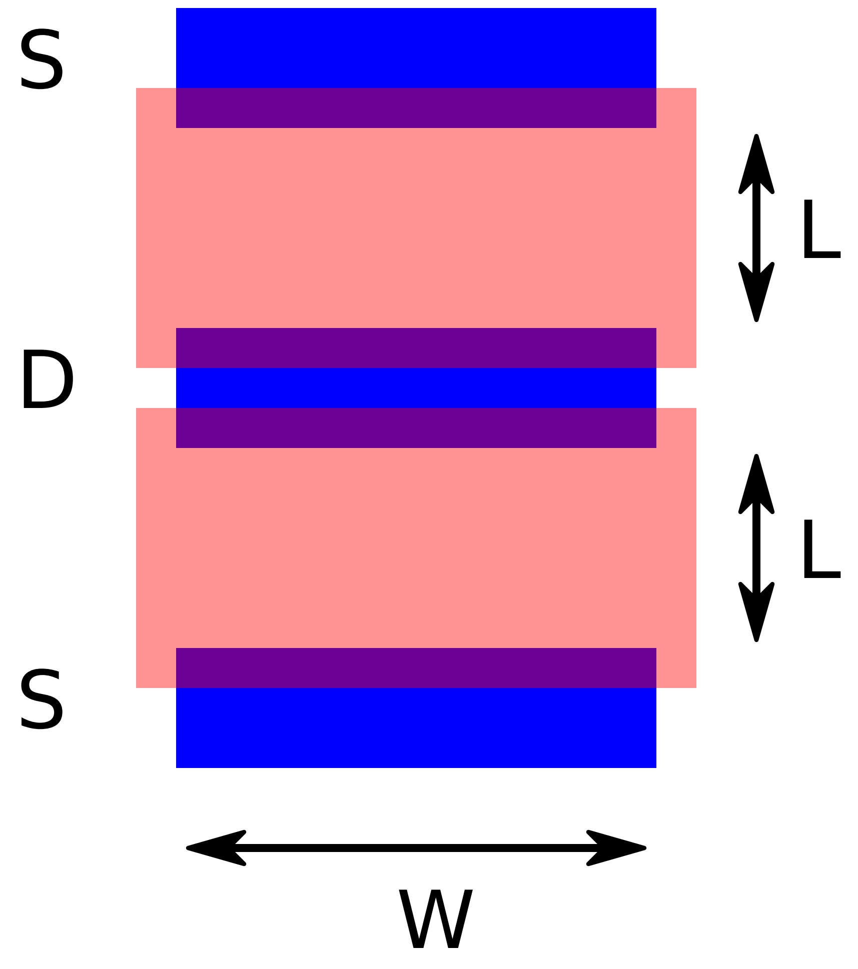
Design question

In which way do the performance parameters of a MOS transistor depend on its design parameters

Design parameters available to the designer

Channel width
Channel length
Number of sections

MOS design



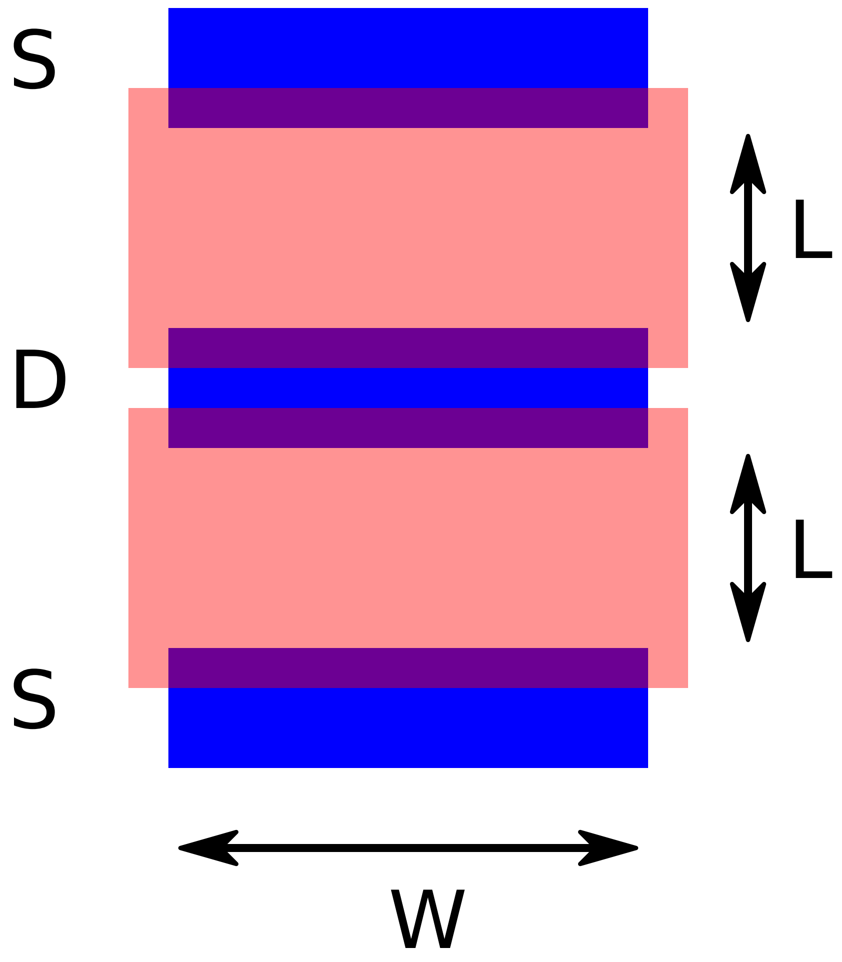
Design question

In which way do the performance parameters of a MOS transistor depend on its design parameters

Design parameters available to the designer

Channel width
Channel length
Number of sections
Drain current

MOS design



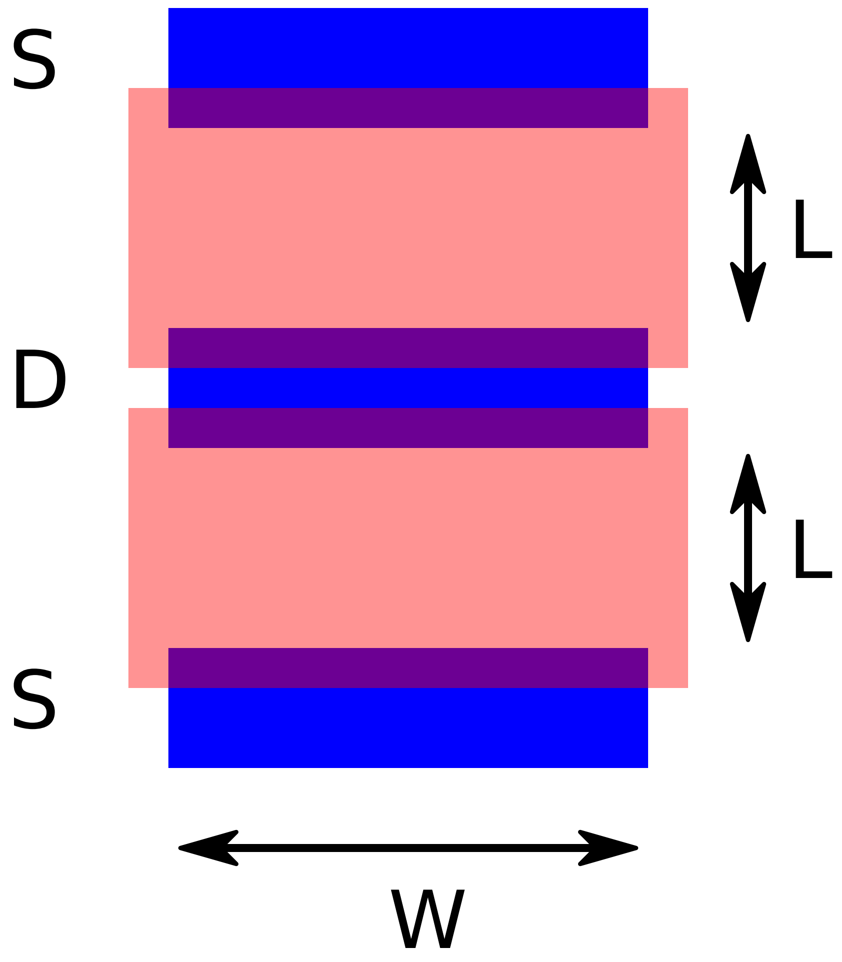
Design question

In which way do the performance parameters of a MOS transistor depend on its design parameters

Design parameters available to the designer

Channel width
Channel length
Number of sections
Drain current
Drain-source voltage

MOS design



Design question

In which way do the performance parameters of a MOS transistor depend on its design parameters

Methods

Design parameters available to the designer

Channel width
Channel length
Number of sections
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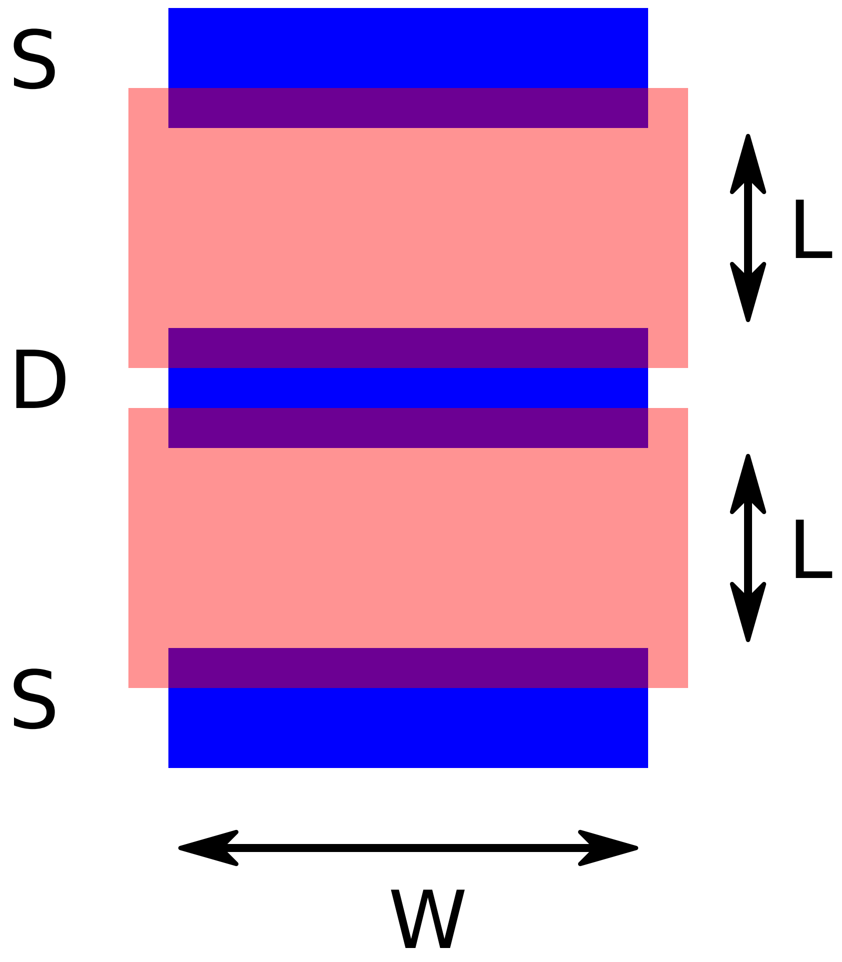
MOS design



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Channel width
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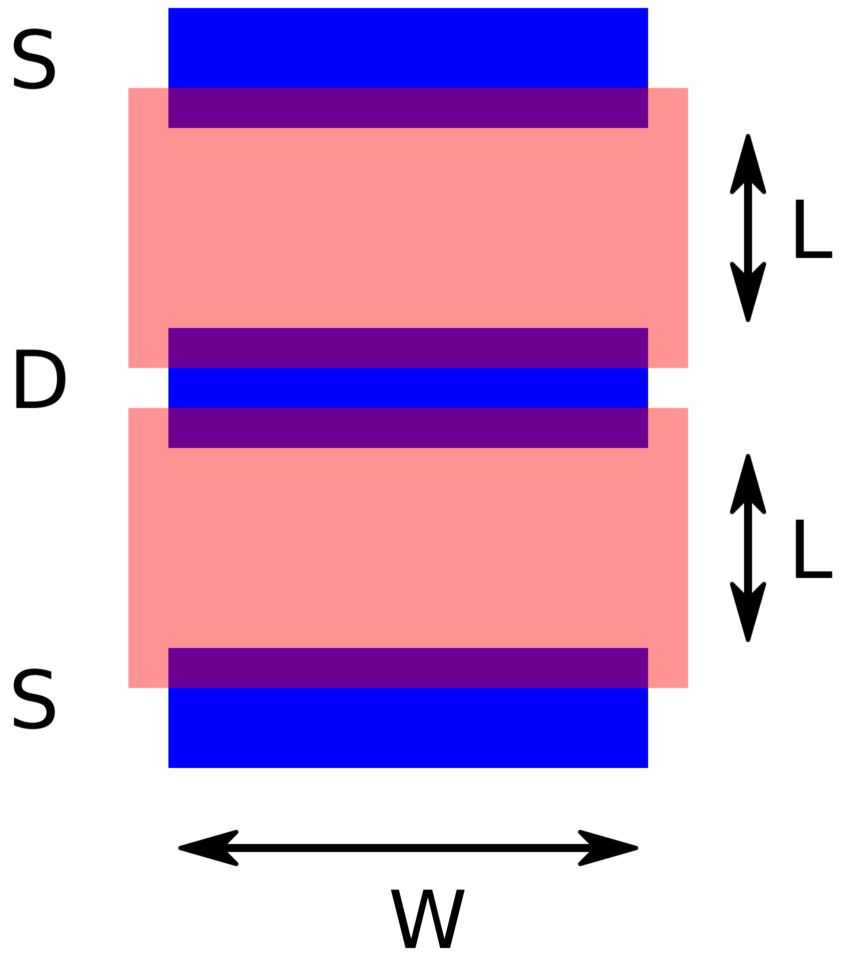
Methods

Use a design manual with graphs and tables and scale devices

MOS design



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Design parameters available to the designer

Channel width
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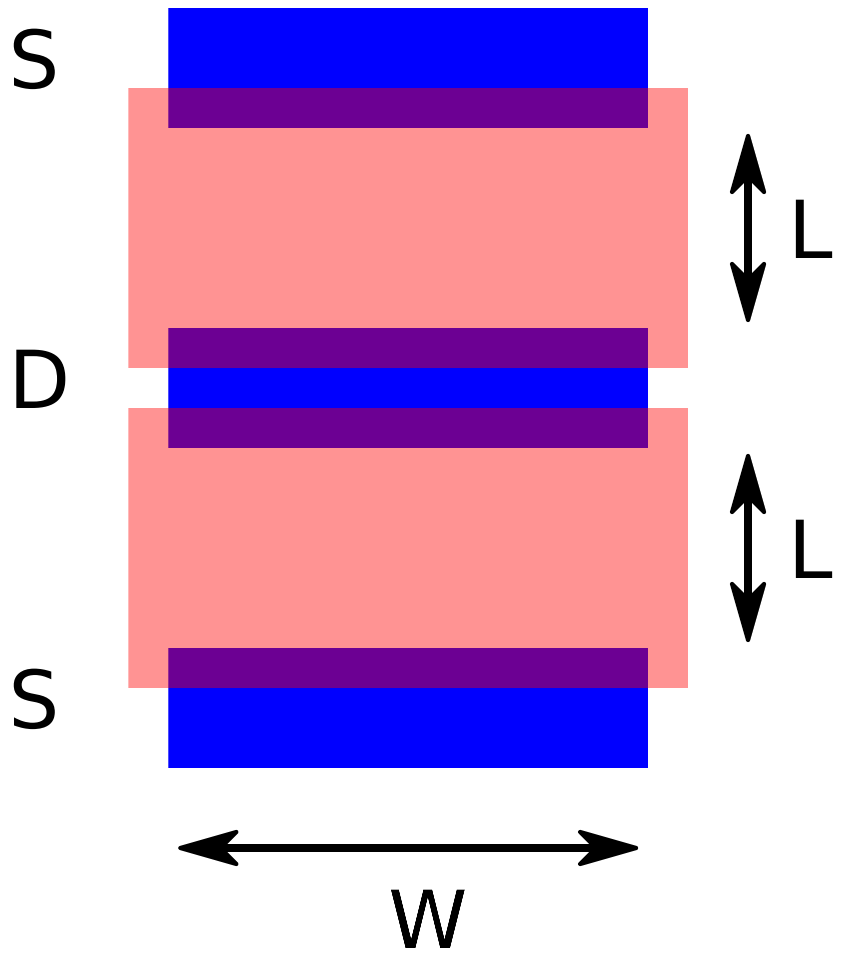
Useful, but generally not all situations will be covered

MOS design



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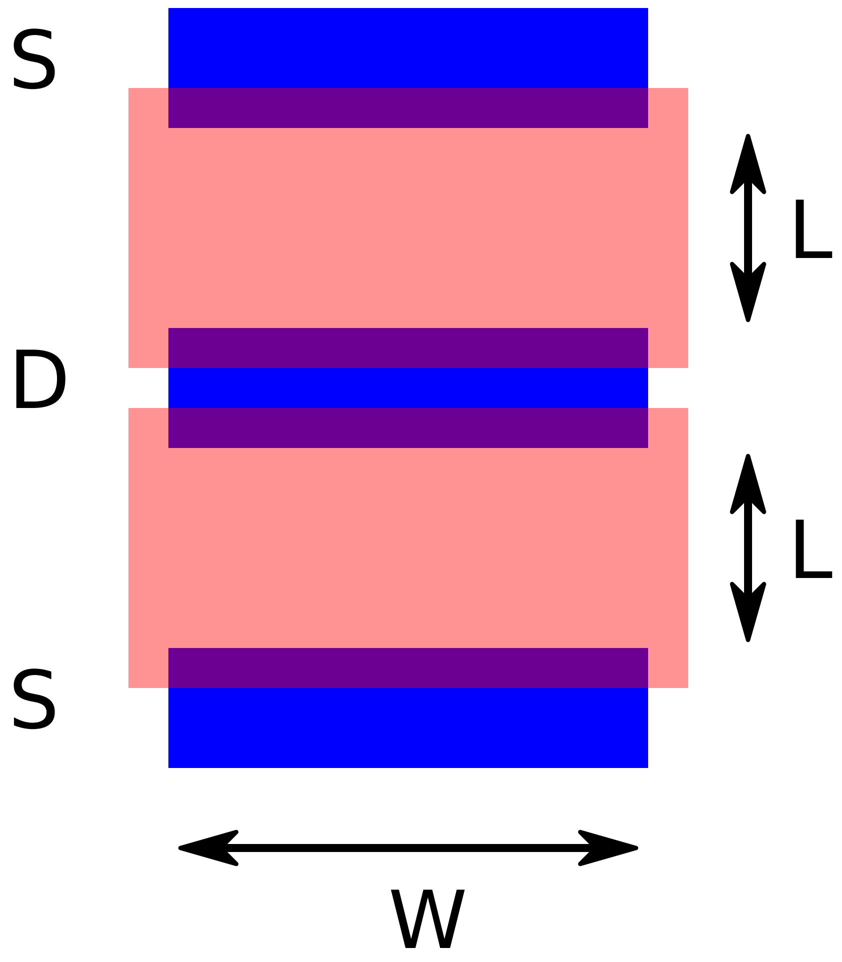
Design a device and study its performance through simulation

MOS design



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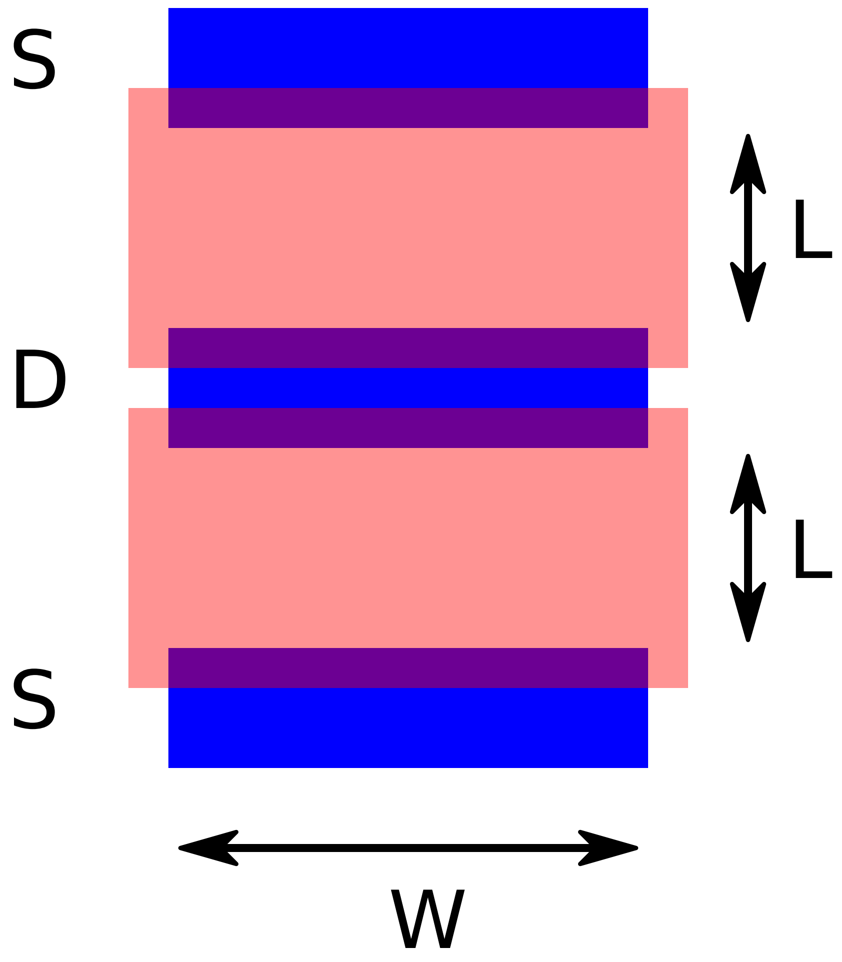
Useful for finding DC operating conditions and small-signal model parameters

MOS design



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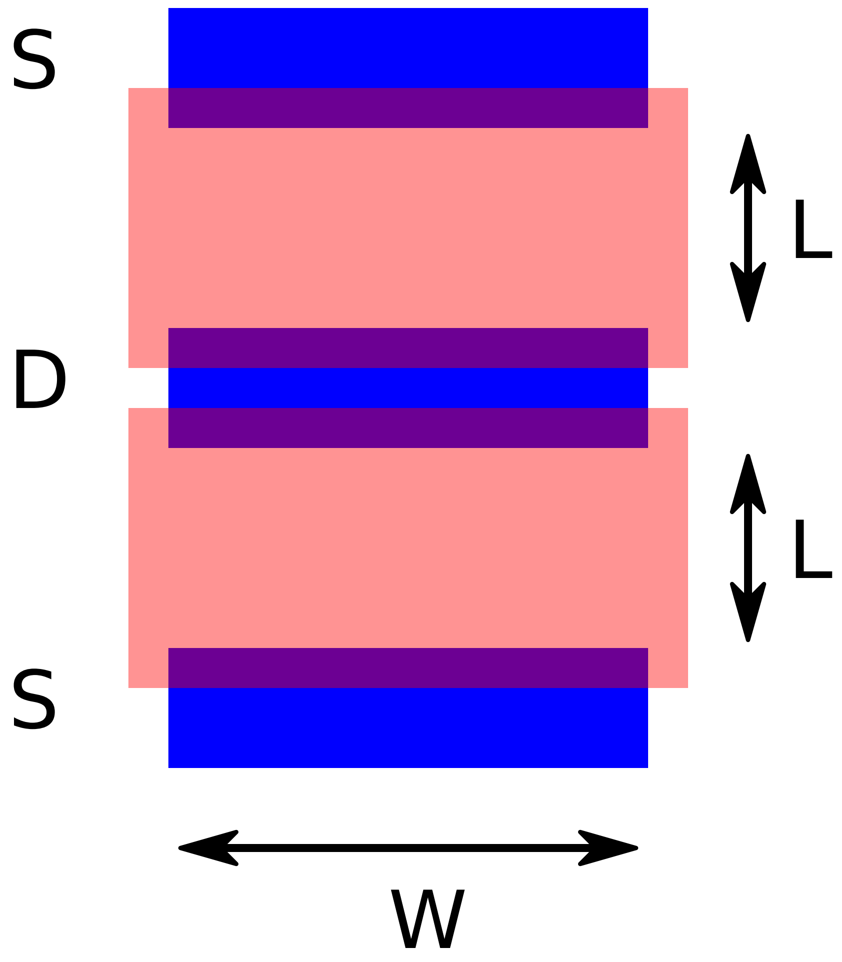

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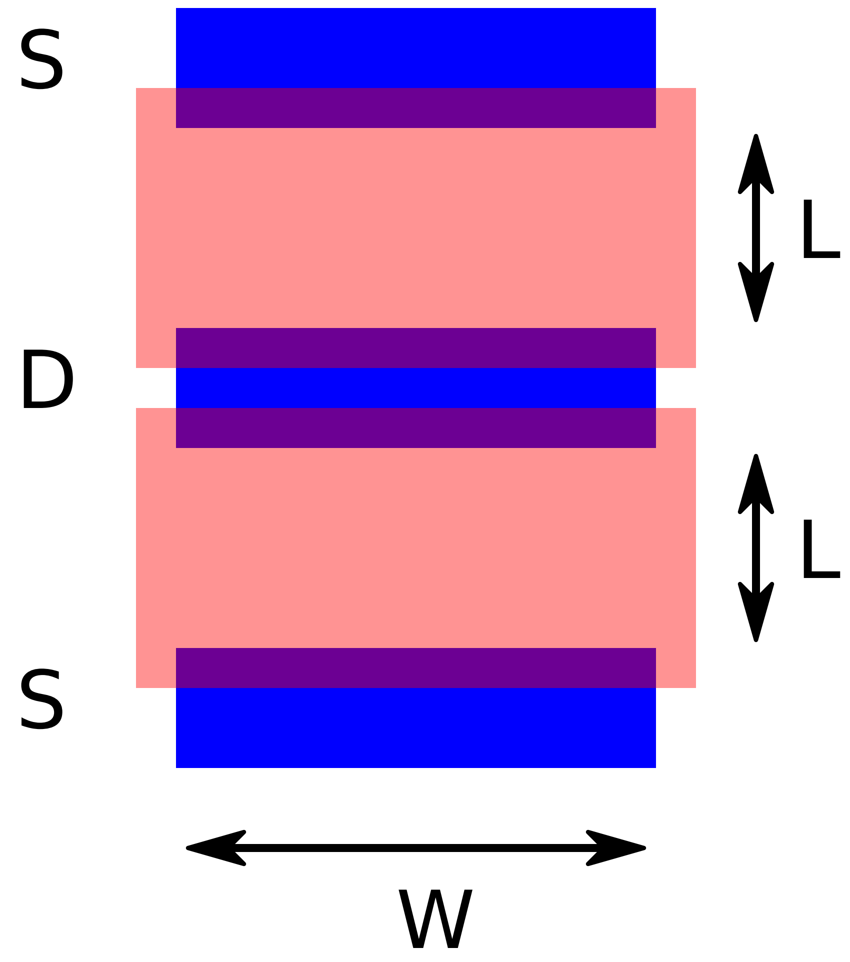
MOS design



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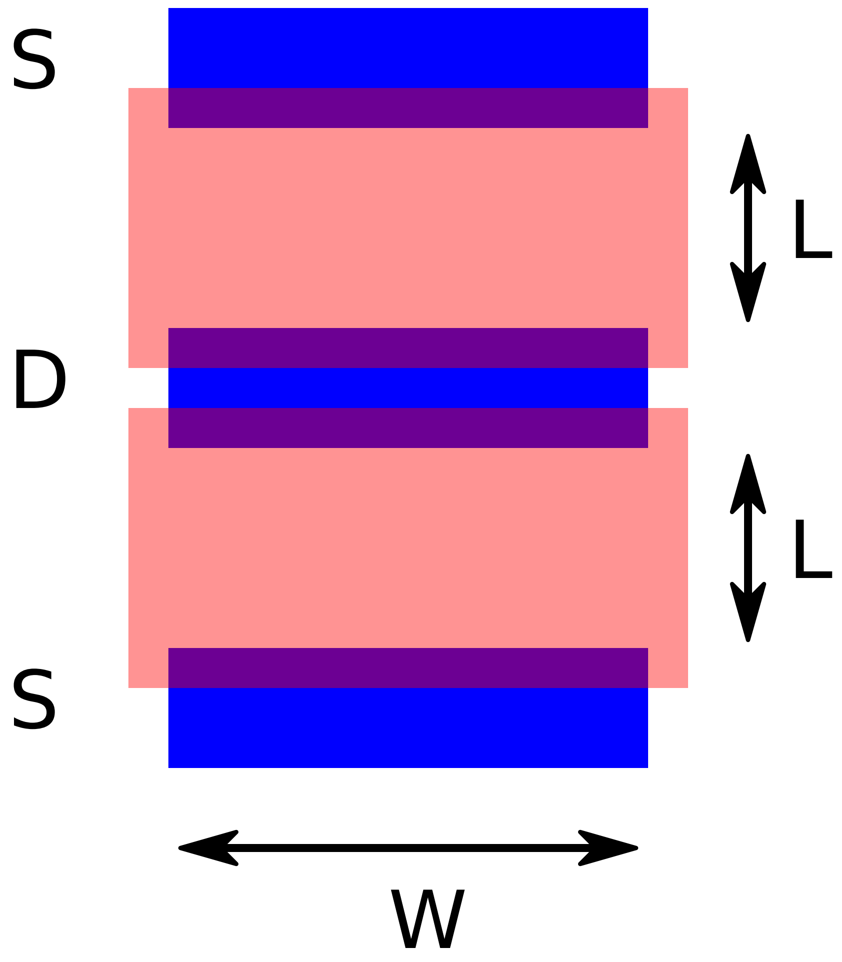
Use simplified design models

MOS design



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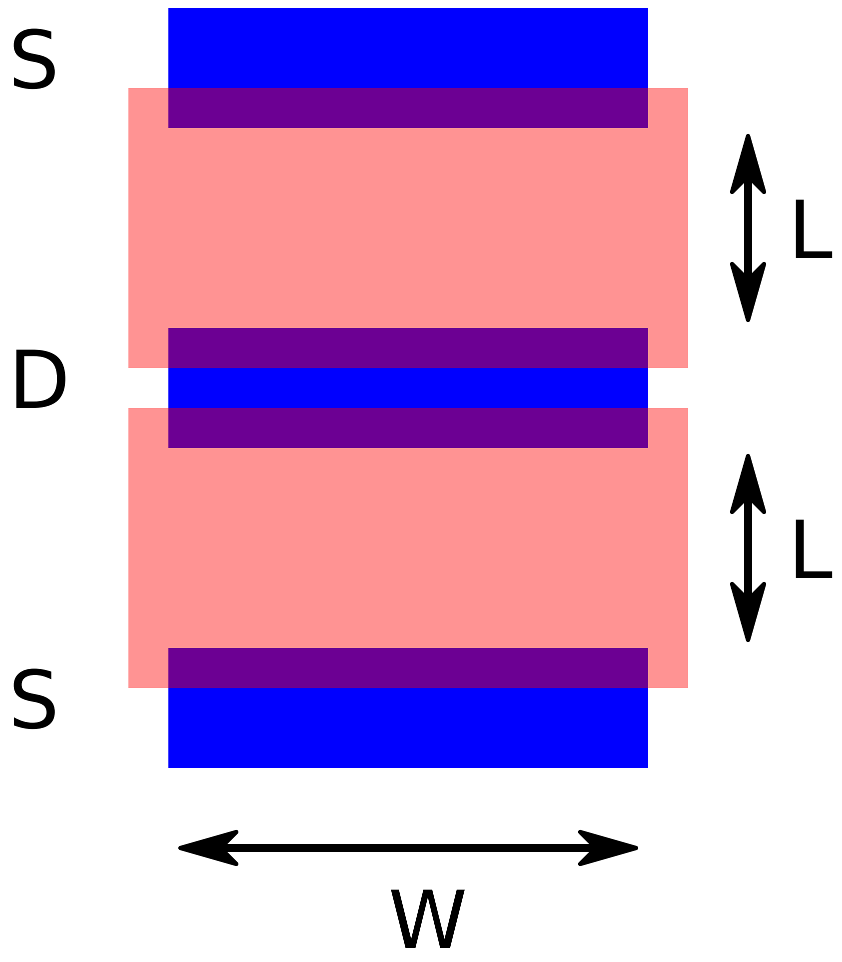
MOS design



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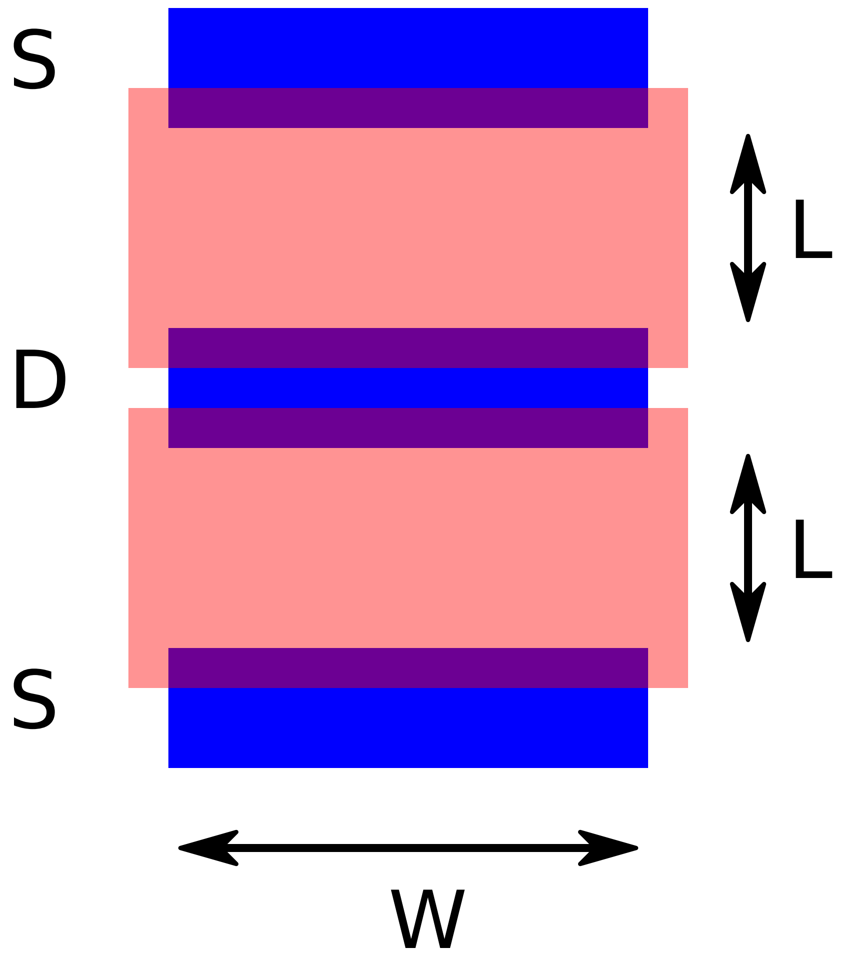
MOS design



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MOS small-signal model

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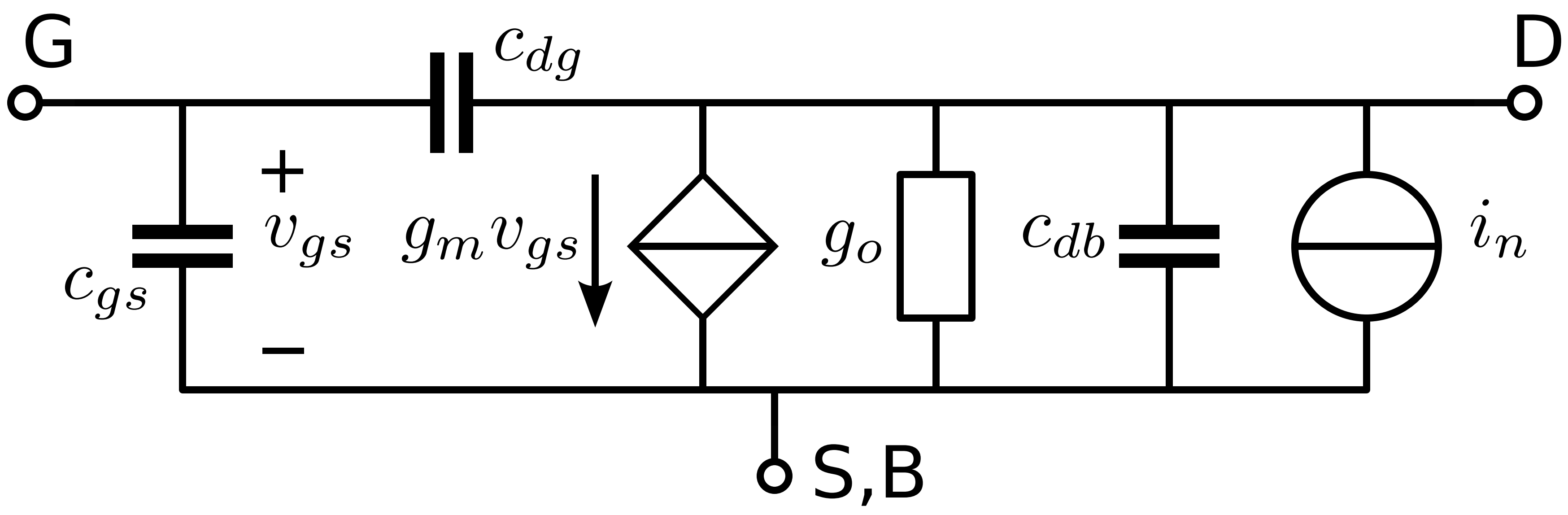

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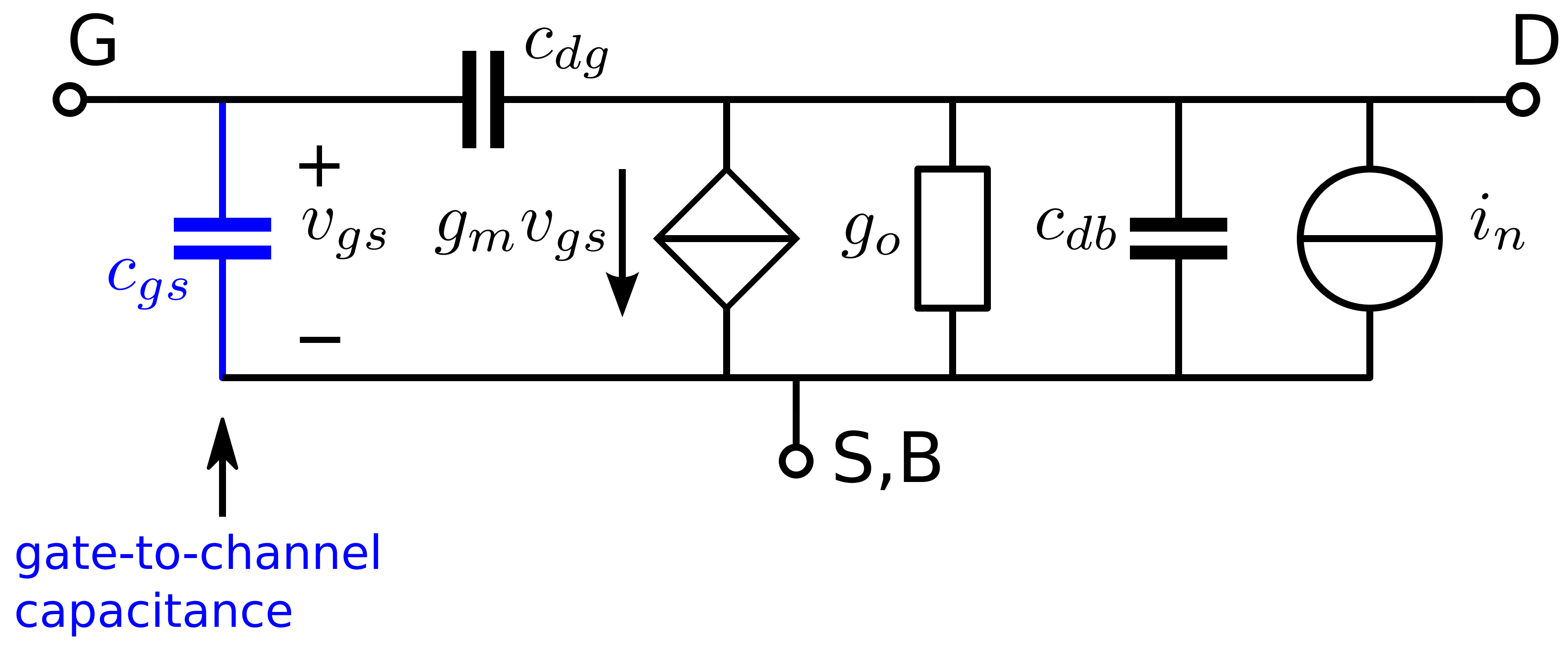

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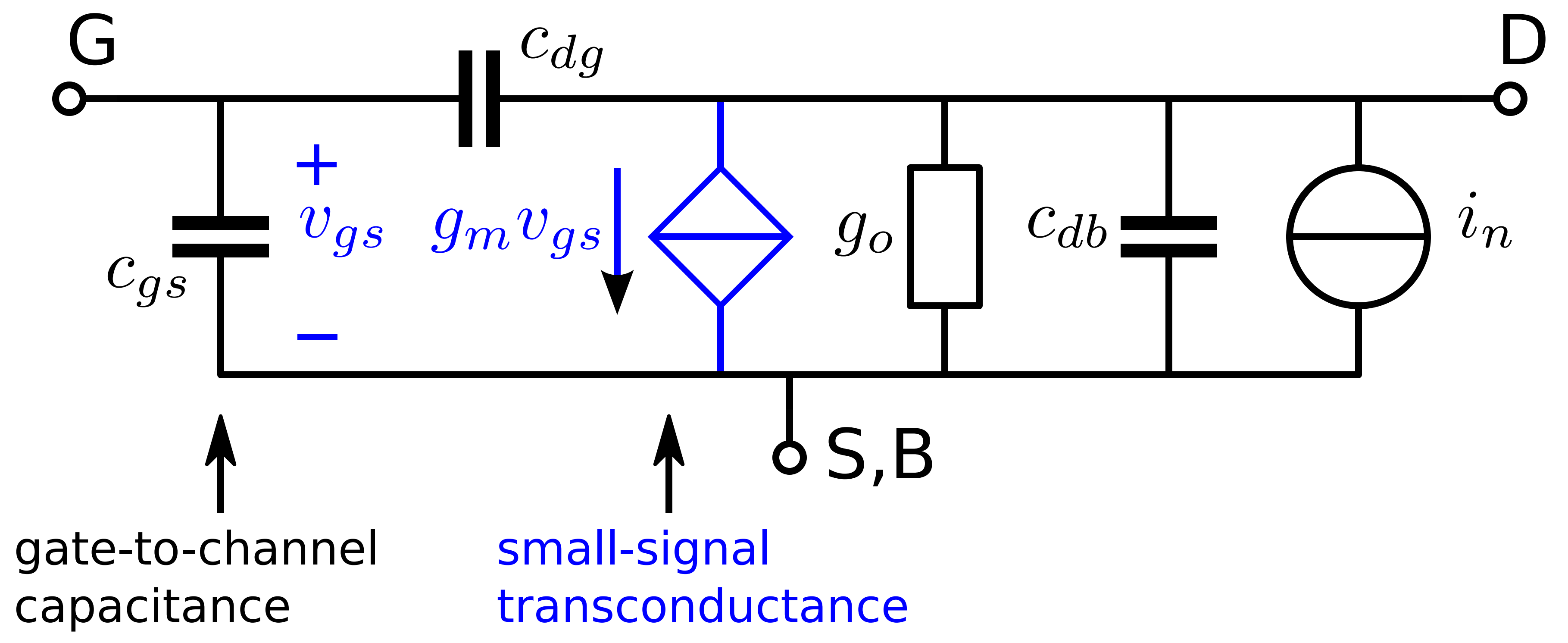

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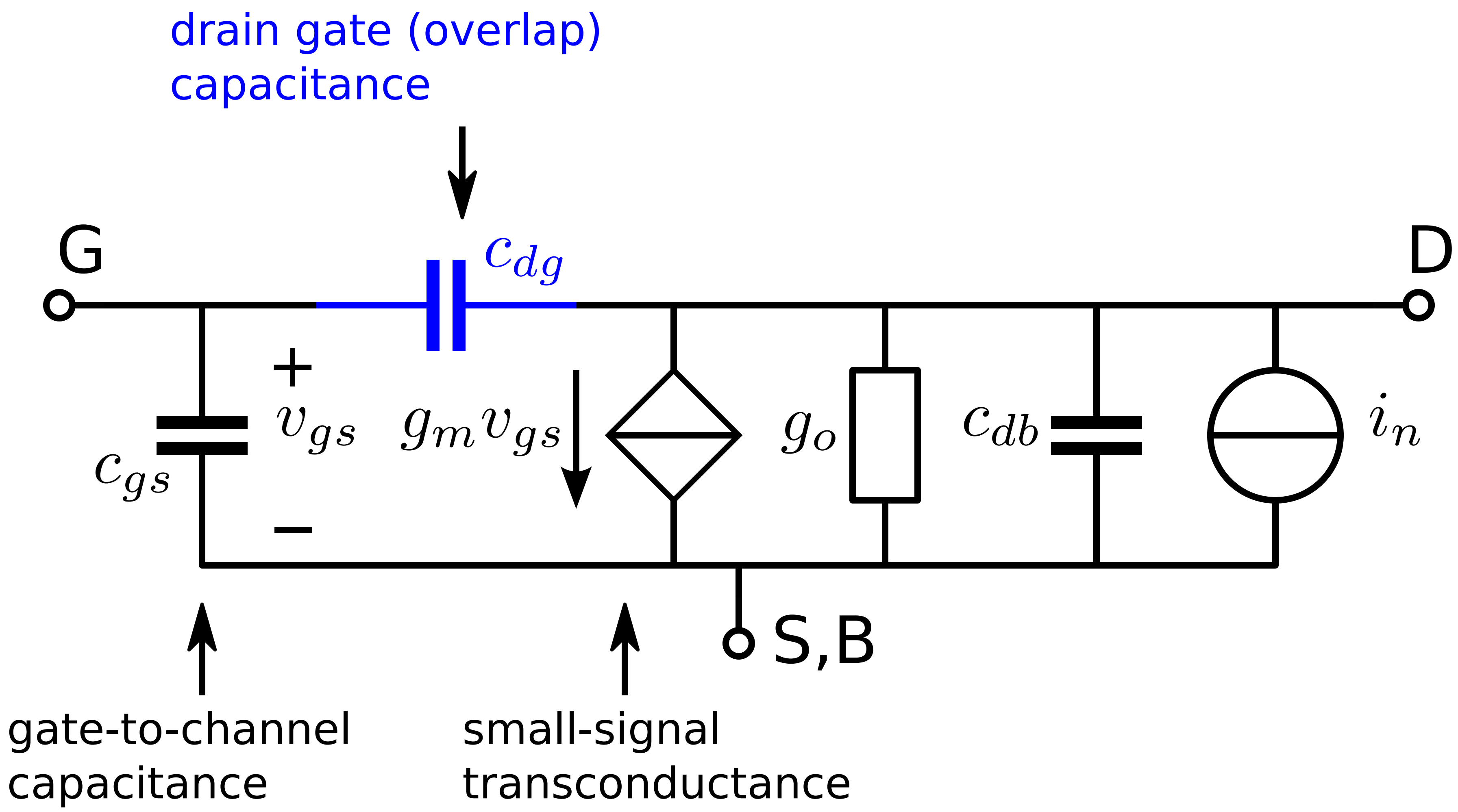

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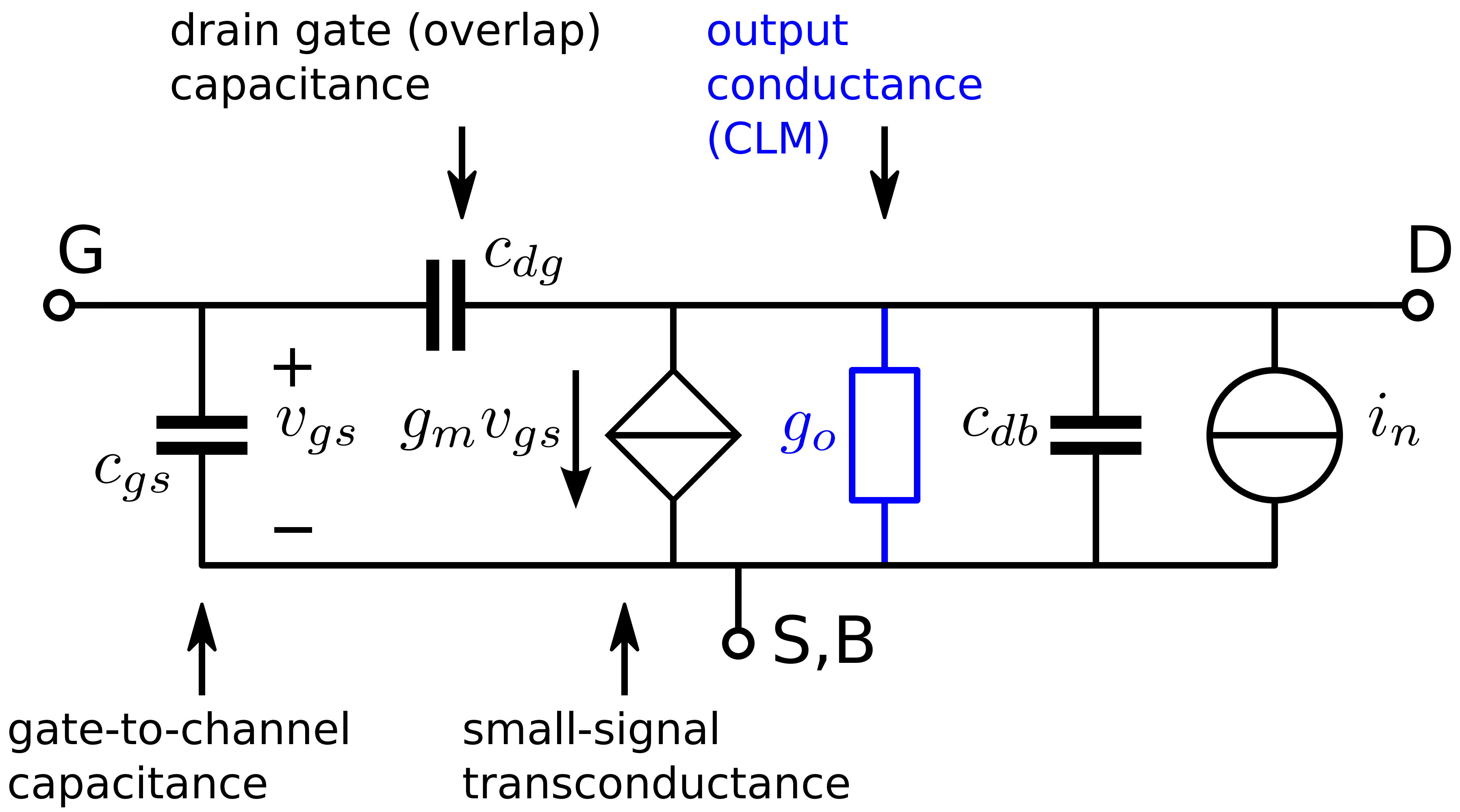

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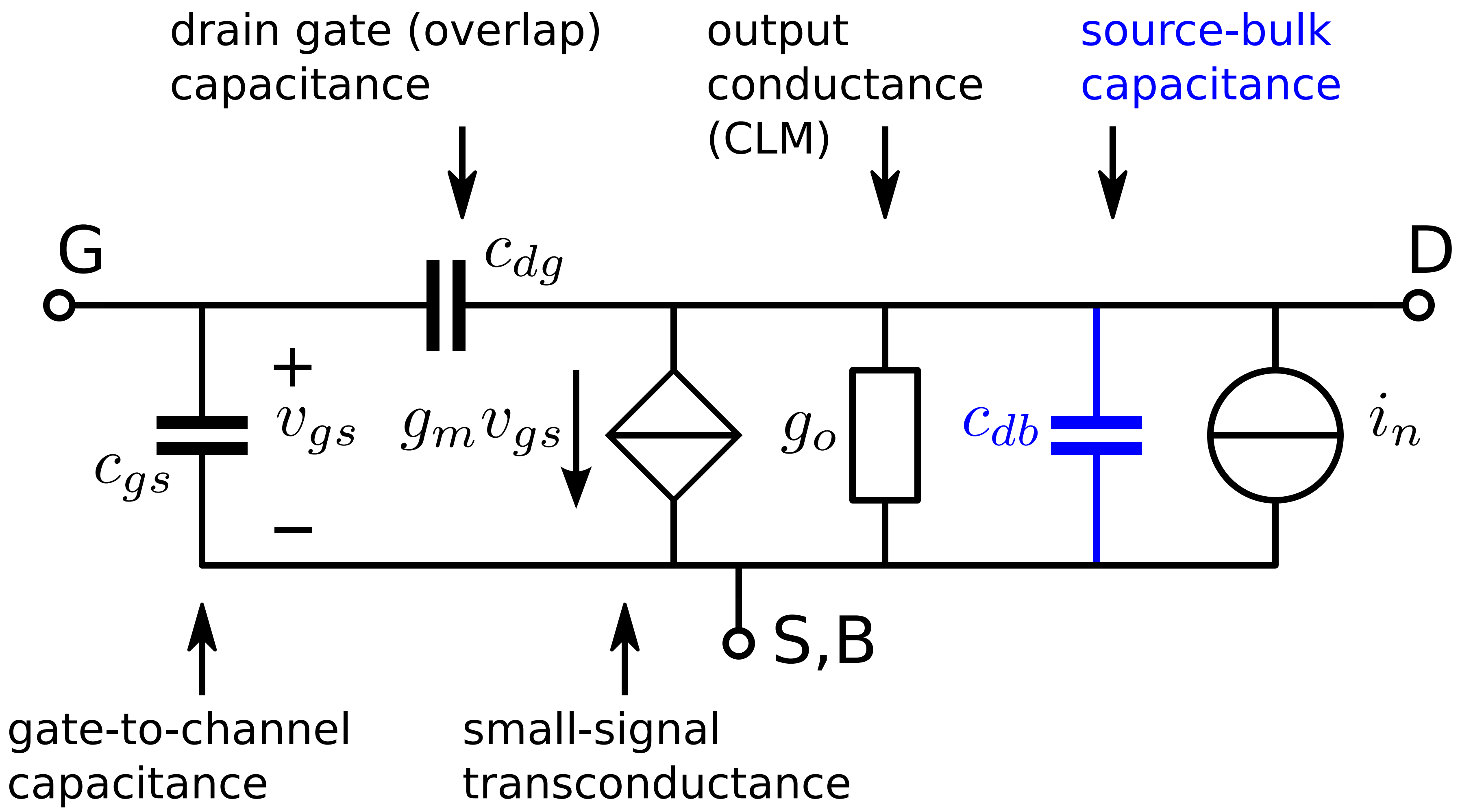

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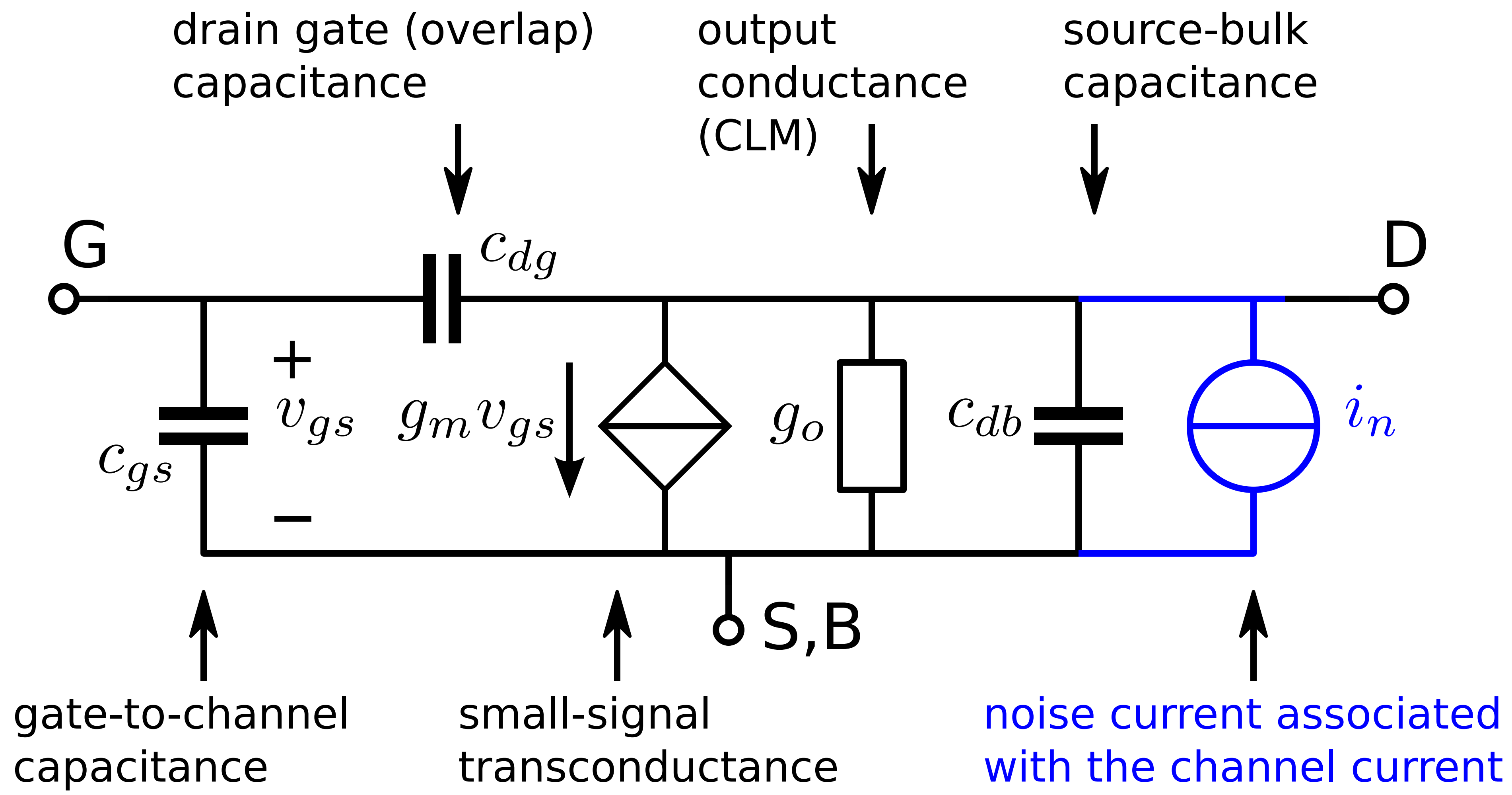

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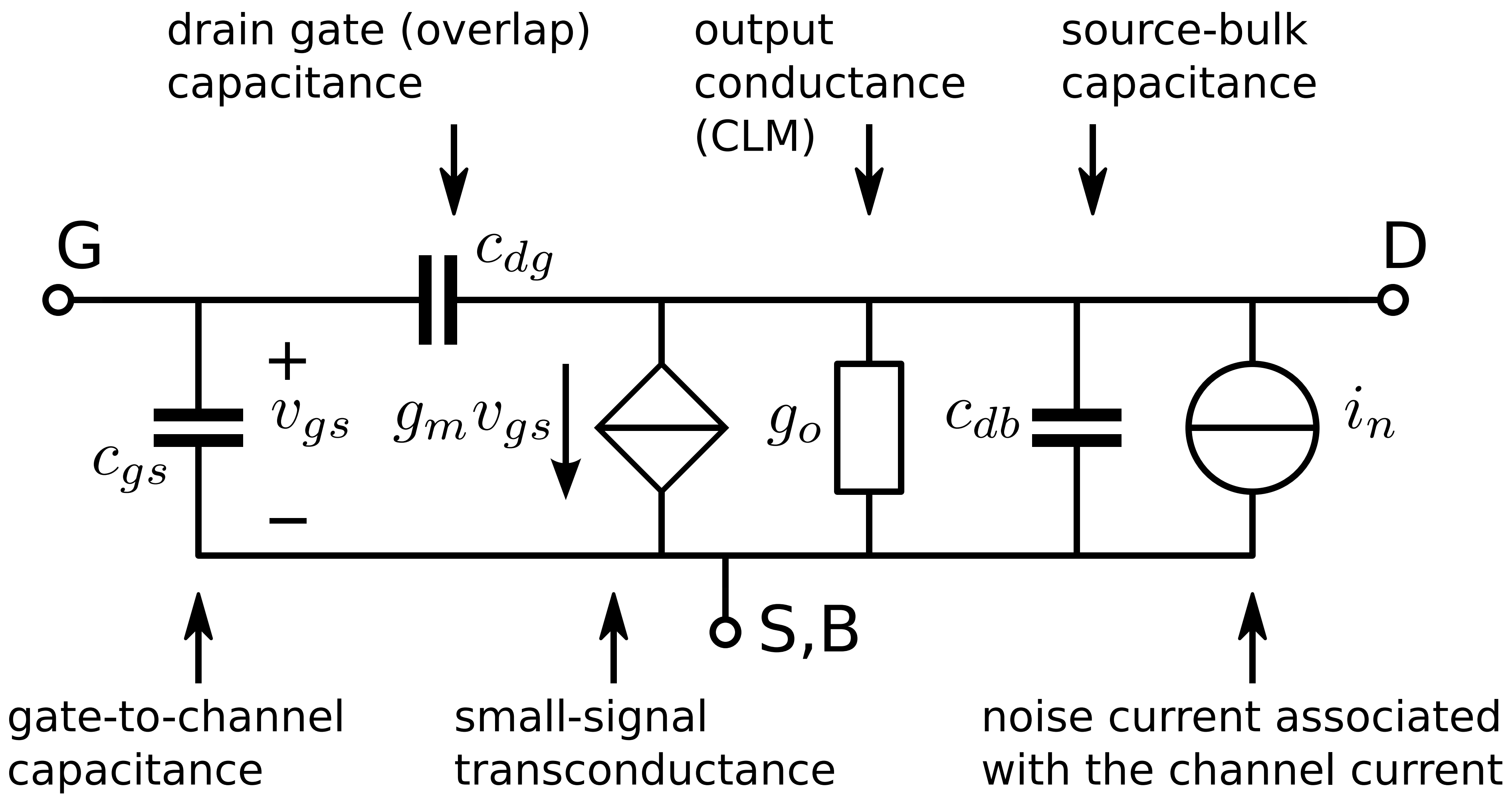
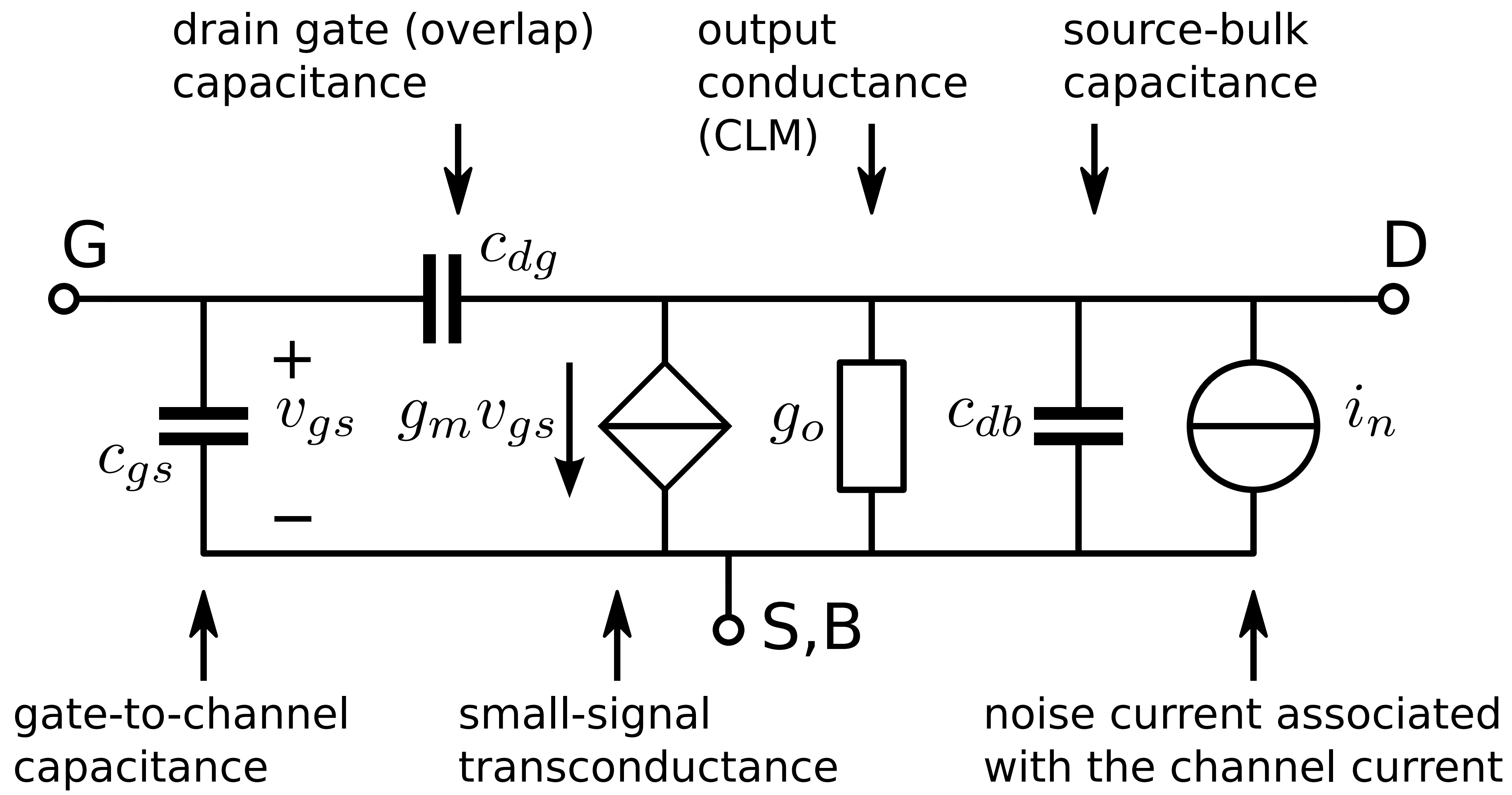


How do these model parameters depend on the device geometry and the operating conditions?

MOS small-signal model



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MOS EKV model

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1995: C.C. **E**nz, F. **K**rummenacher and E.A. **V**ittoz

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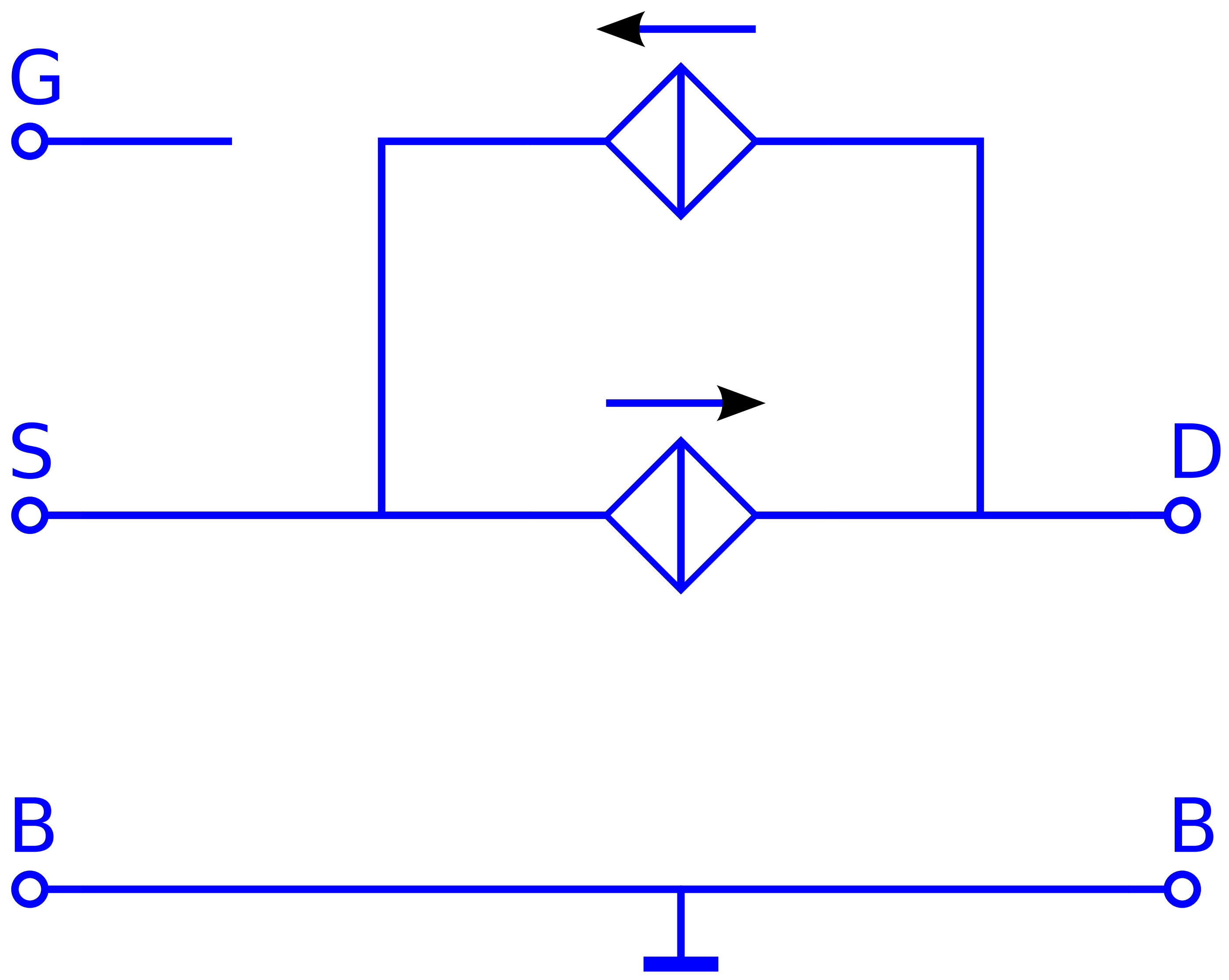

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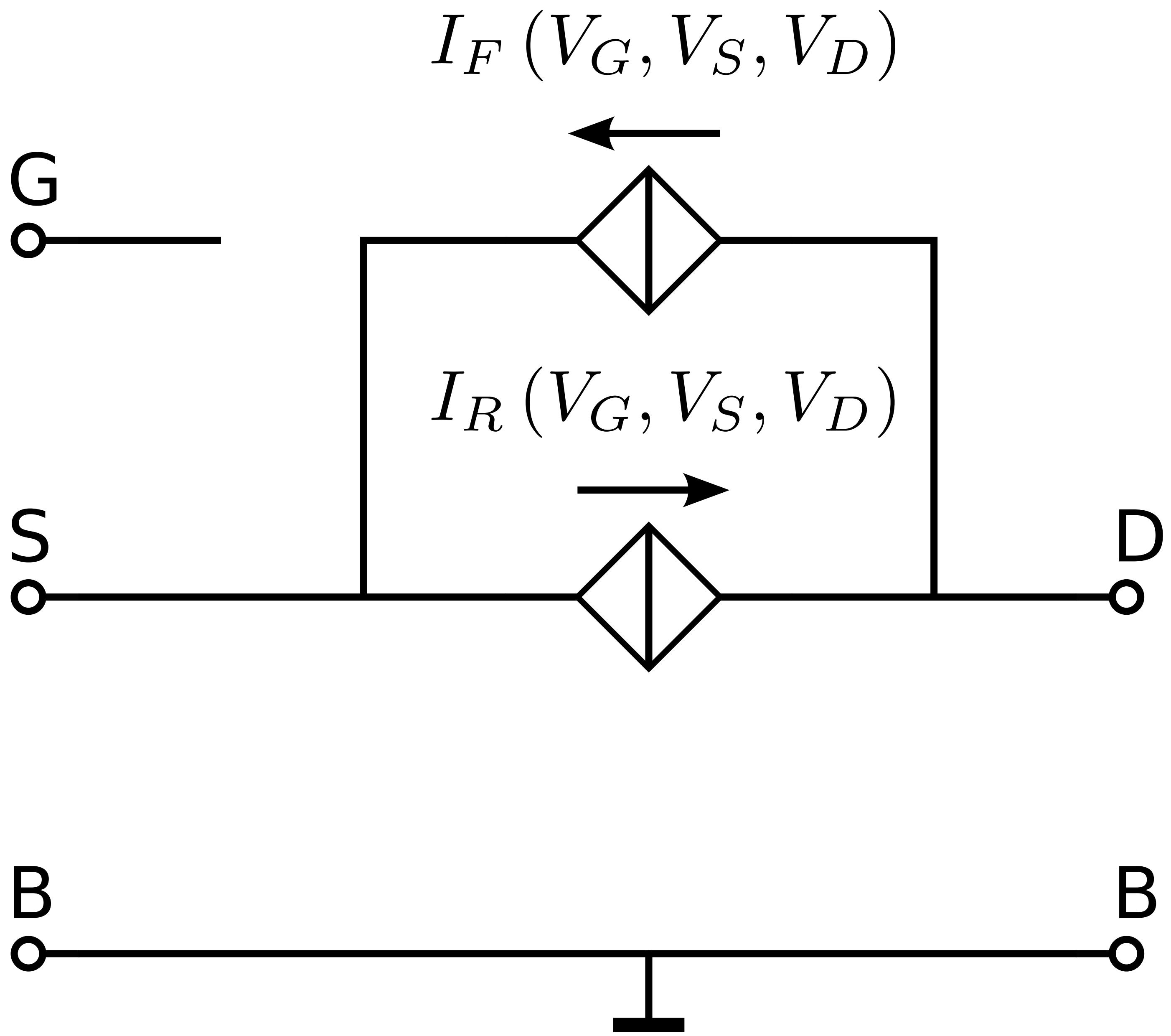

Models all operating regions
from weak inversion to strong inversion

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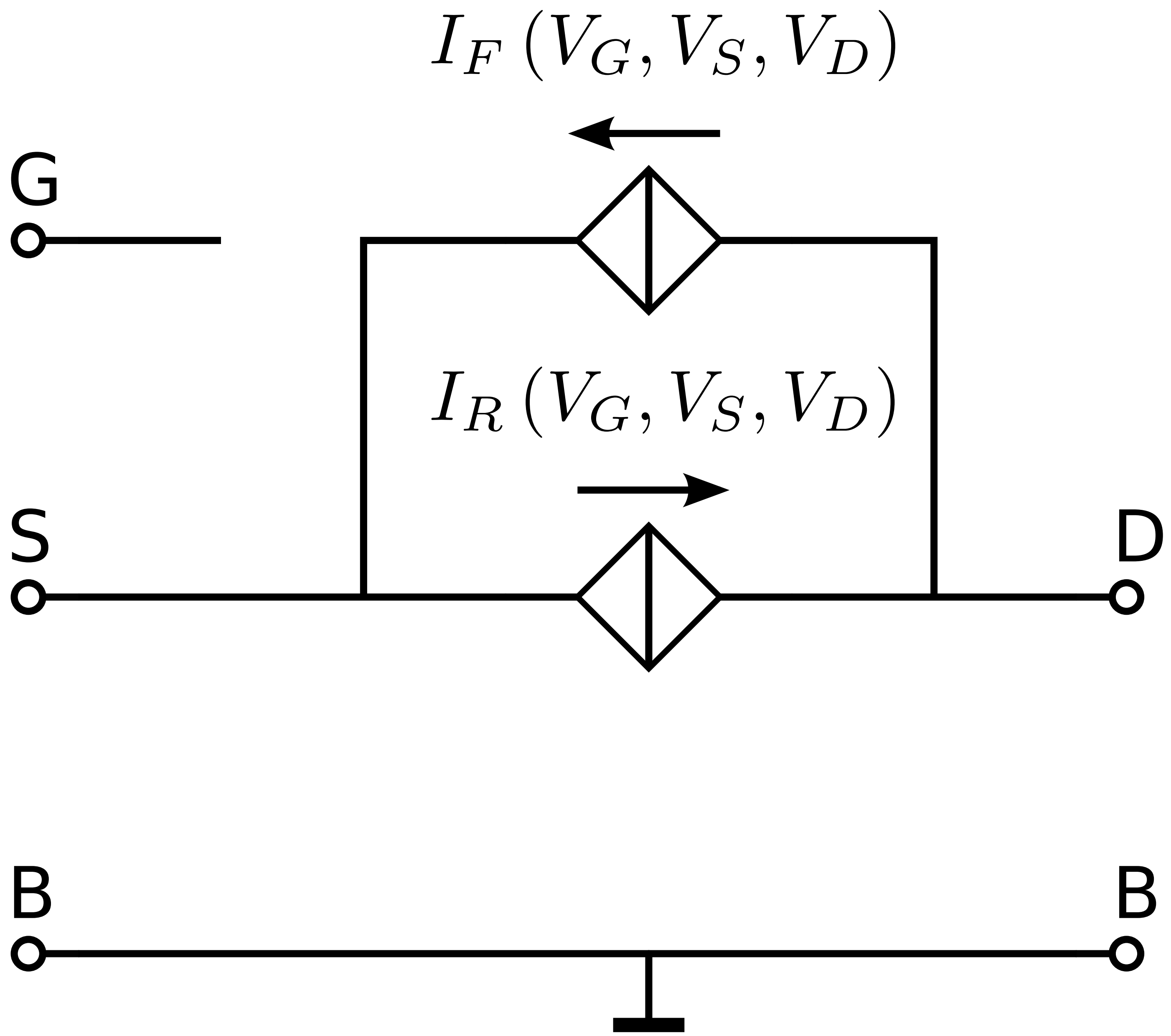

Models all operating regions
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Gate, source and drain voltages
with respect to the bulk

MOS EKV model

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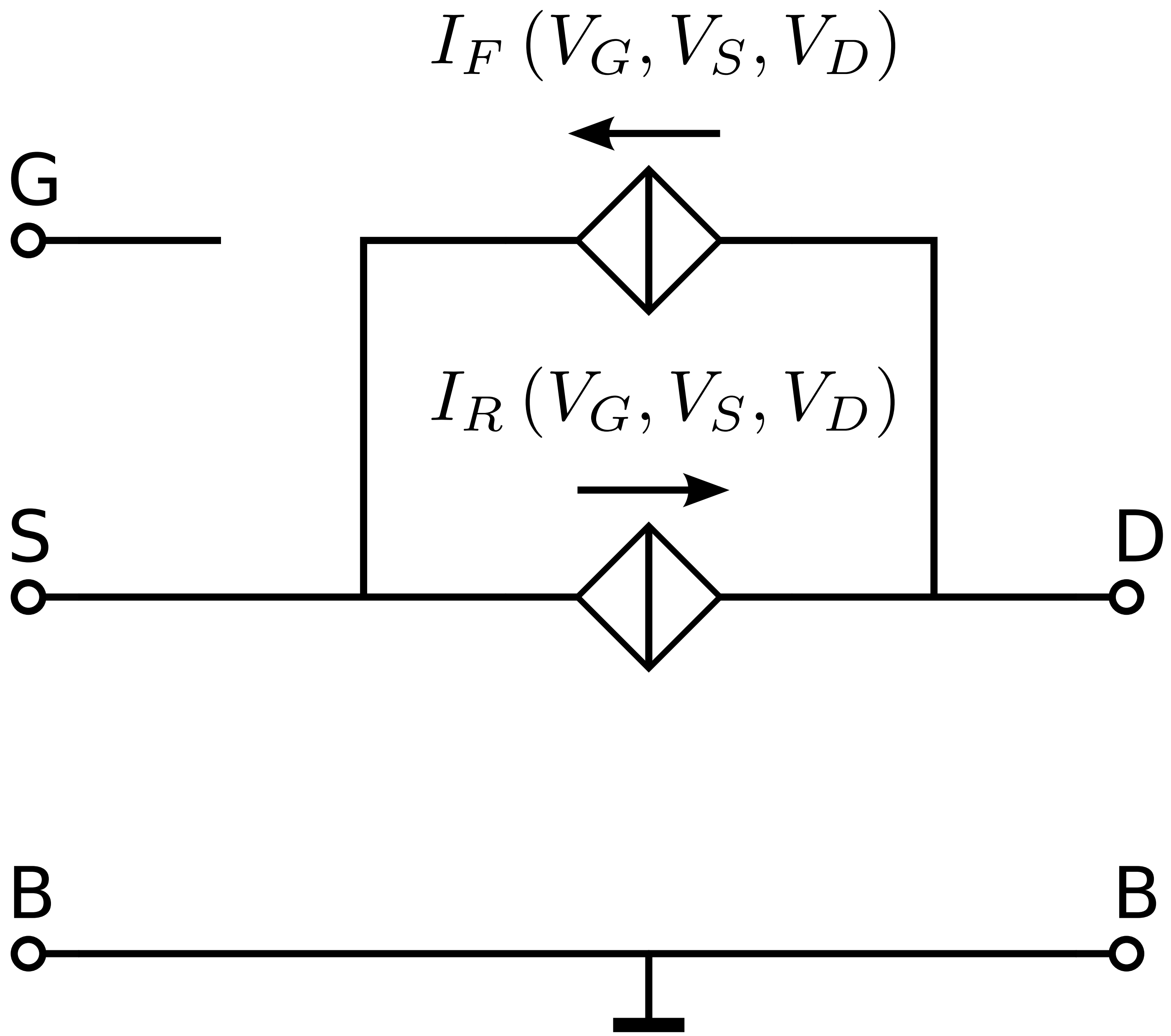

Models all operating regions
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Gate, source and drain voltages
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Symmetrical charge-controlled model

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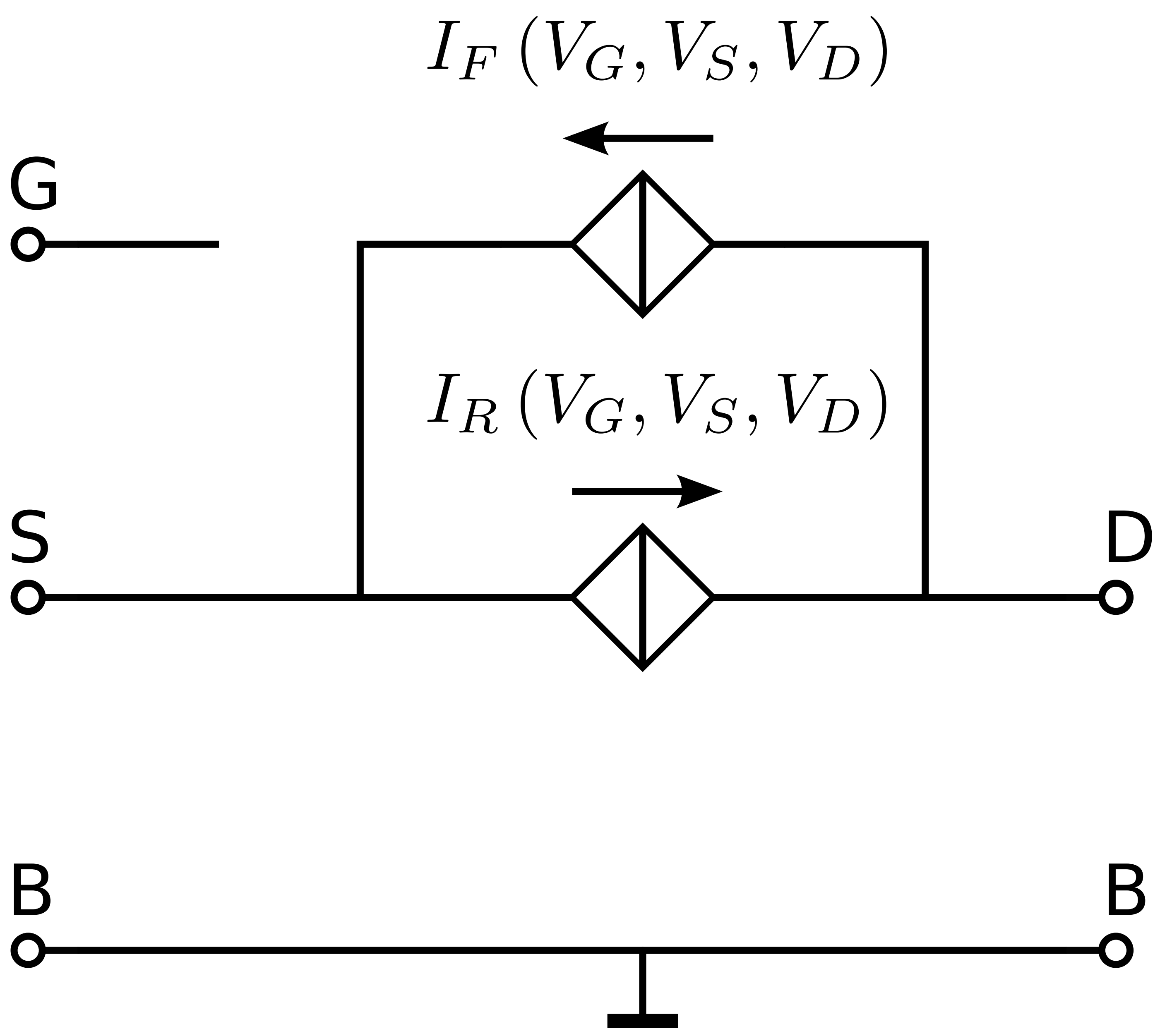

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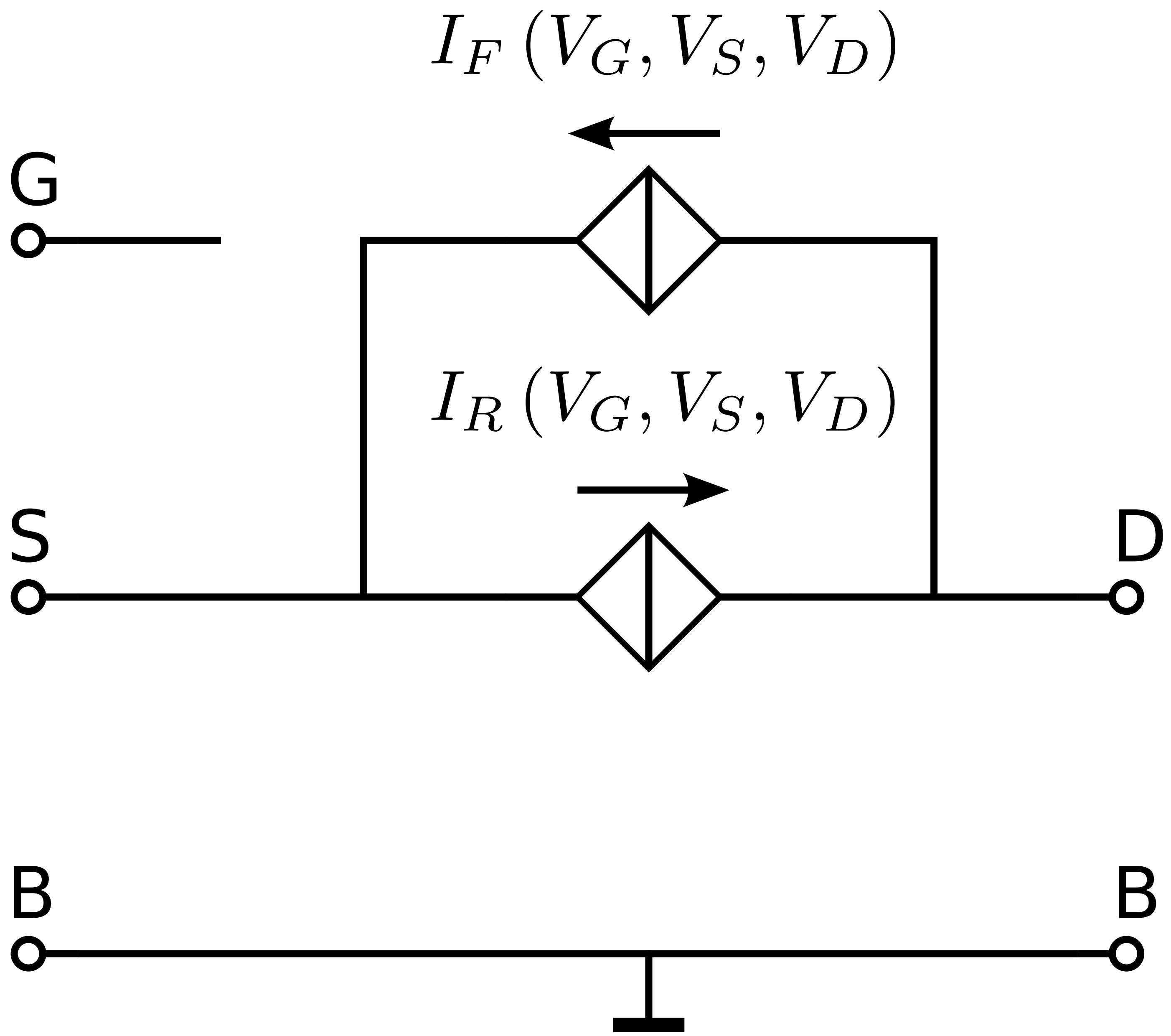

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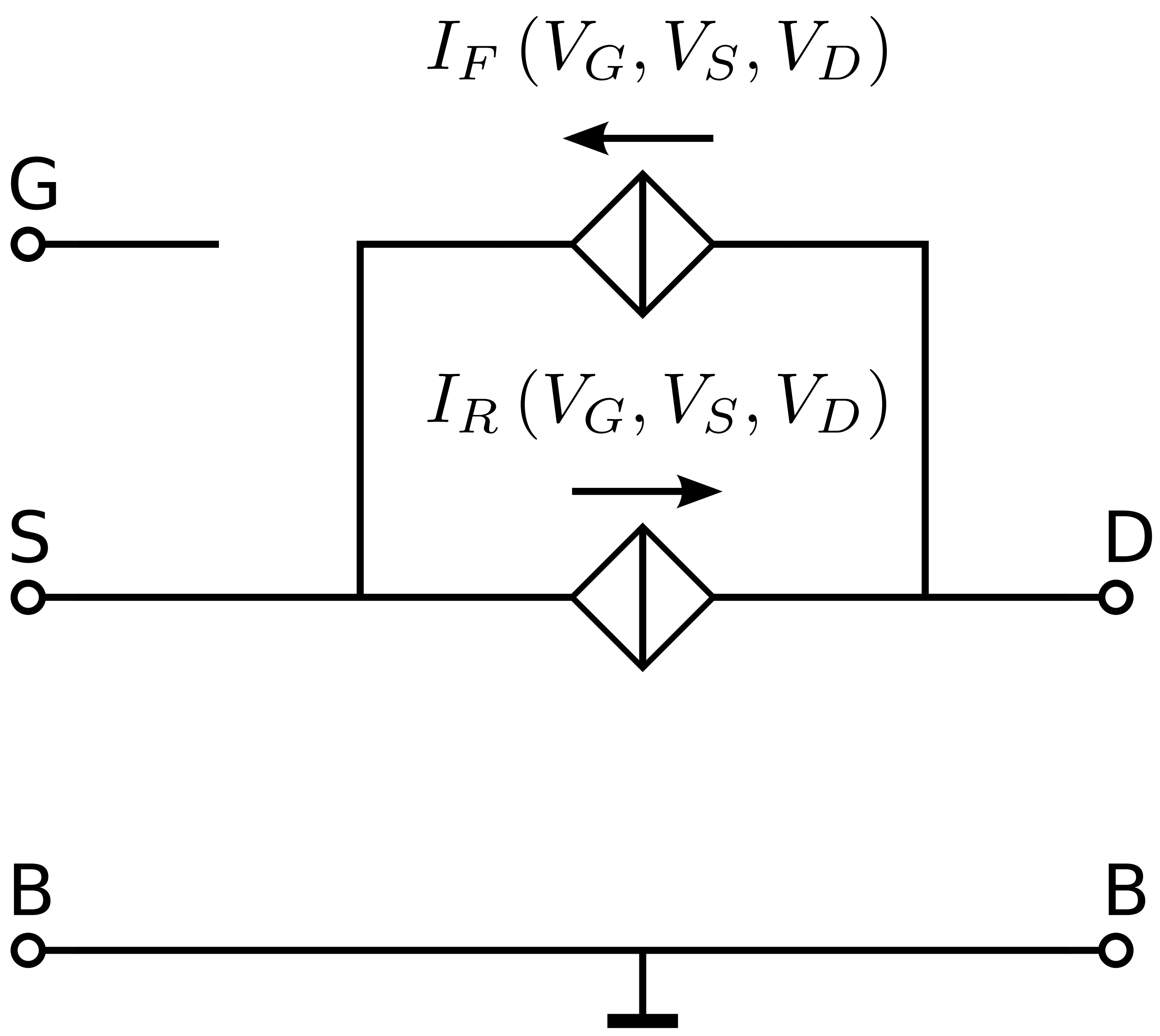

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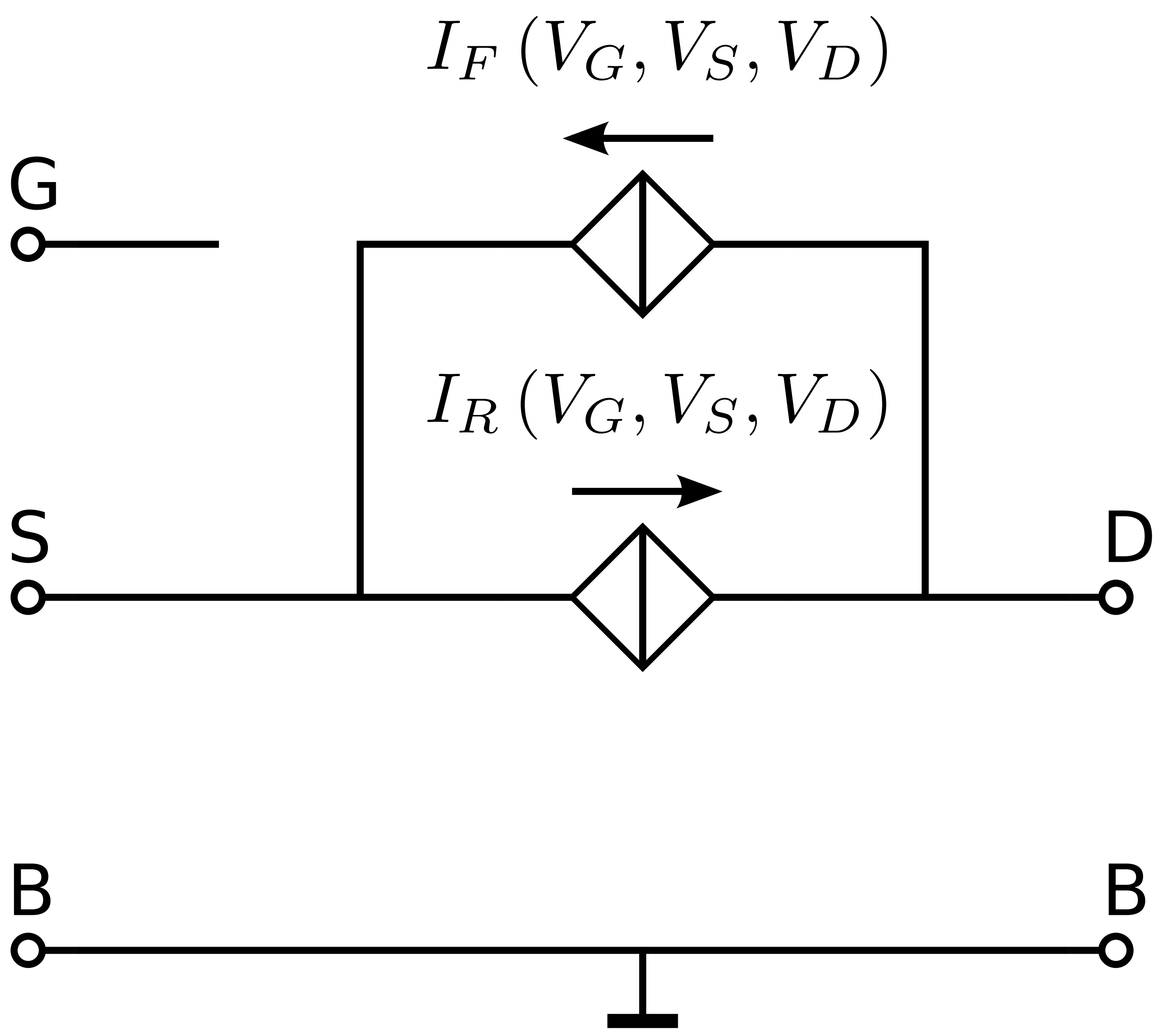

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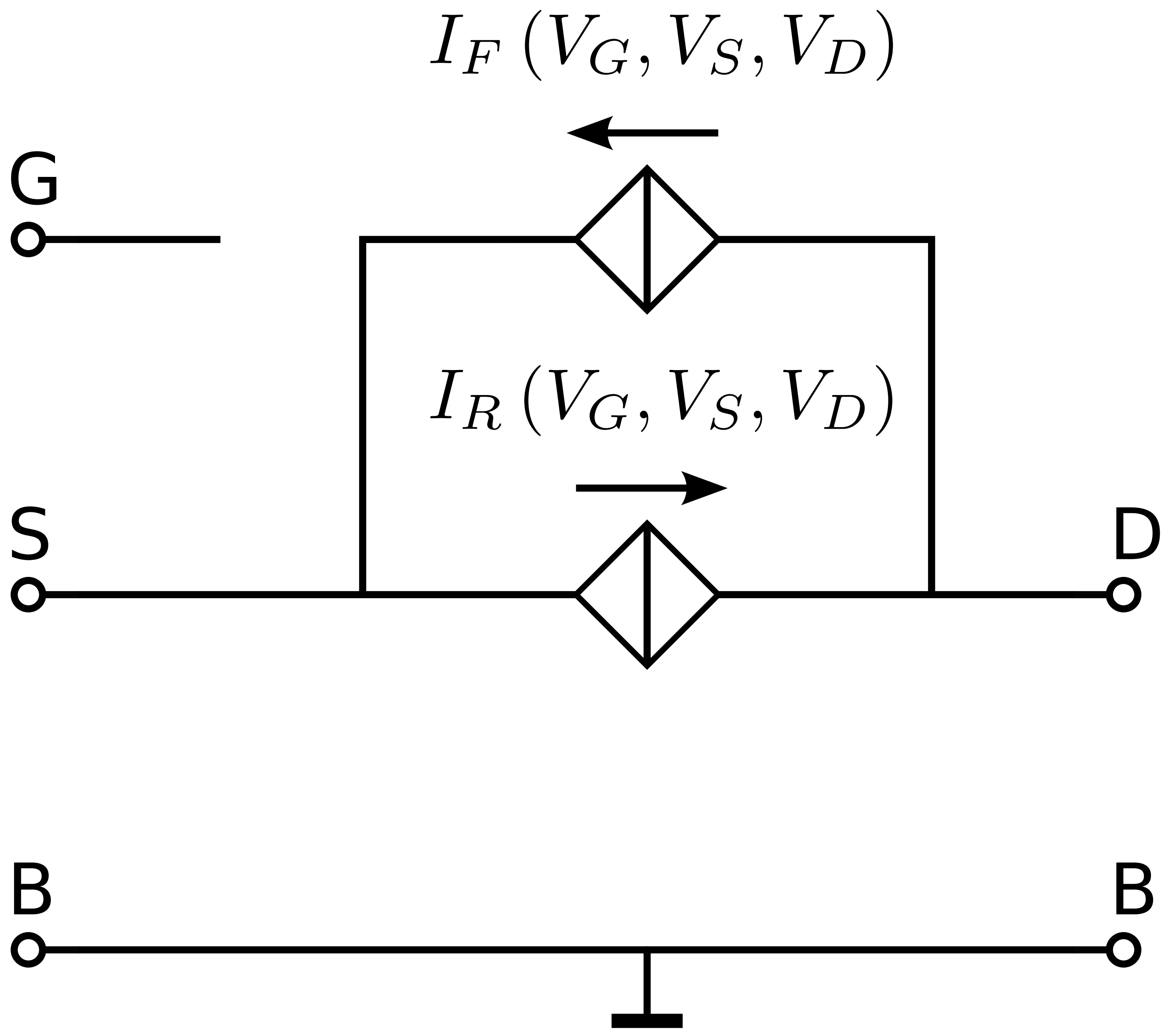

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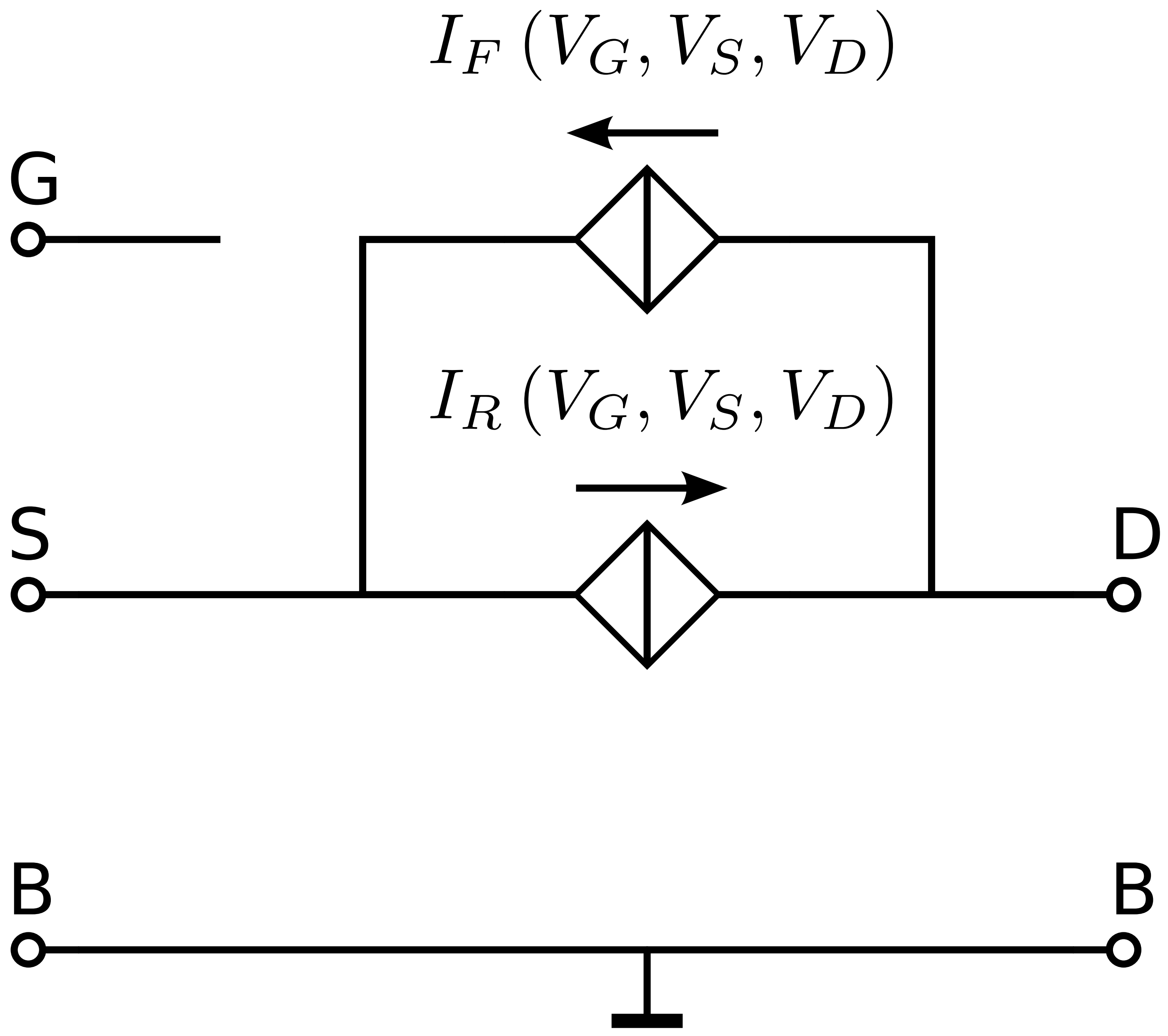

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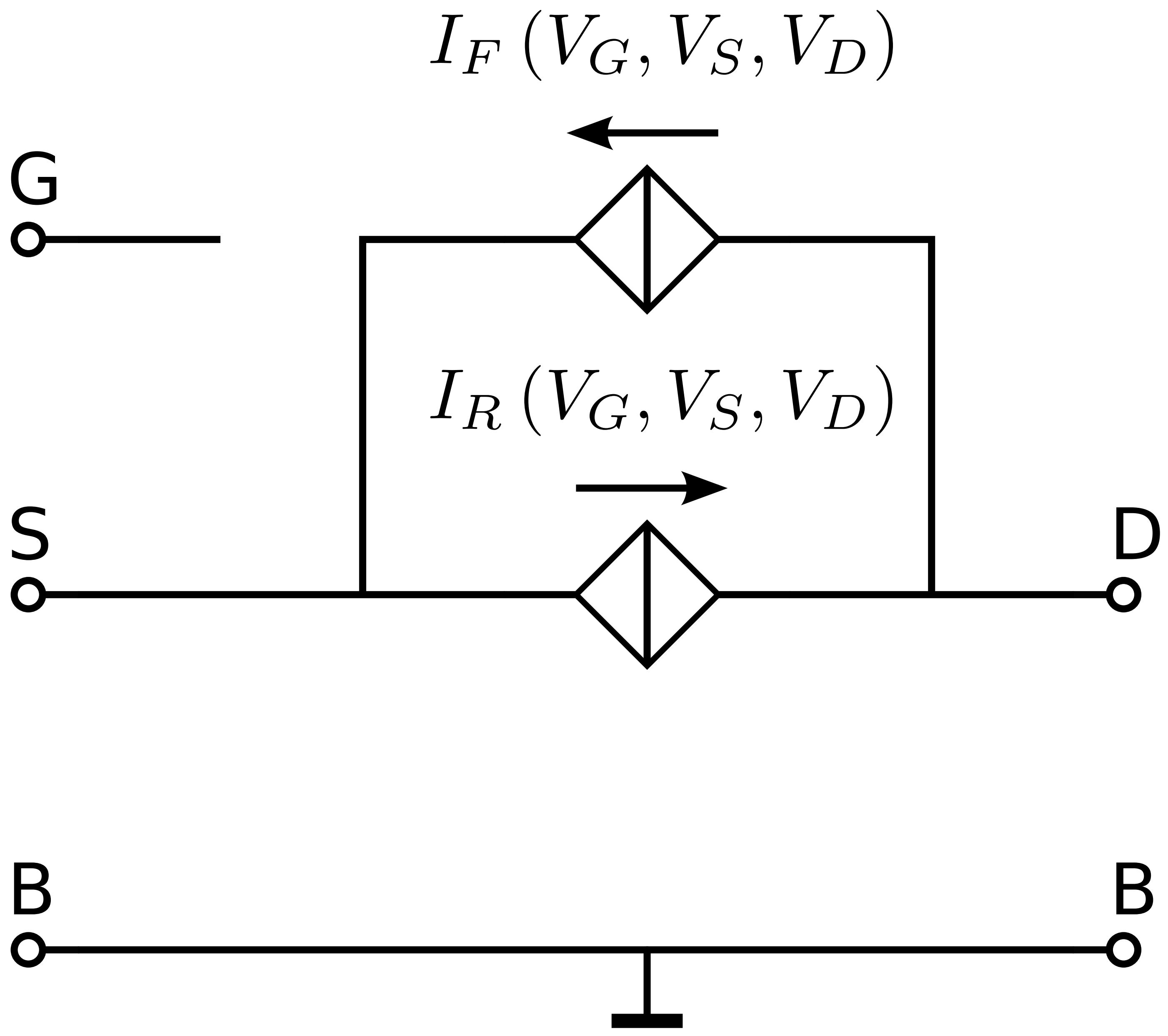

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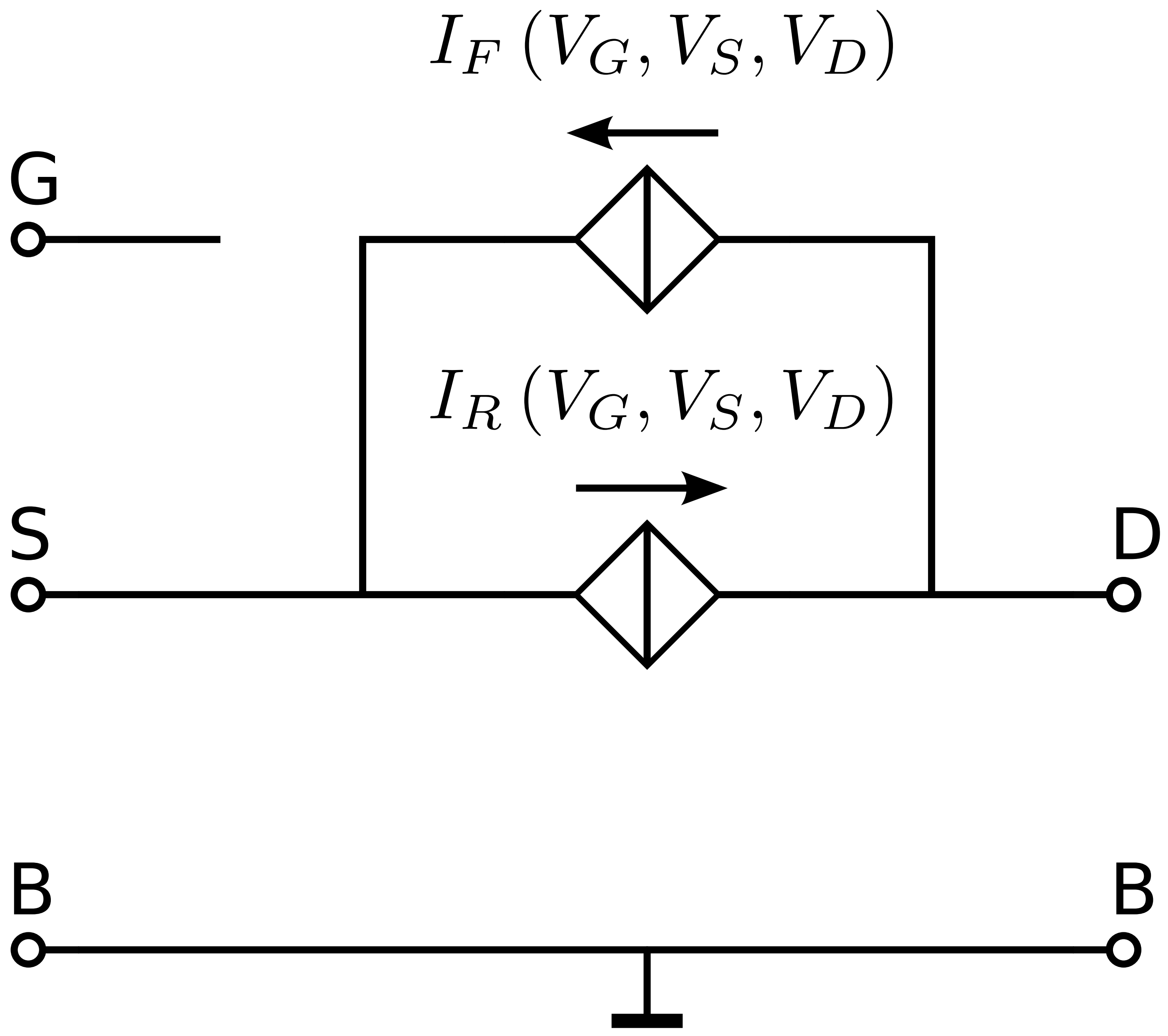

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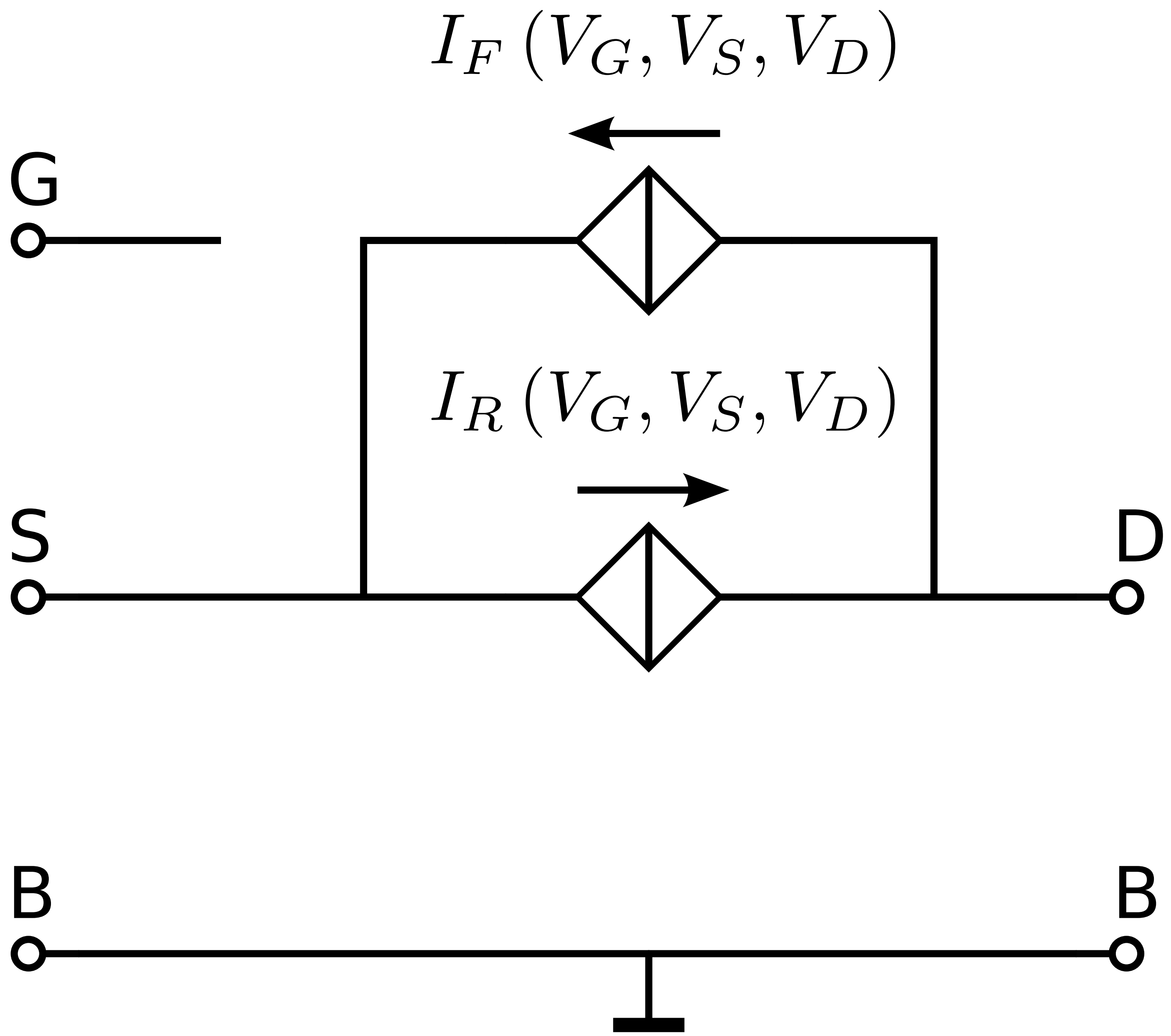

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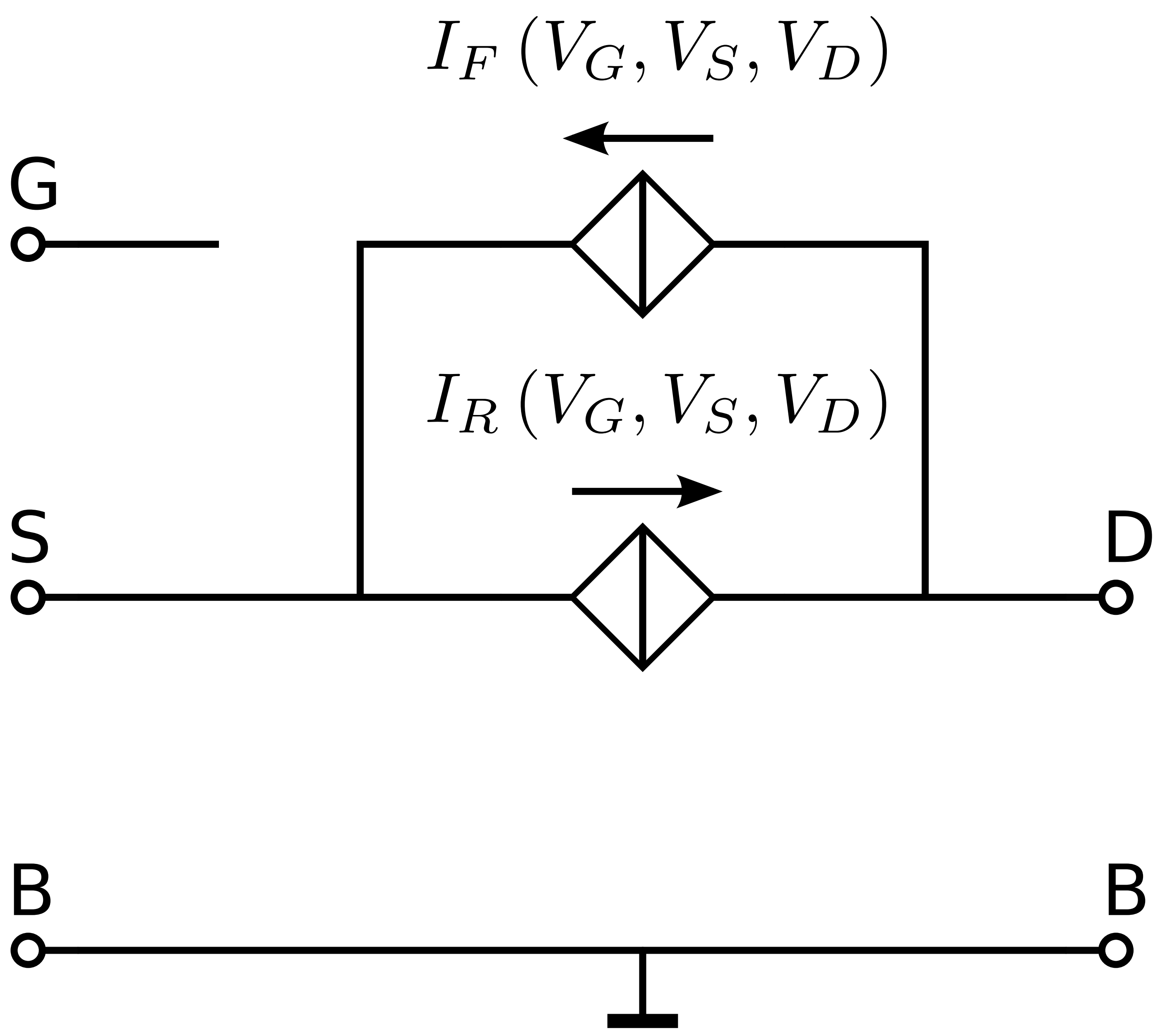

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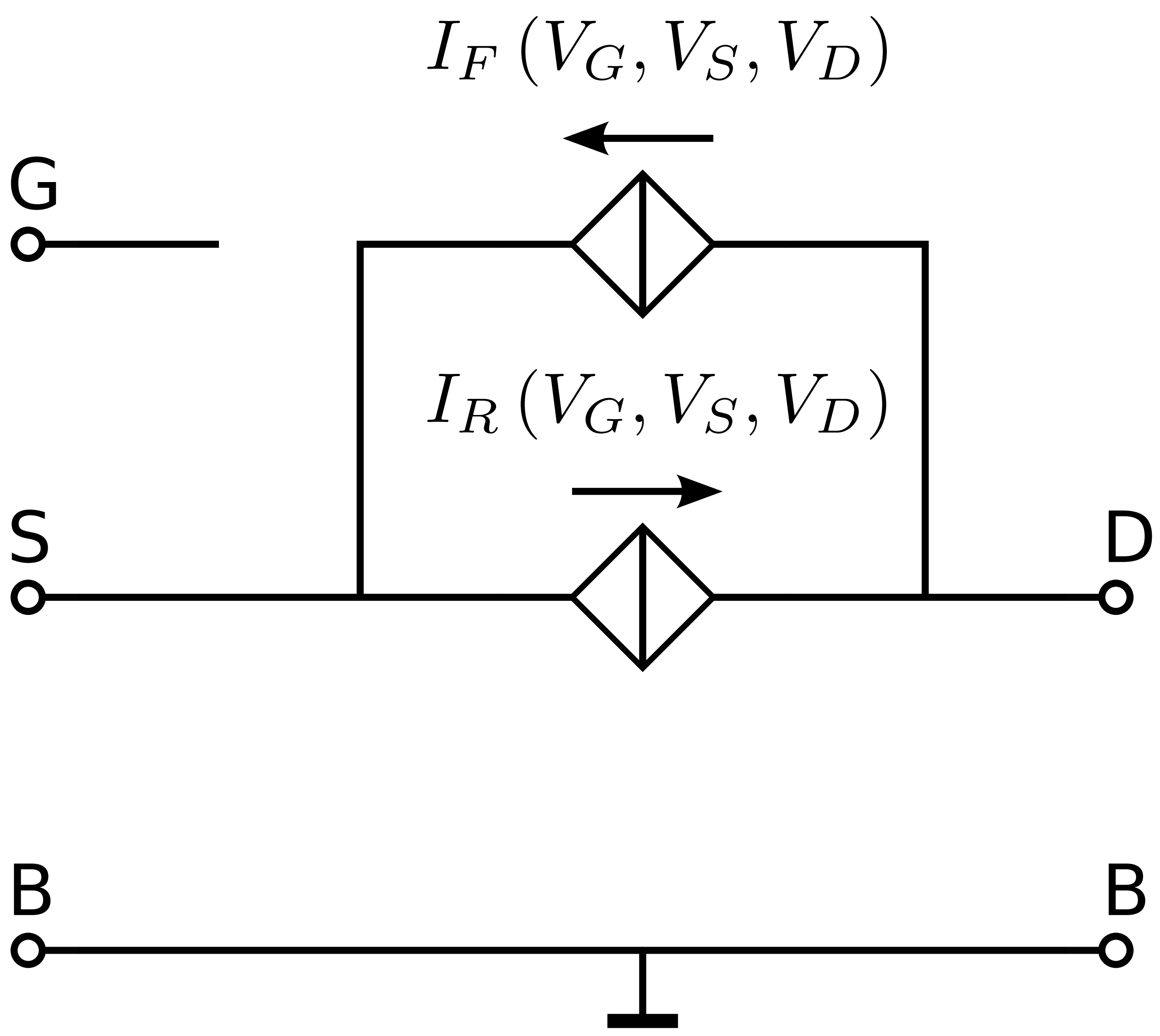

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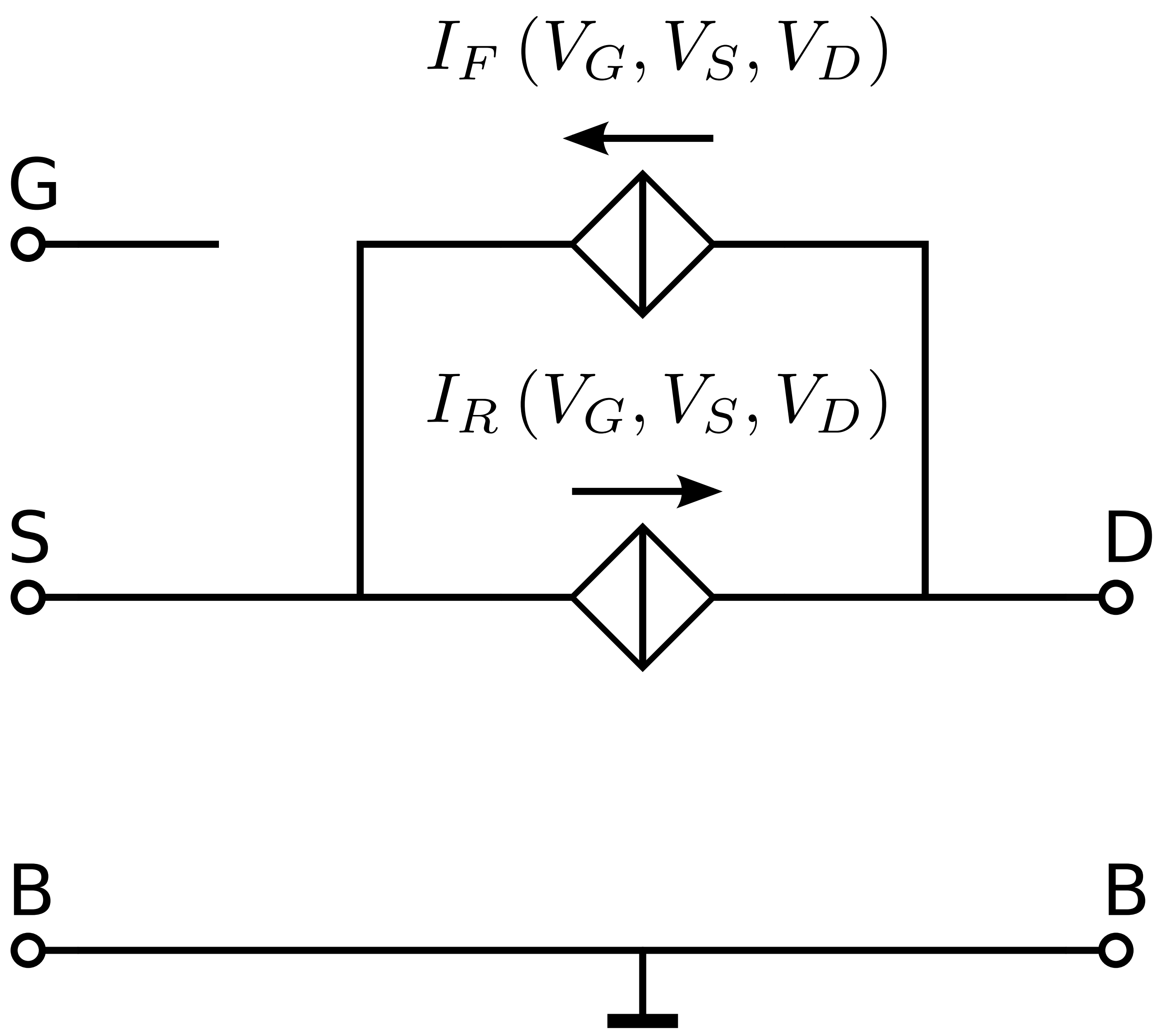

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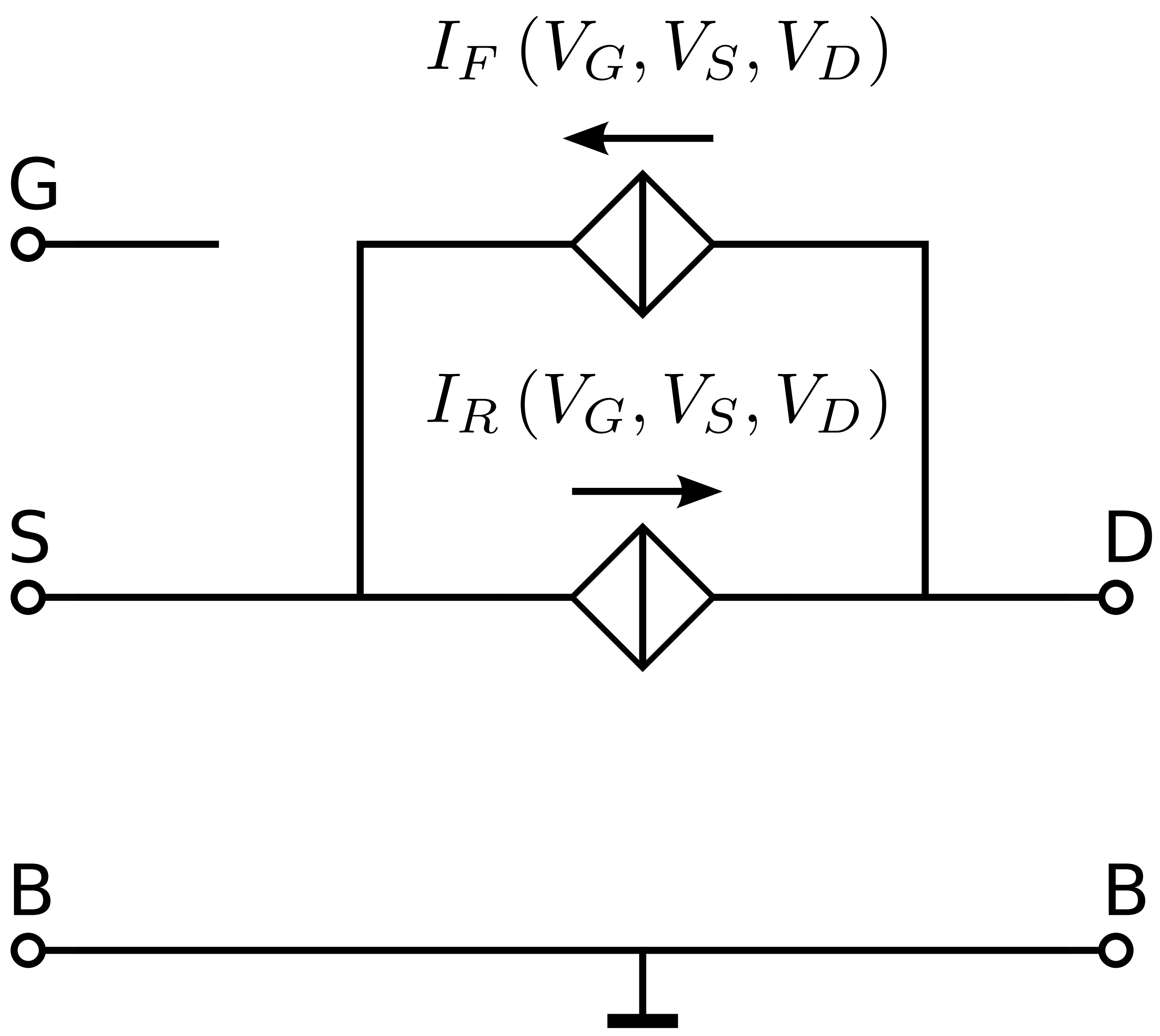

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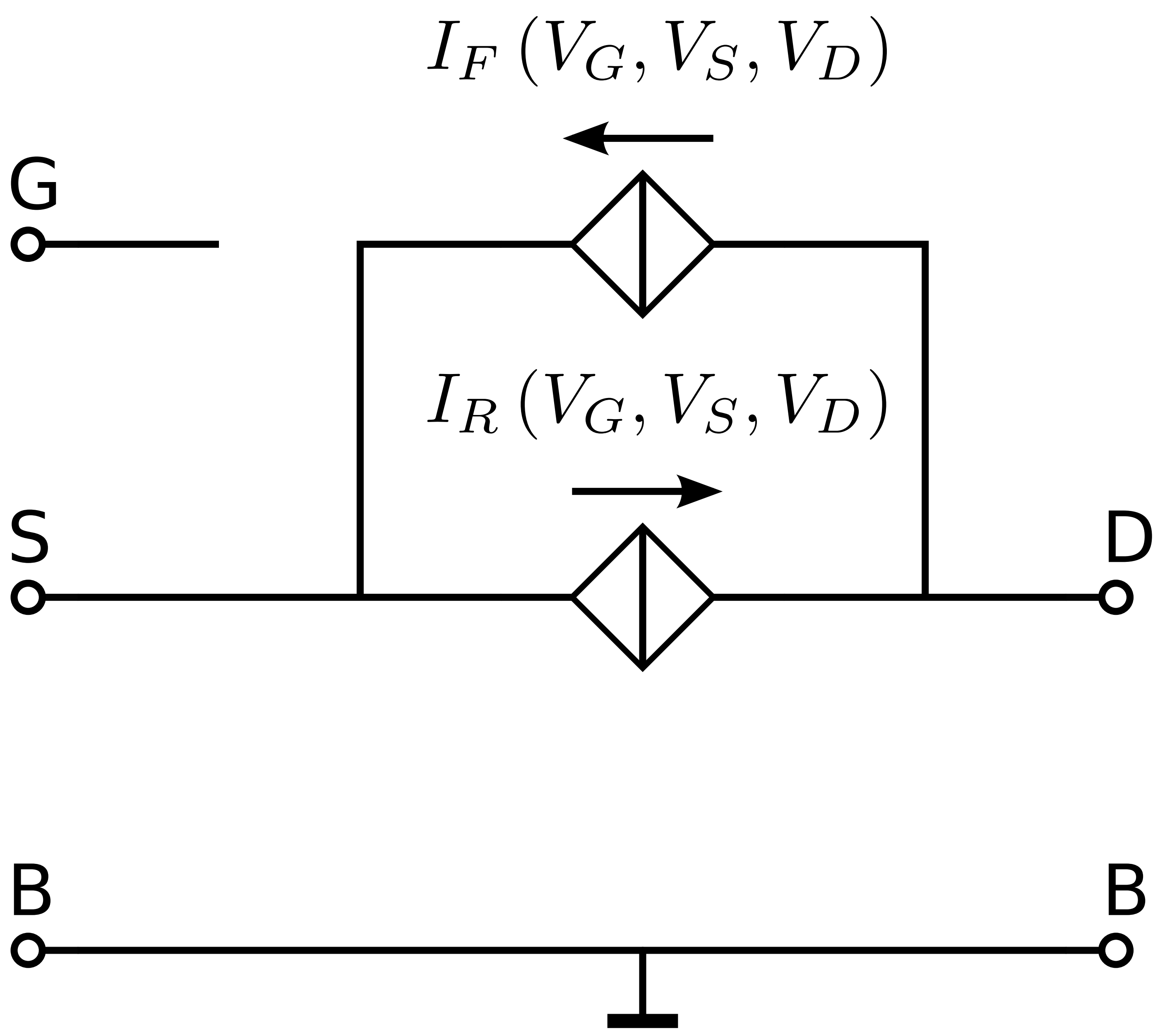

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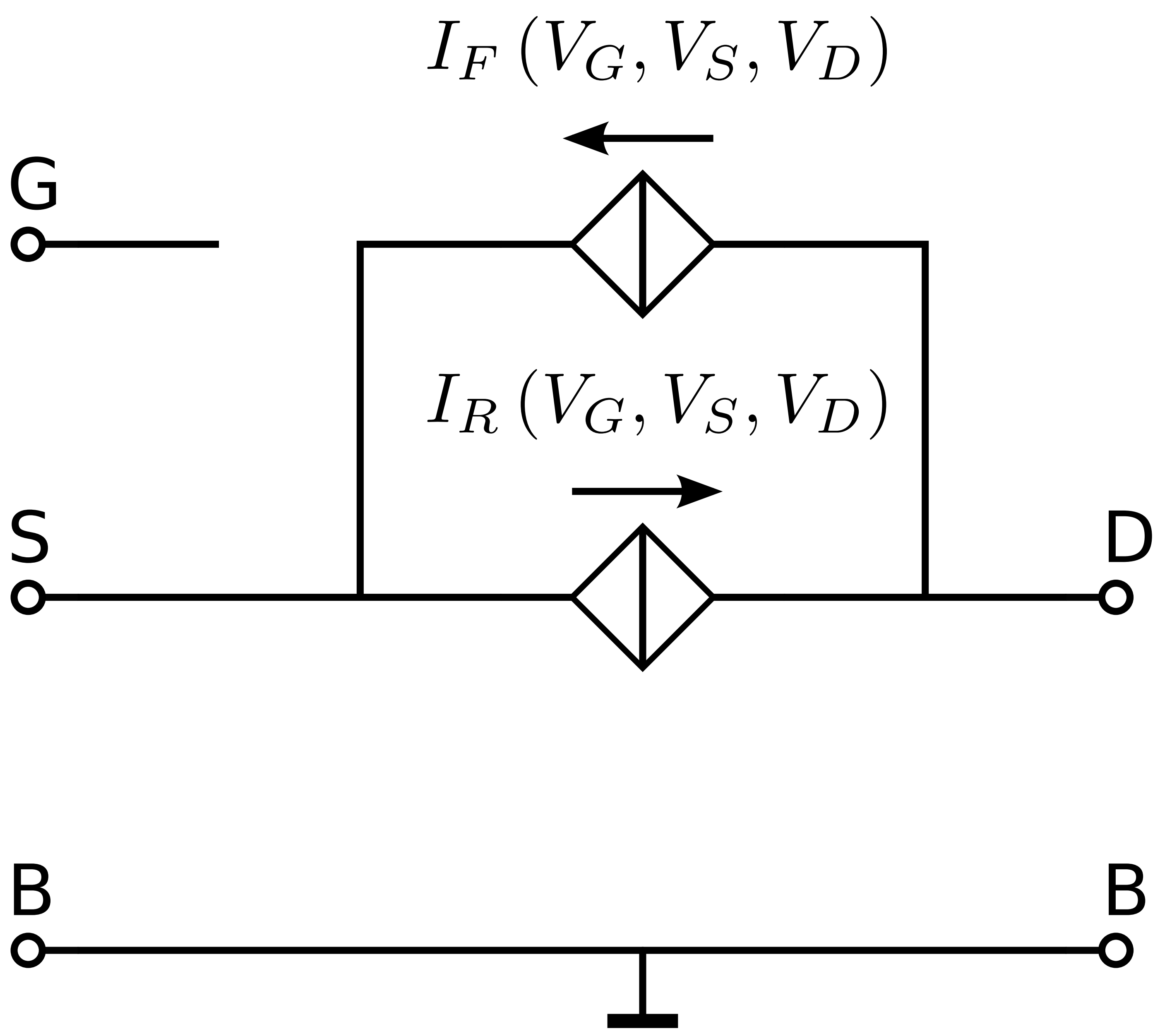

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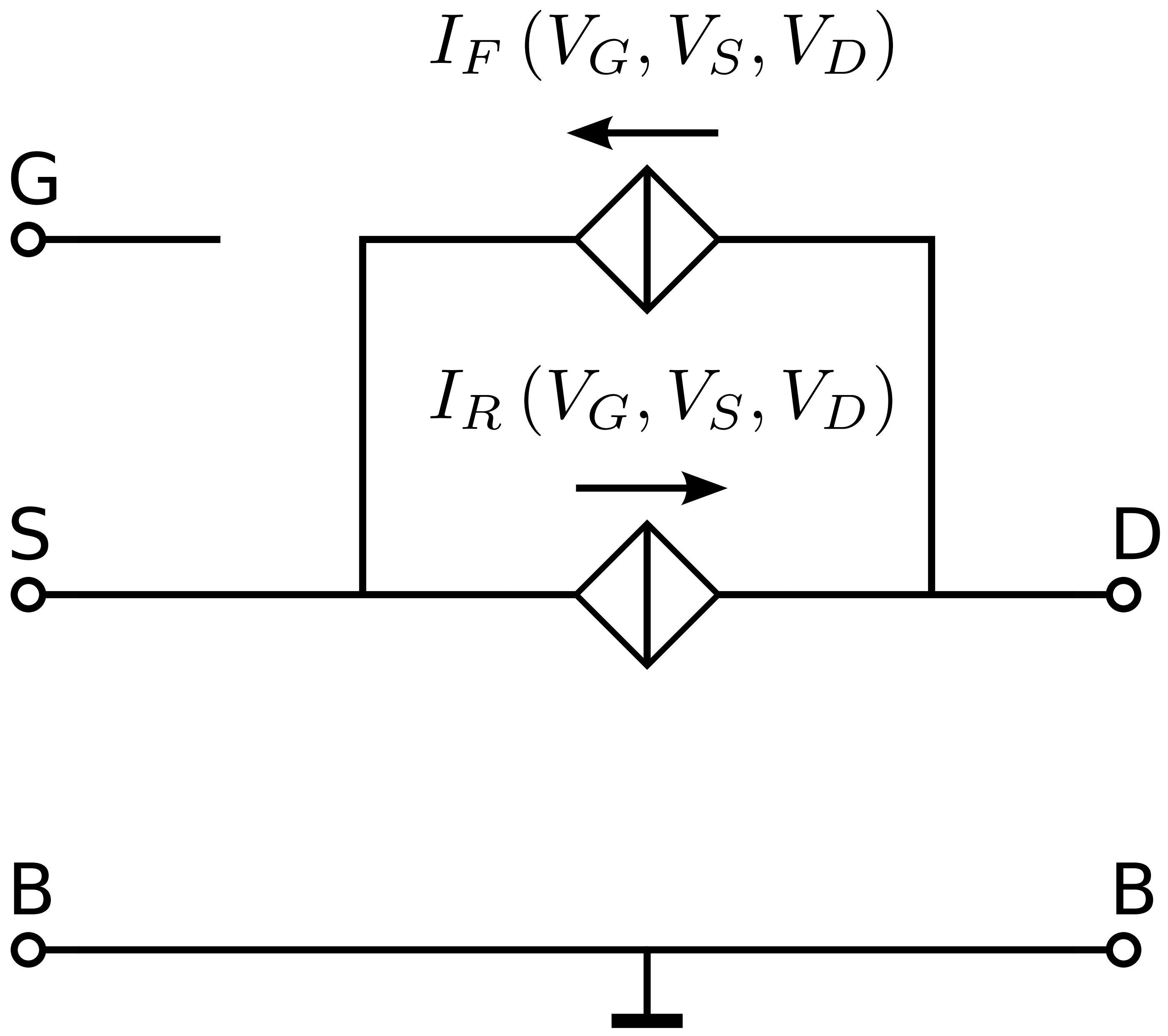

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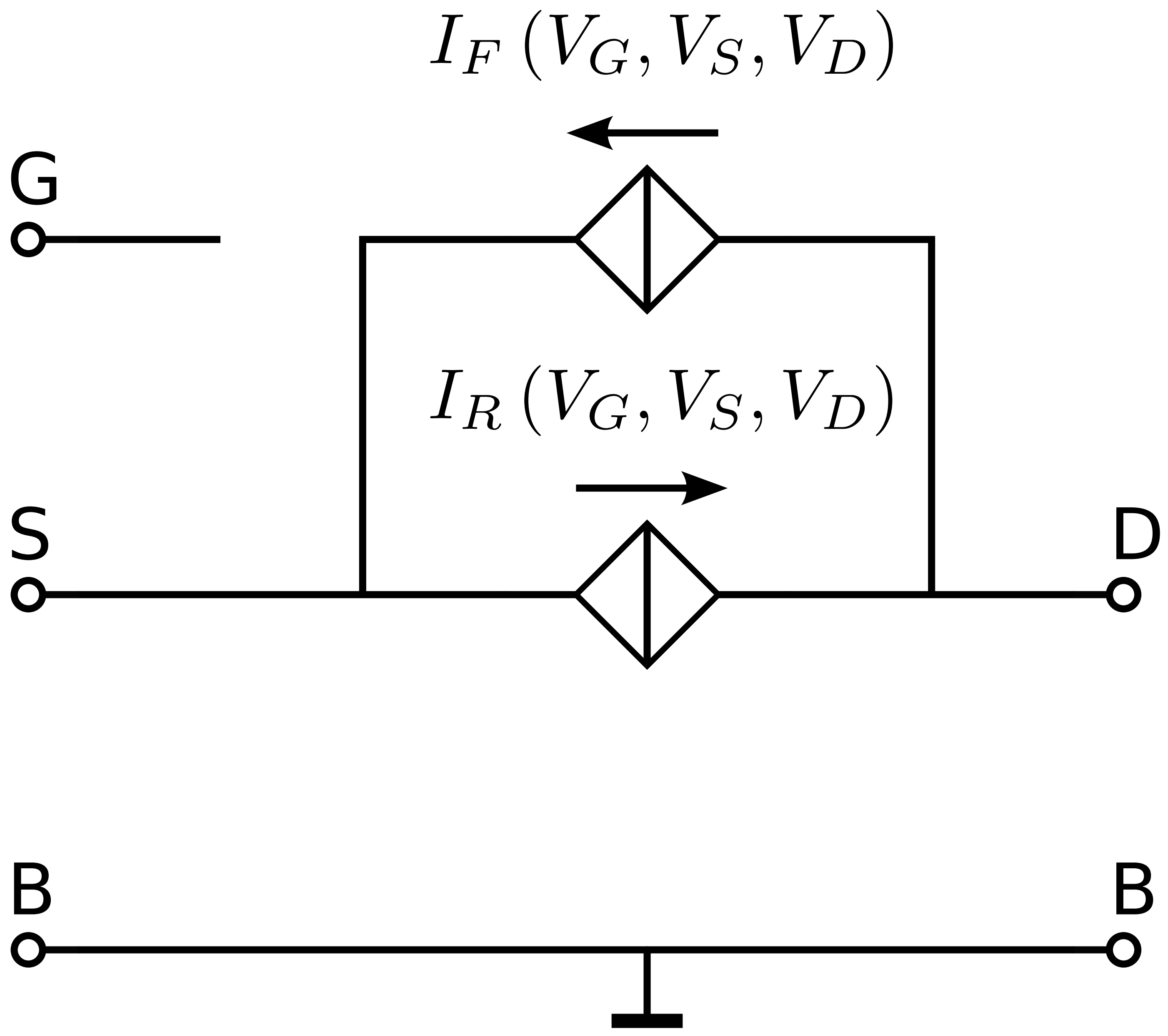

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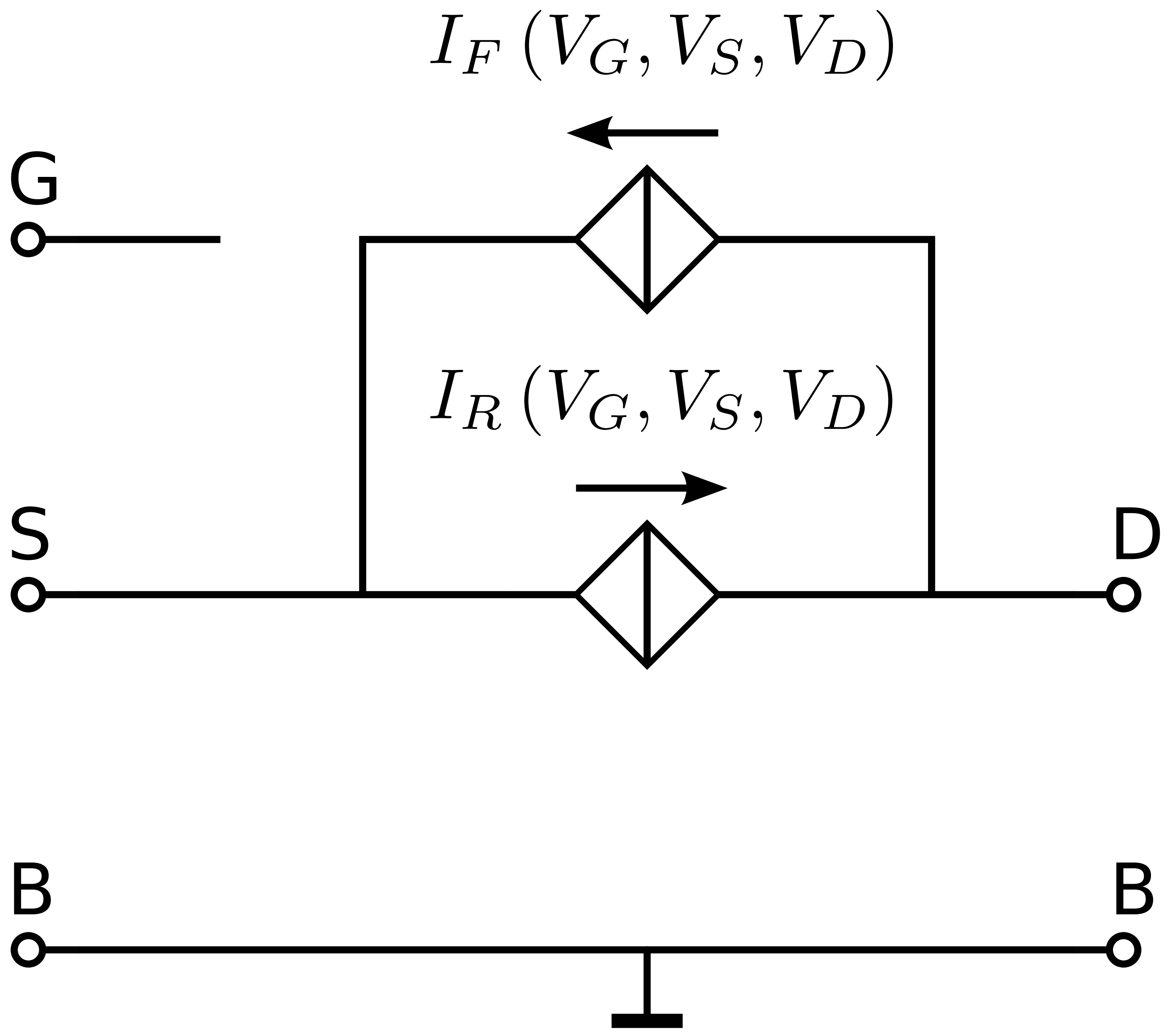

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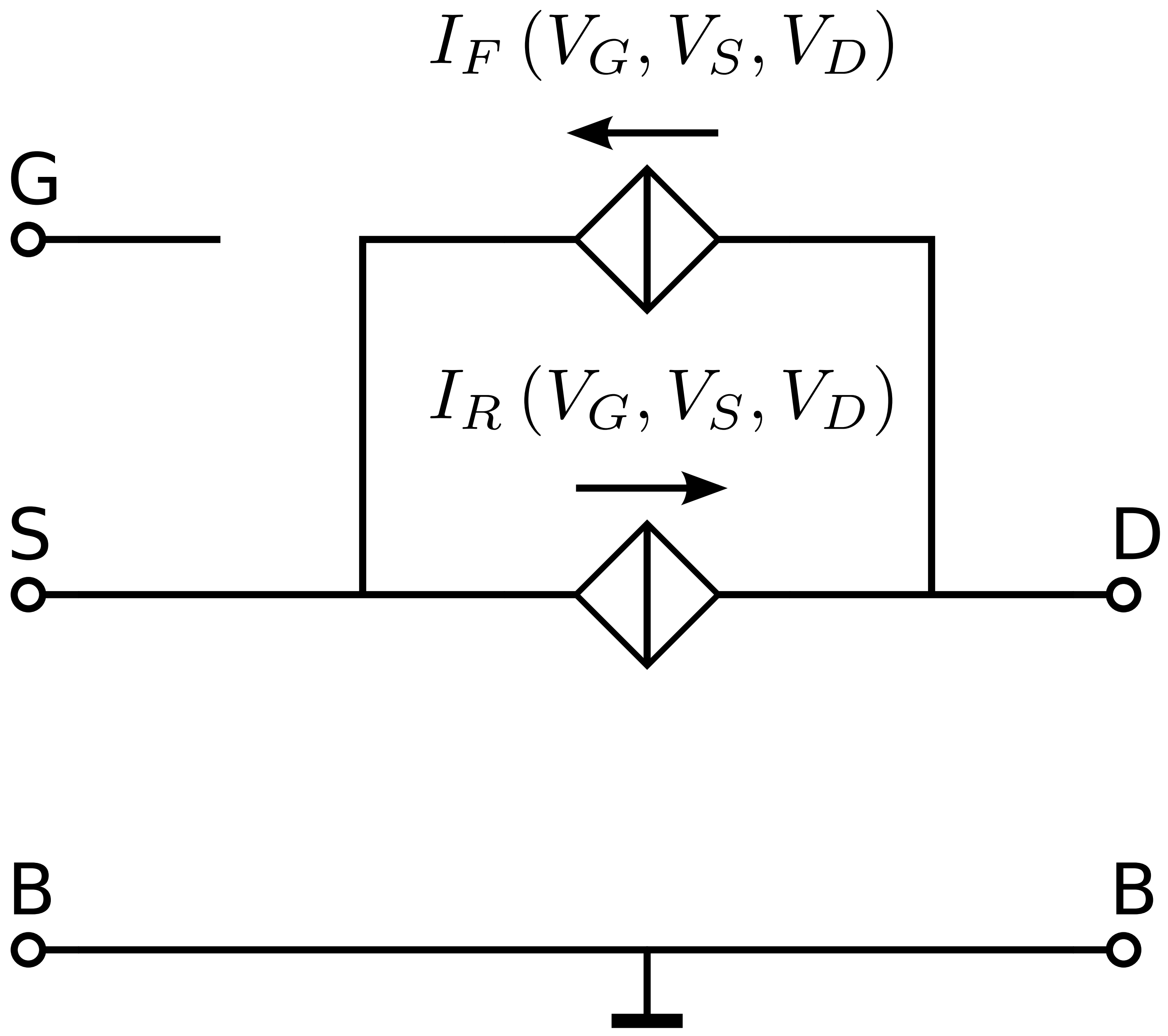

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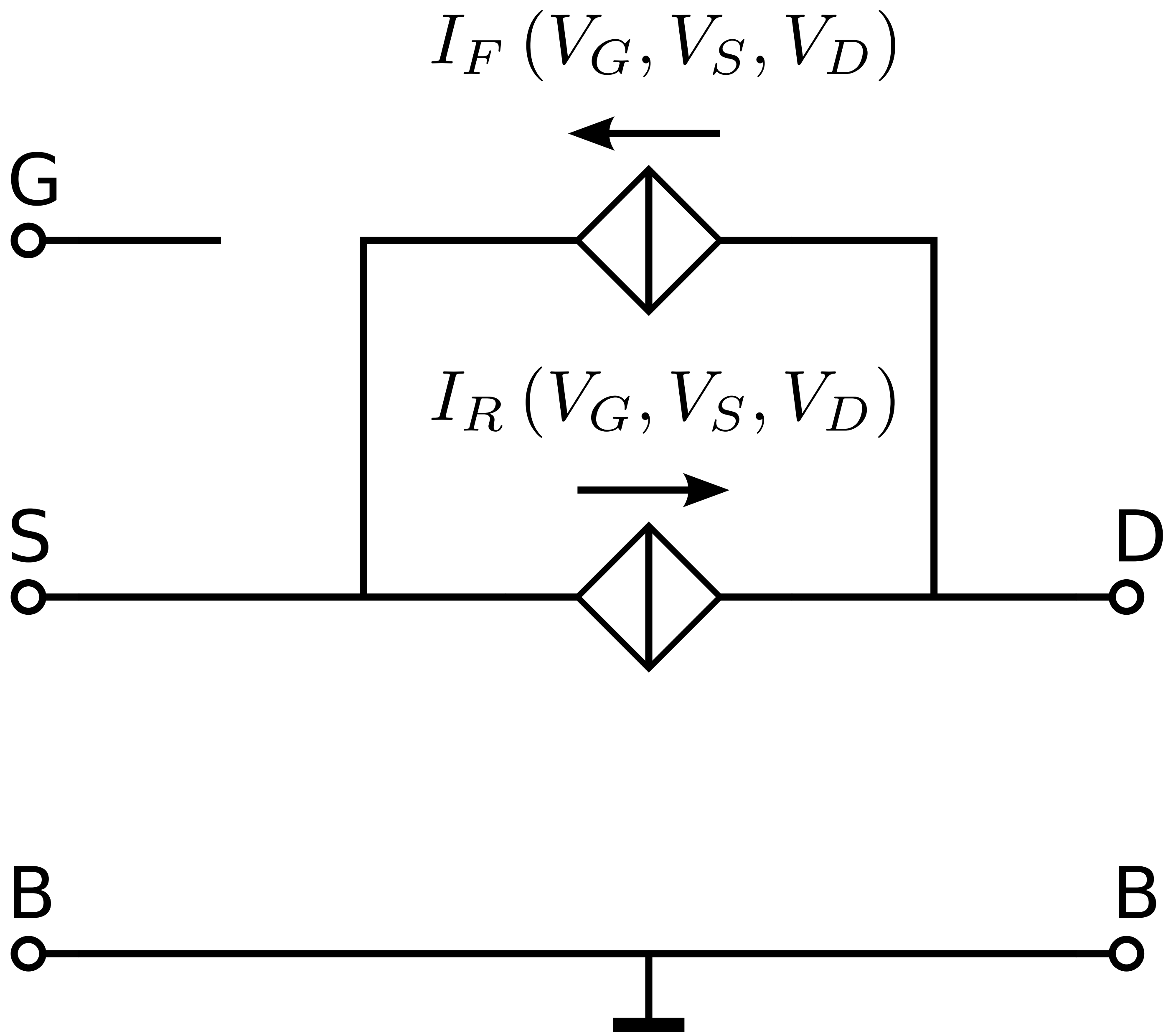

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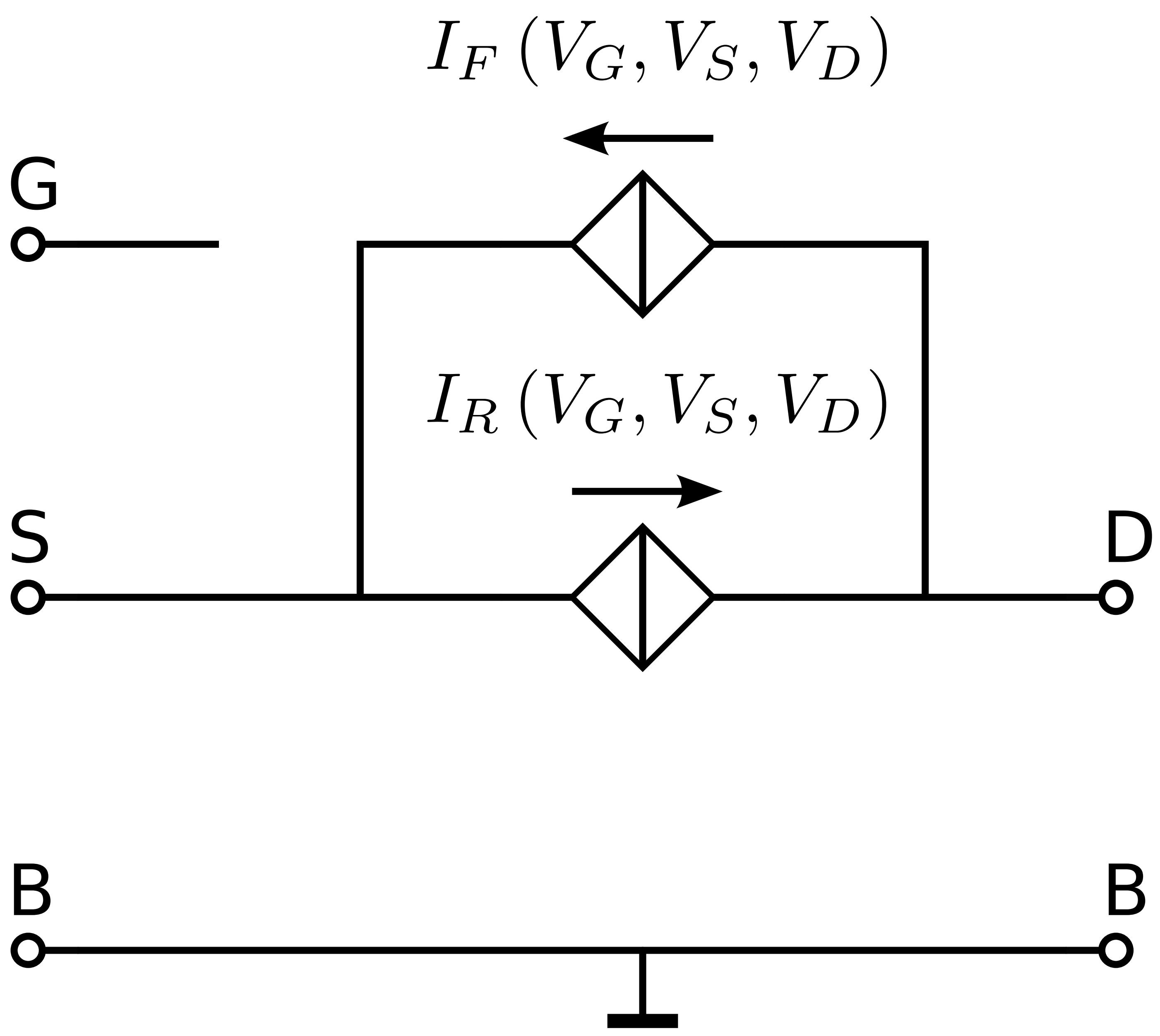

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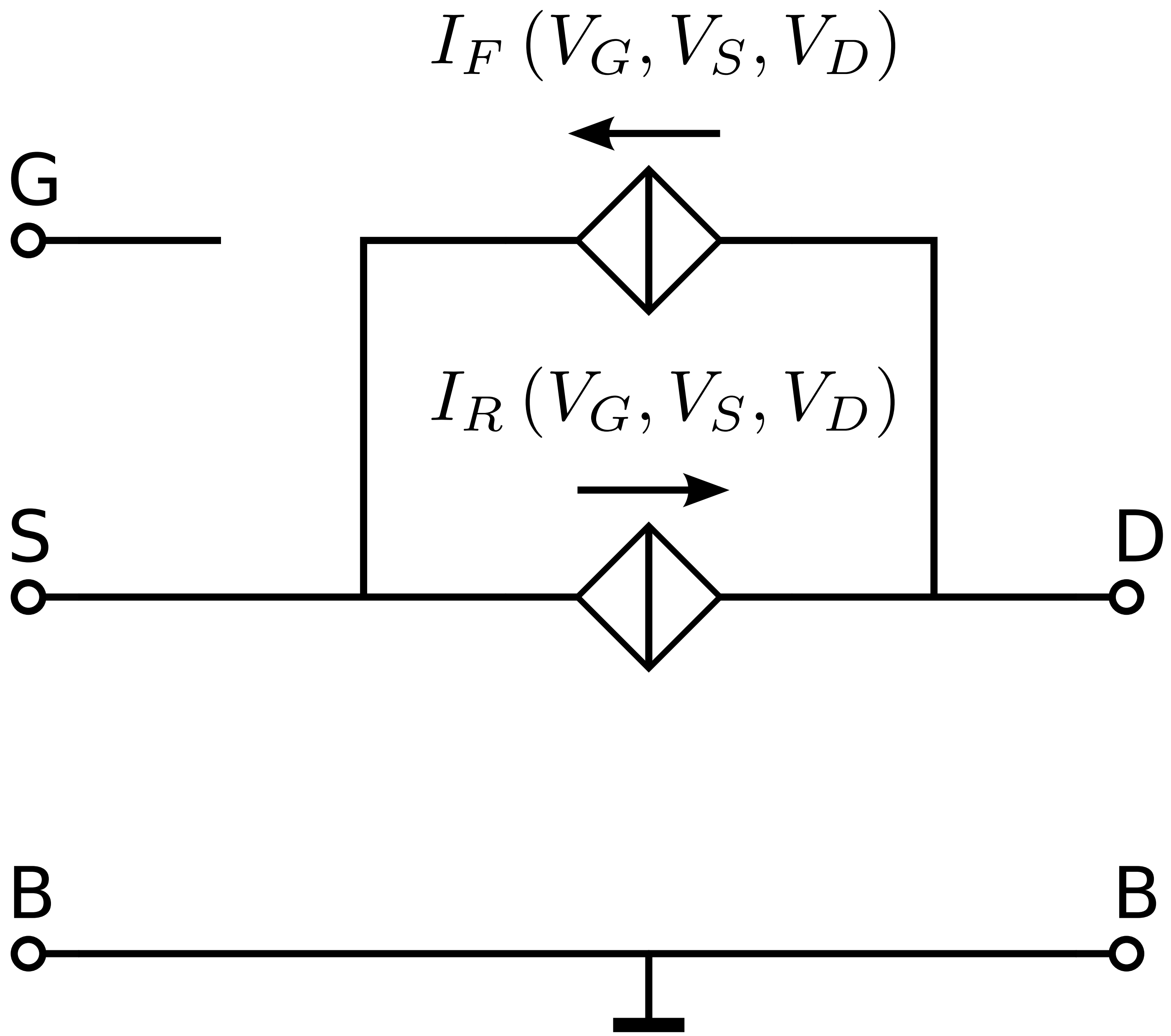

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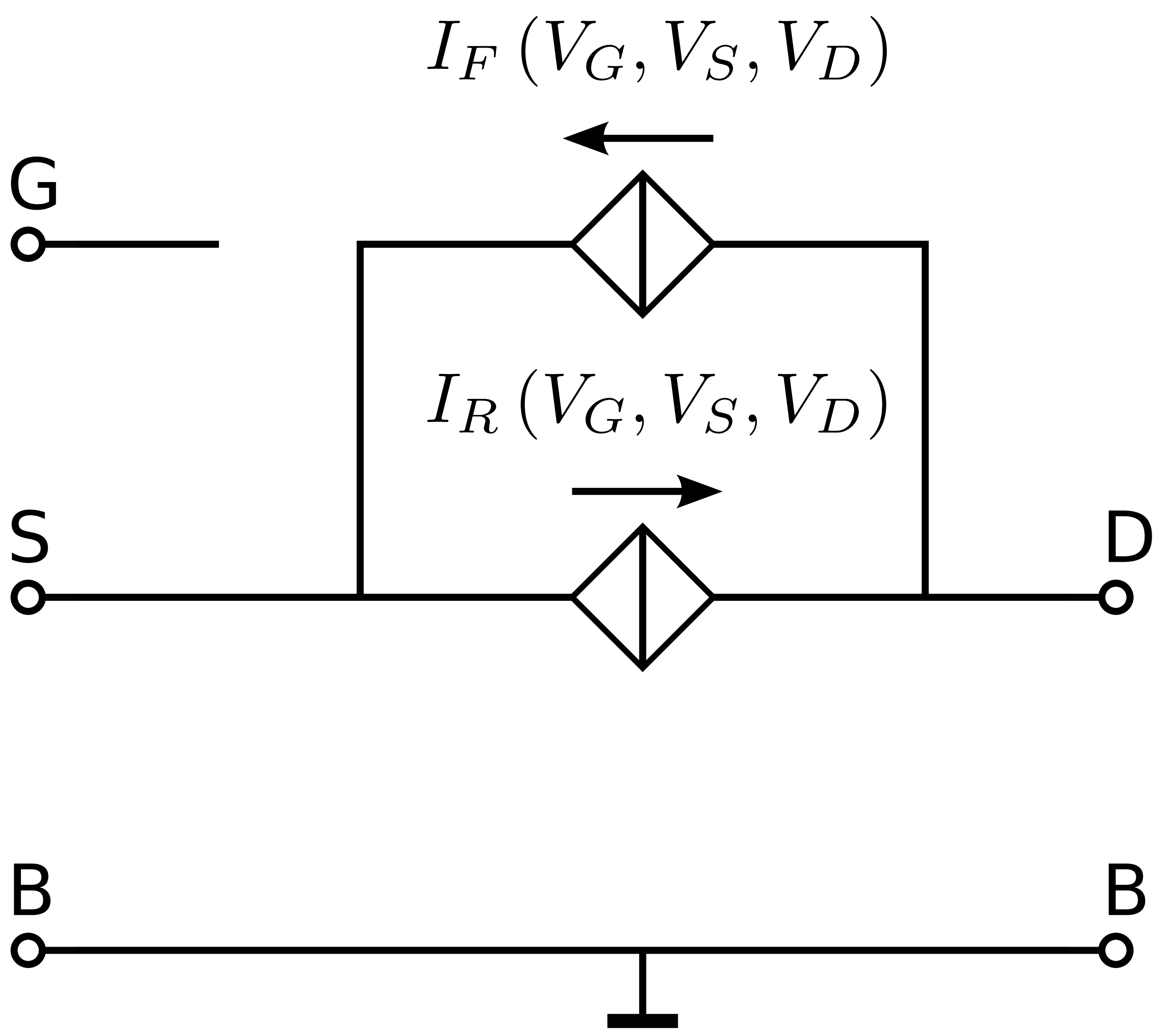

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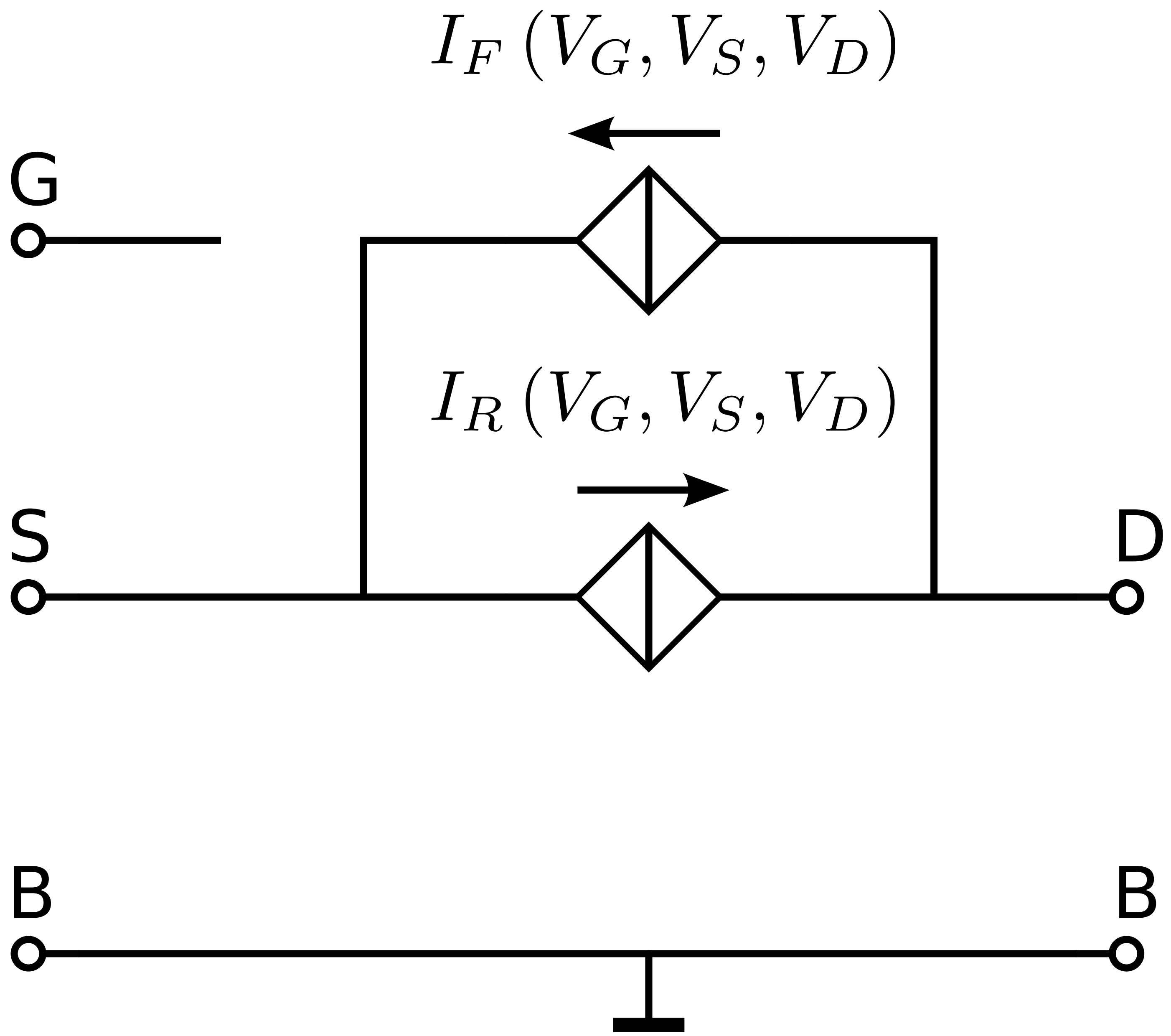

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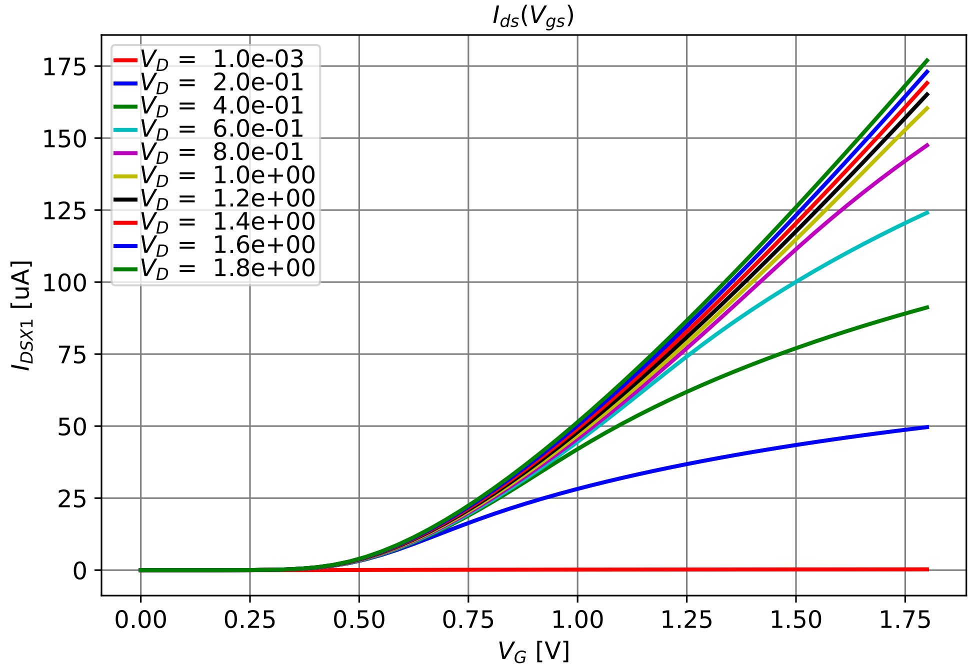

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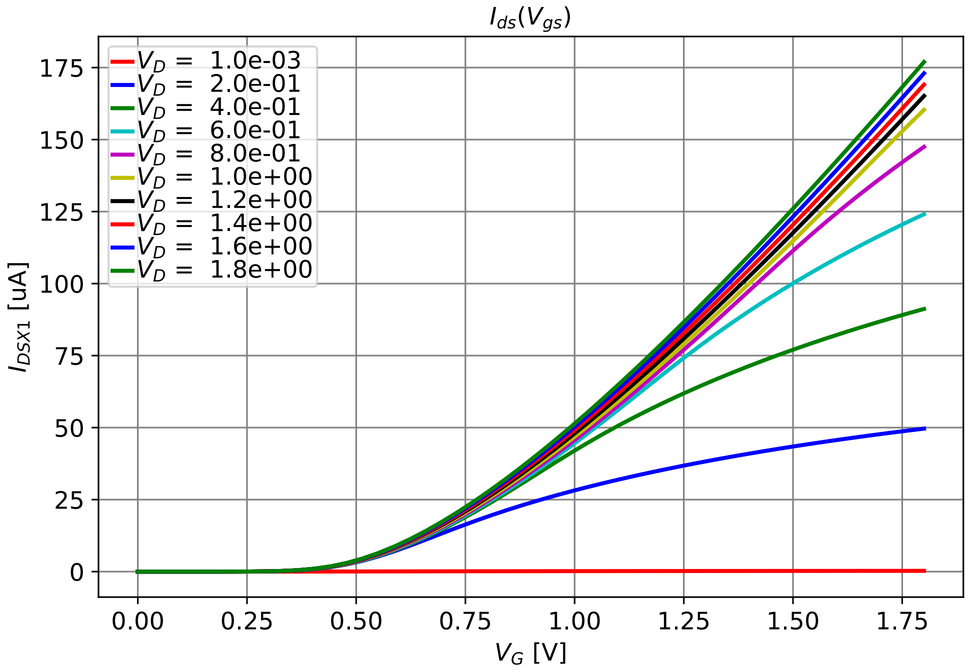

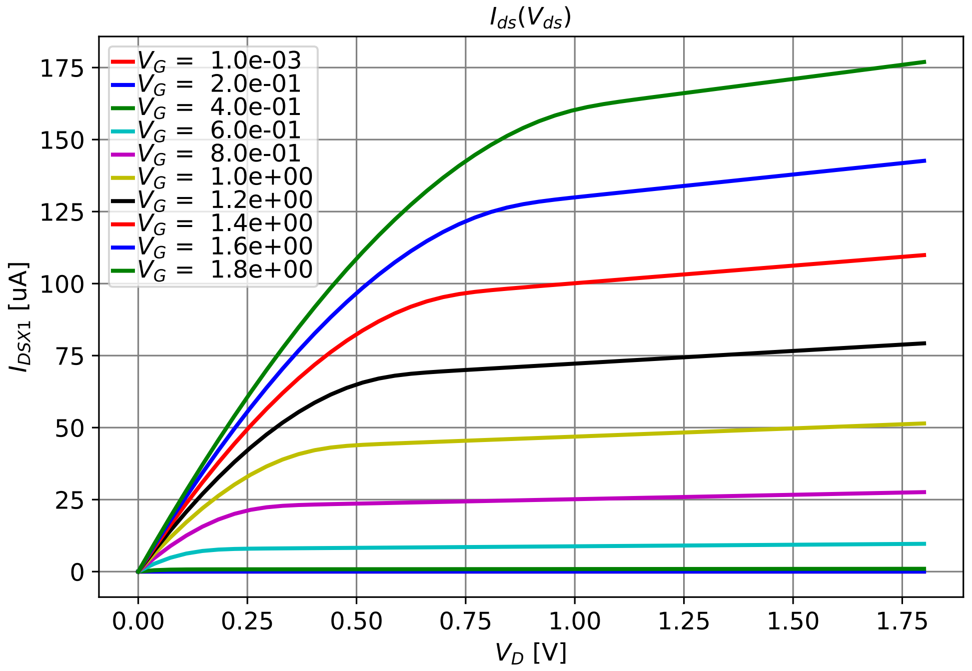

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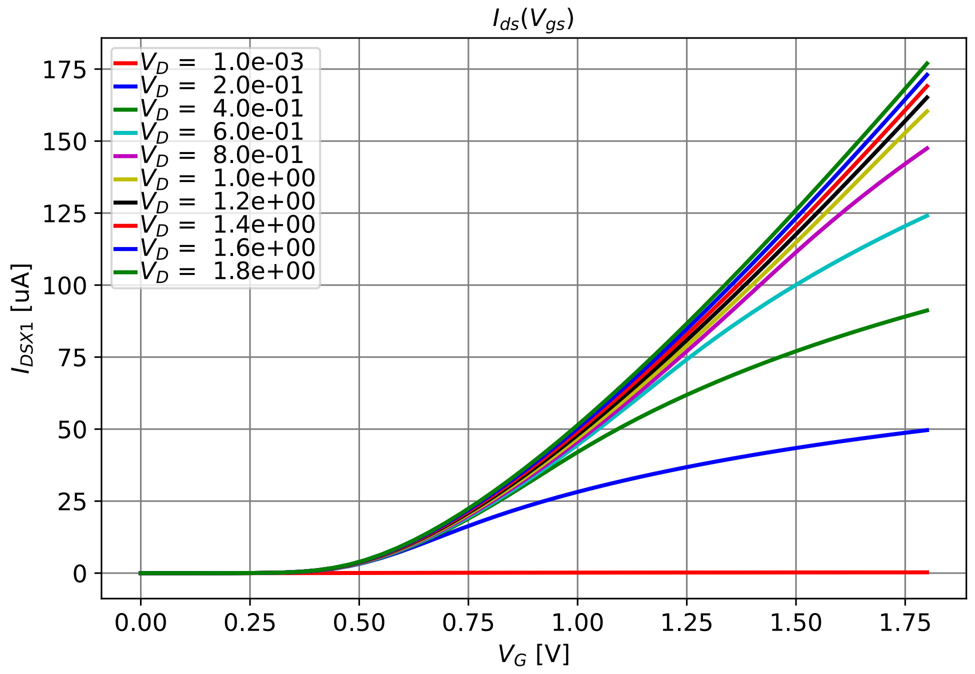

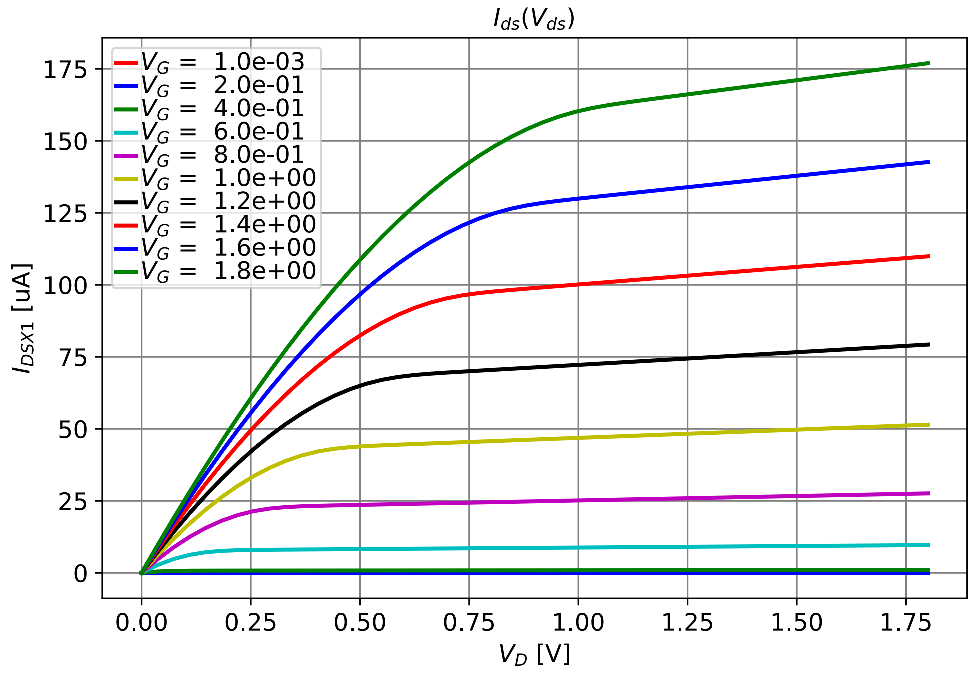

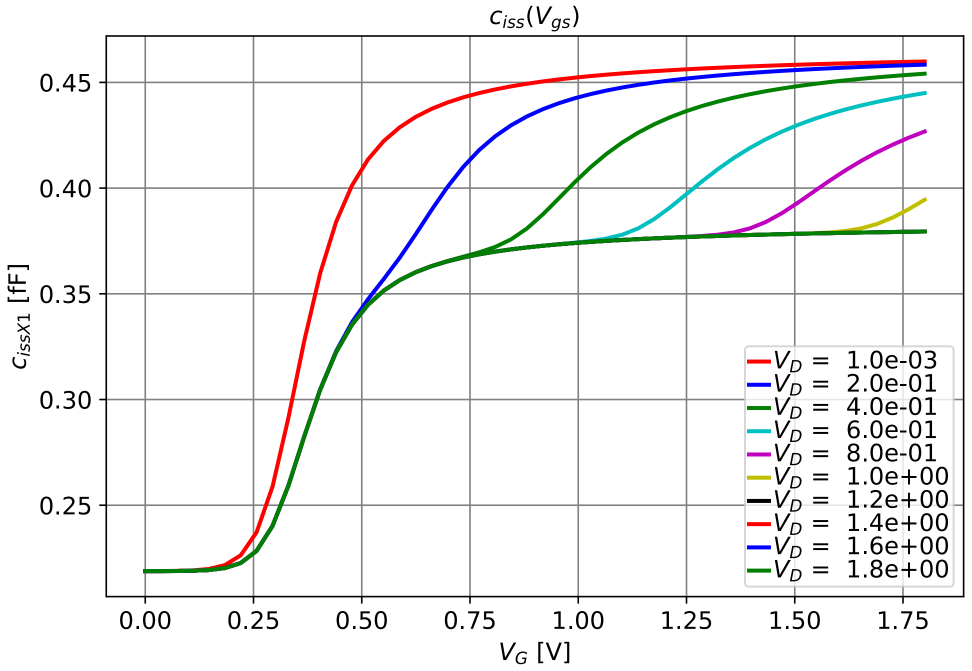

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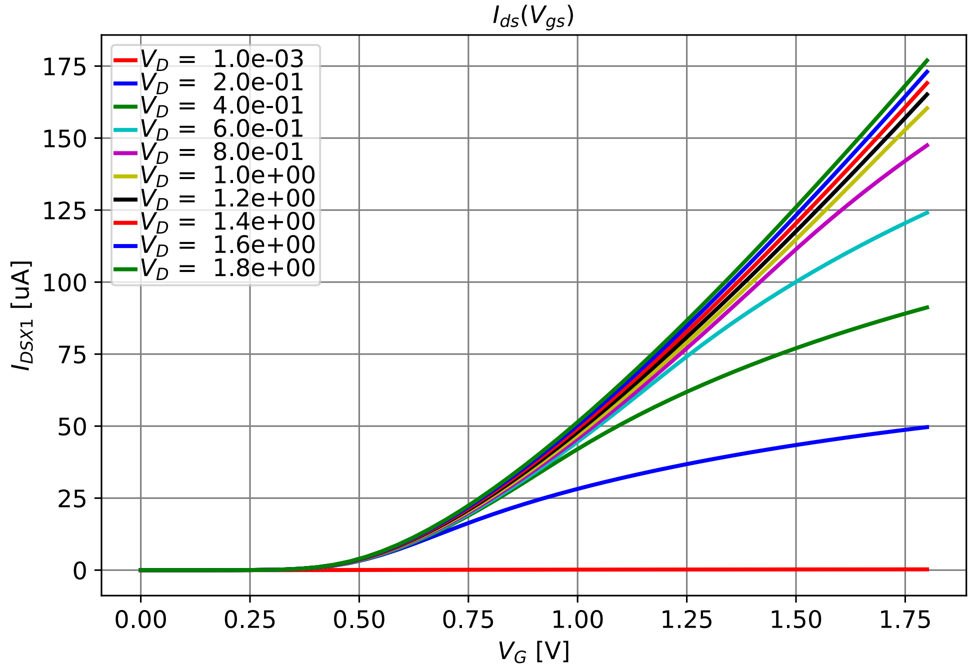

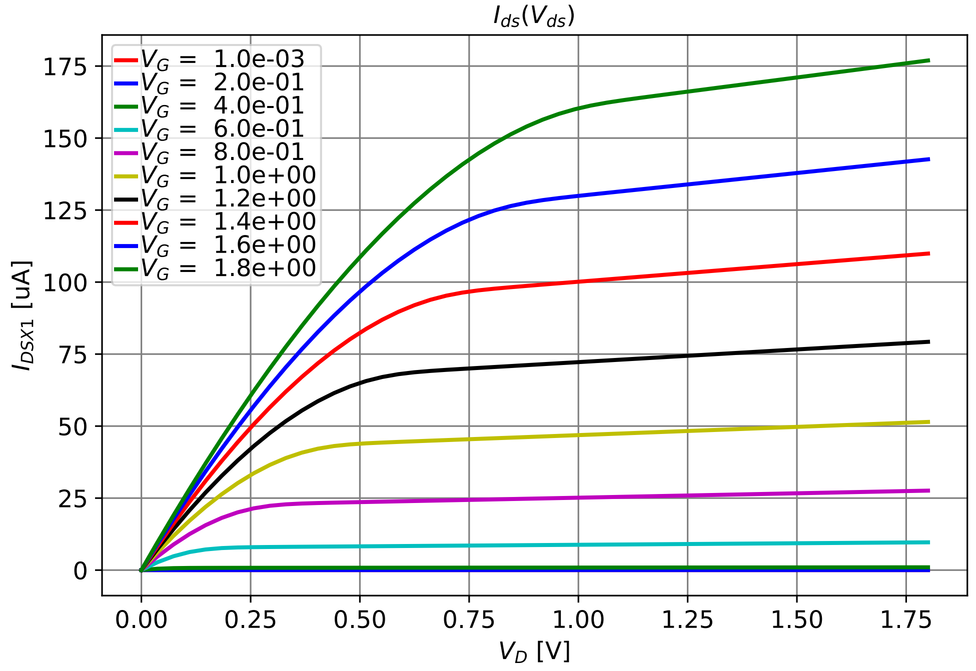

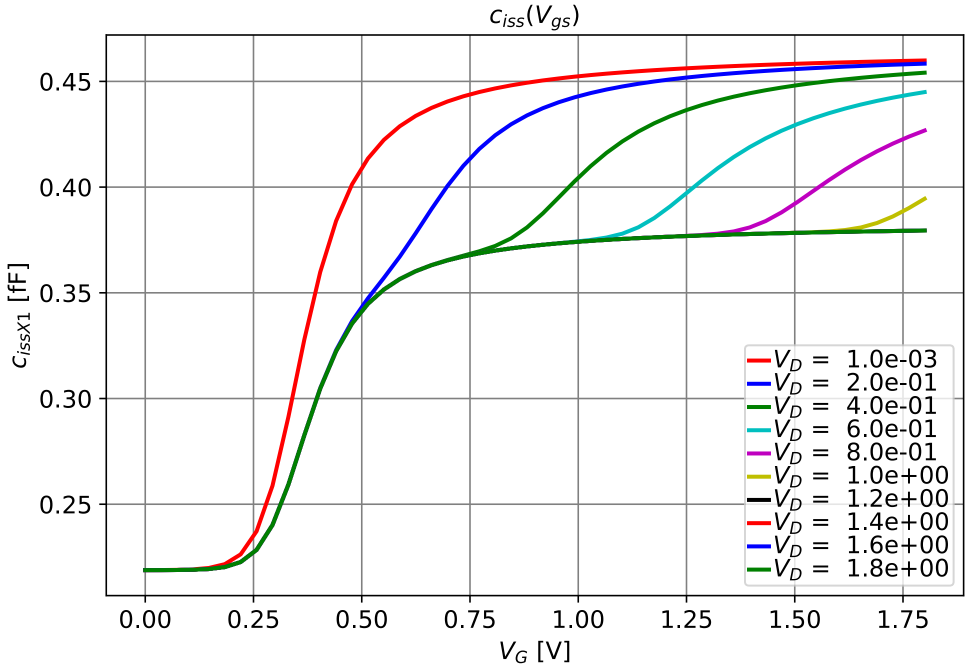

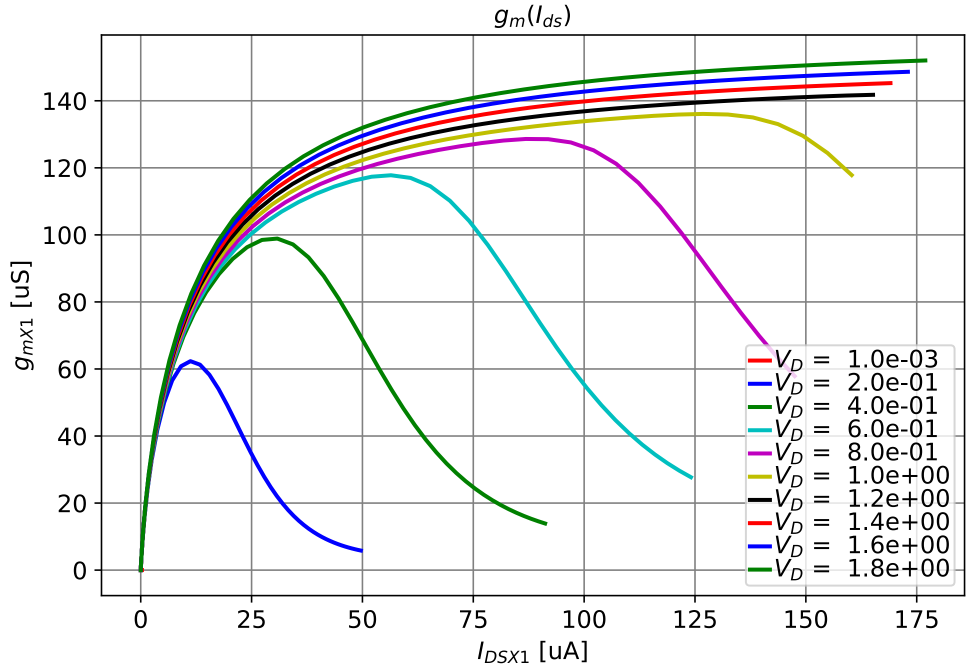

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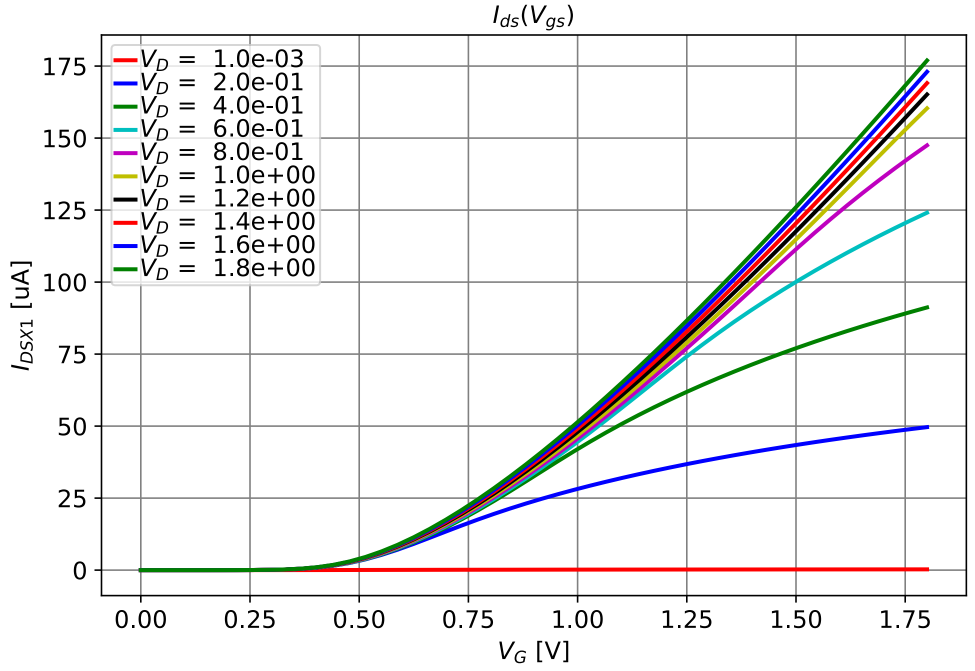
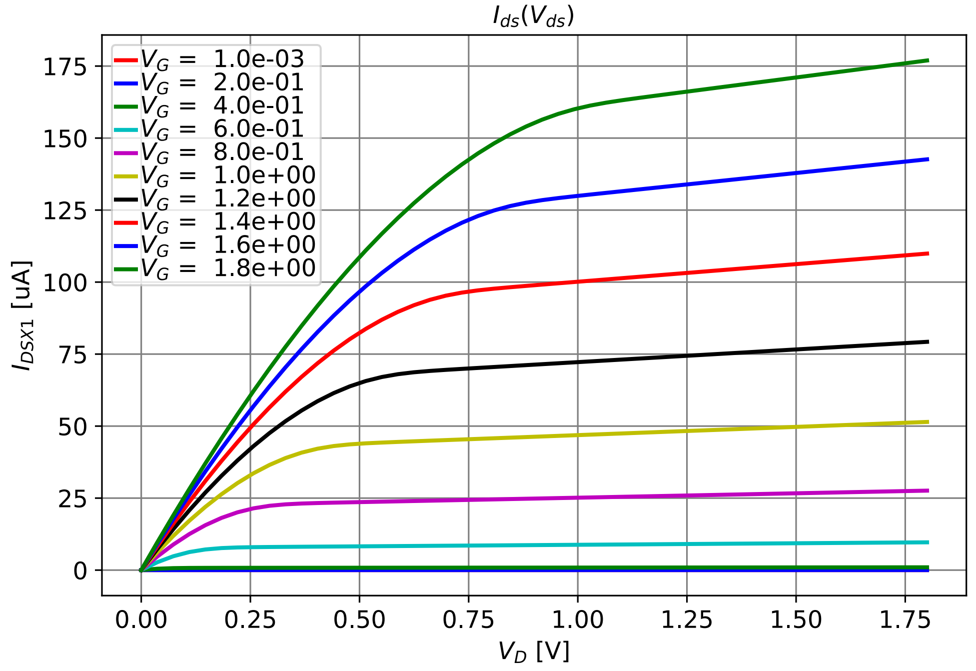
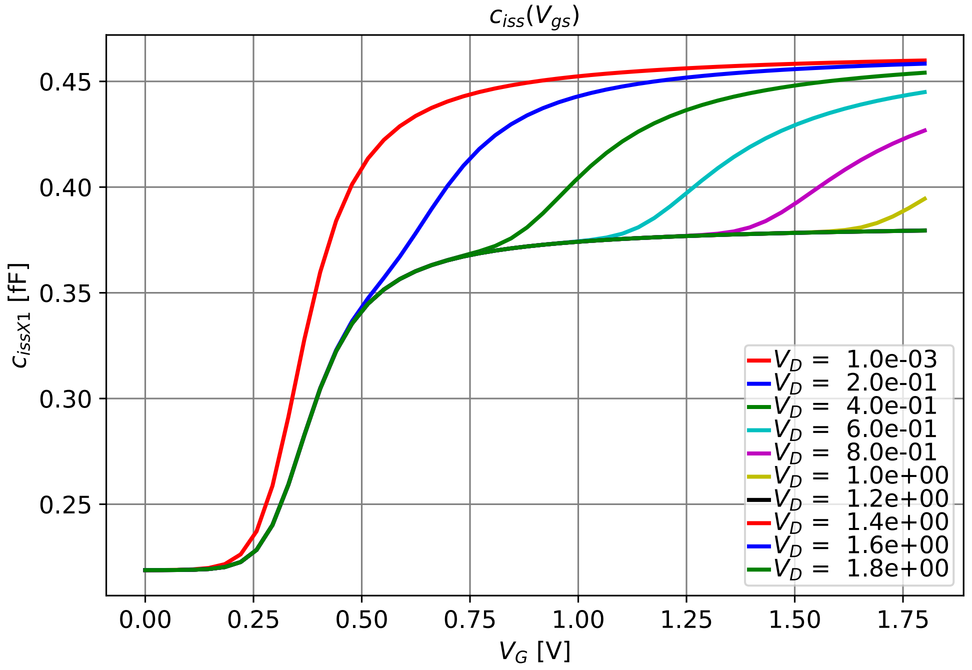
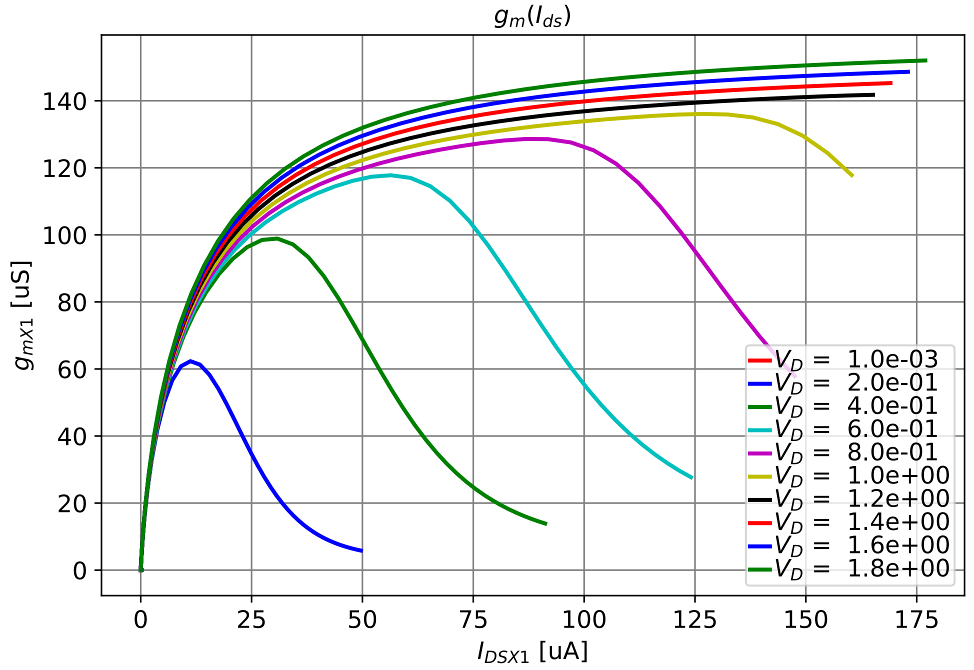
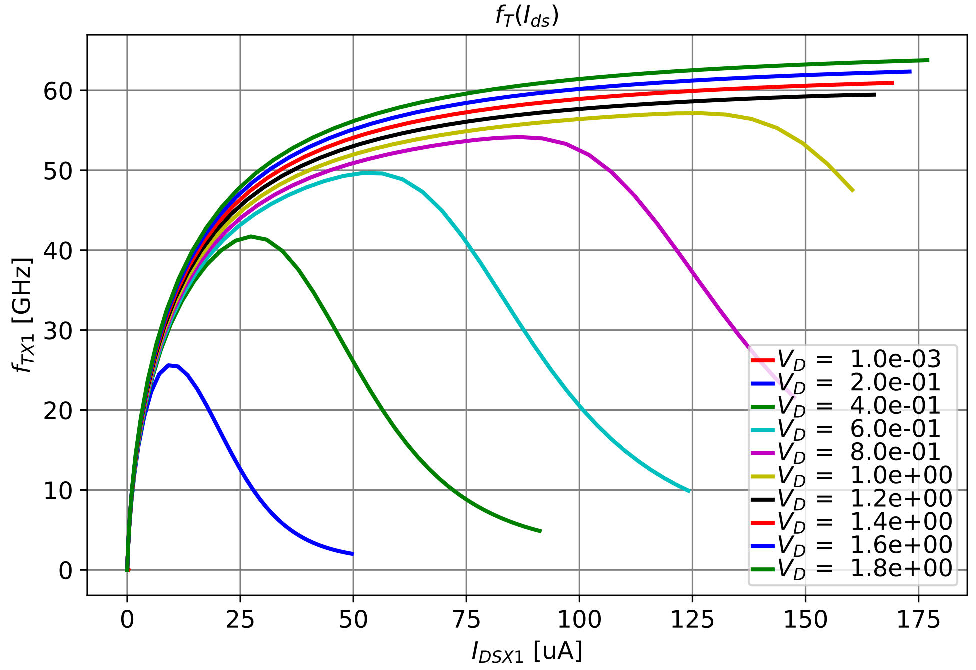
Tradeoffs and Optimization in Analog CMOS Design



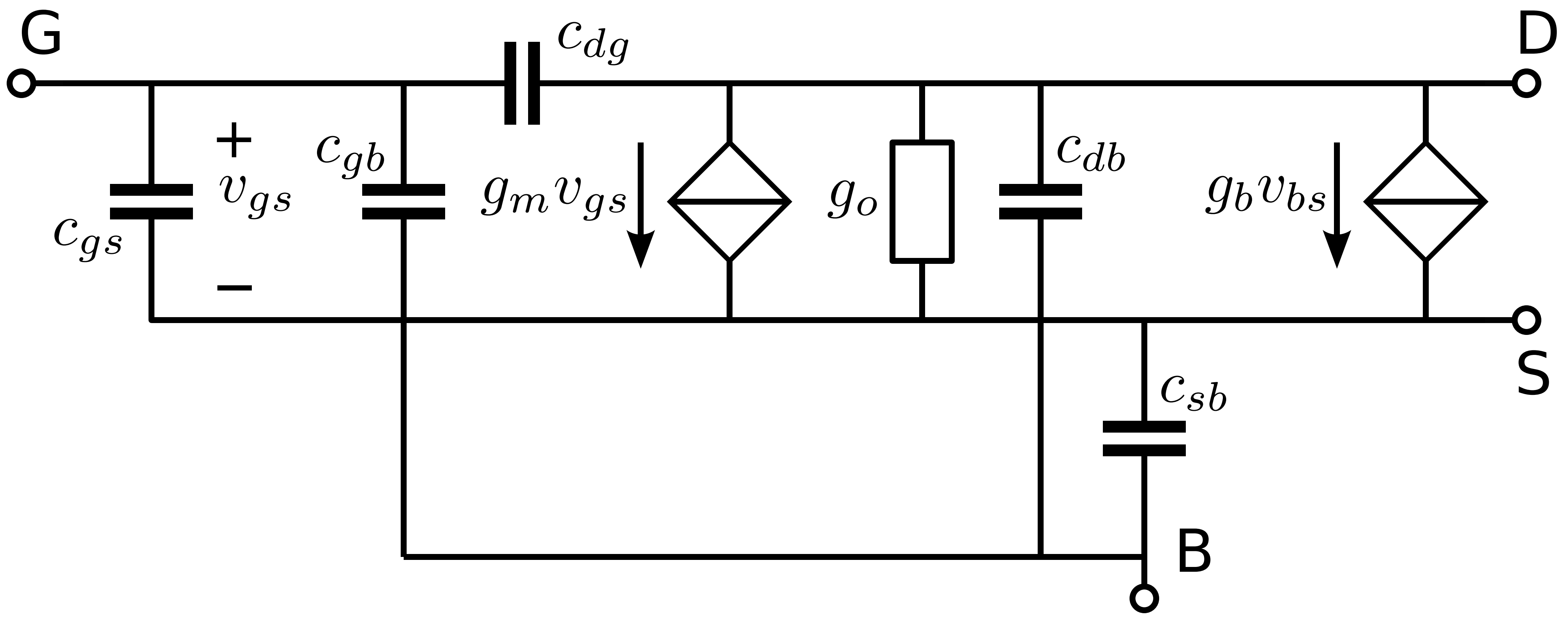








SLiCAP MOS small-signal model

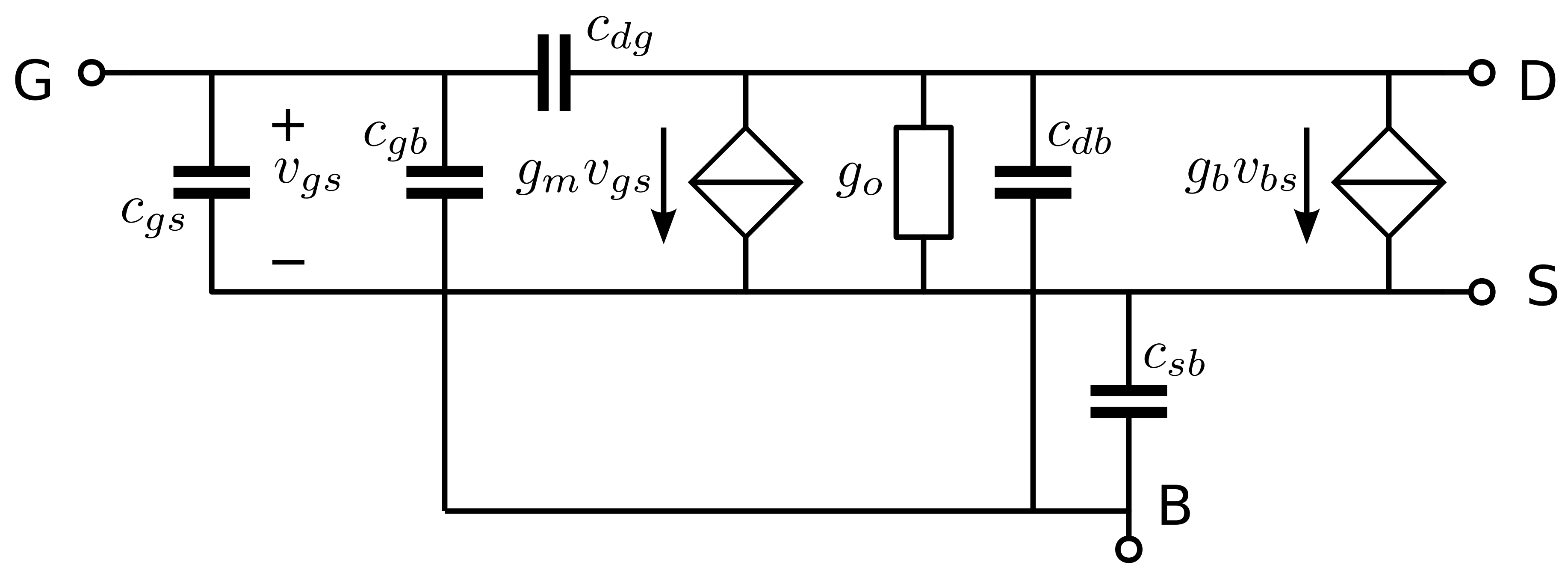


Physical constants CMOS18 technology parameters, and subcircuits with device equations in SLiCAP.lib

SLiCAP MOS small-signal model

`Technology parameters:` $n, C_{OX}, I_0, V_{AL}, \Theta, E_{CRIT}, C_{GSO}, C_{GBO}, C_{JBO}$

`Circuit parameters:` ID, W, L



$$IC = \frac{ID}{I_0}\frac{W}{L}$$

$$IC_{CRIT} = \frac{1}{\left(4nU_T\left(\Theta + \frac{1}{E_{CRIT}L}\right)\right)^2}$$

$$g_m = \frac{ID}{4nU_T\sqrt{IC\left(1+\frac{IC}{IC_{CRIT}}\right)+0.5\sqrt{IC\left(1+\frac{IC}{IC_{CRIT}}\right)+1}}}$$

$$g_b = g_m\left(n-1\right)$$

$$g_o = \left(g_{DIBL}\right) + \frac{ID}{V_{AL}L}$$

$$c_{gs} = \frac{2-x}{3}C_{OX}WL + C_{GSO}W$$

$$c_{dg} = C_{GSO}W$$

$$c_{gb} = \frac{1-x}{3}C_{OX}WL\frac{n-1}{n} + C_{GBO}2L$$

$$c_{db} = C_{JB0}L_{DS}W$$

$$c_{sb} = C_{JB0}L_{DS}W$$

$$x = \frac{\sqrt{IC+0.25}+1.5}{\left(\sqrt{IC+0.25}+0.5\right)^2}$$